AMCS1013

Problem Solving and Programming

Assignment

2025/26

Programme	:	
Tutorial Group	:	
Tutor Name	:	Mr. Lim Kah Lok
Date Submitted to Tutor	:	9 September 2025

Team Members:

No	Student Name	Student ID
1.	Chia Koh Suan	
2.	Siow Pei En	

No.	Team Member	Task(s) Completed (in detailed including mention the programming and reporting tasks)	Overall Contribution (%)
1.	Chia Koh Suan	Programming: I completed the entire customer module. My teammate helped with integration of file handling functions and data management components. Reporting: - Draw the flowchart for the function of the entire customer module. - Design the structure chart with teammates together. - Write two of the test data for program testing and outputs from the customer module. - Source code for Customer module	50%
2.	Siow Pei En	Programming: I completed the entire staff Module and Data Module Reporting: - Brief description - Losing out the constants and variables - Draw flowchart for staff module and data module - Write the program testing for staff - Source code for staff module and data module - Design the structure chart with teammates together.	50%
		Total	100

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1.0 BRIEF DESCRIPTION / PURPOSE

1.1 PURPOSE OF THE PROGRAM — BOOK ORDERING SYSTEM

- The purpose of this program is to provide a **digital platform for customers to order books online. At the same time, staff are also allowed to manage this system. It reduces manual work and enhances efficiency to perform the tasks.**

- Main Objectives

1. **Staff:**
    - Manage the book catalogue such as add new books, update book info and remove books
    - Monitor and update stocks
    - Generate daily and monthly sales report
    - Generate **order reports** for tracking customer orders
2. **For Customers:**
    - Browse and view the book catalogue.
    - Add books into a shopping cart.
    - Place and process orders
    - View order history

1.2 Module Explanation and Task of Each Number

Member A: Siow Pei En

Responsibilities:

1. Data Module

   Function: loadBook(), saveBook(), displayBook(), loadOrder(), saveOrder(), loadCustomer(), saveCustomer(), getCurrentDate()

   Purpose: read and save the data.

2. Main Module

   Function: main(), mainMenu()

   Purpose: Entry point of the program. It initializes the system, loads data and display the first interface.

1. Staff Login Module

   Function: staffLogin()

   Purpose: Allows staff to log into the system using valid username and password.

2. Staff Menu Module

   Function: staffMain(), staffMenu()

   Purpose: Provides the main menu for staff.

3. Book Catalogue Module

   Function: bookcatalogue()

   Purpose: display available book catalogue for staff and cutomers

4. Menu Management Module

   Function: menuManage(), updateBook(), removeBook(), addBook()

   Purpose: Enables staff to manage the book

5. Stock Management Module

   Function: stock()

   Purpose: allow staff to monitor and update the stock for each book

6. Sales Report Module

   Function: salesReport()

   Purpose: allow staff to generate daily and monthly sales report by entering date

7. Order Reporting Module

   Function: orderReport()

   Purpose: allow staff to generate customer’s order report

Member B: Chia Koh Suan

Responsibilities:

1. Navigation Module

   Function: clearScreen(), backToMenu()

   Purpose: Provides supporting utilities for smooth navigation.

2. Customer Login and Registration Module

   Function: customerLogin(), customerRegistration()

   Purpose: allow user register a new account and login into system using valid username and password

3. Customer Menu Module

   Function: customerMenu()

   Purpose: Provides the main menu for customers

4. Browse Book Module

   Function: browseBooks()

   Purpose: display available book catalogue for customers

5. Order Module

   Function: order_Entry(), addToCart(), ViewShoppingCart()

   Purpose: Handles customer ordering process

6. Checkout Module

   Function: processCheckout()

   Purpose: Handles customer making payment process

7. Order History Module

   Function: orderHistory()

   Purpose: Displays past orders made by a customer

8. Clearing Cart Module

   Function: clearShoppingCart()

   Purpose: automatically clear the shopping cart while the making payment process is done.

2.0 CONSTANTS & VARIABLES

2.1 Table for Constants

Constant	Data type	Value	Purpose
MAX_CART_ITEMS	**int**	**20**	**Define the maximum number of items in shopping cart**
MAX_BOOKS	**int**	**100**	**Define the maximum number of books in book catalogue**
MAX_CUSTOMER	**int**	**100**	**Define the maximum number of registered customer**

2.2 Table for Variables

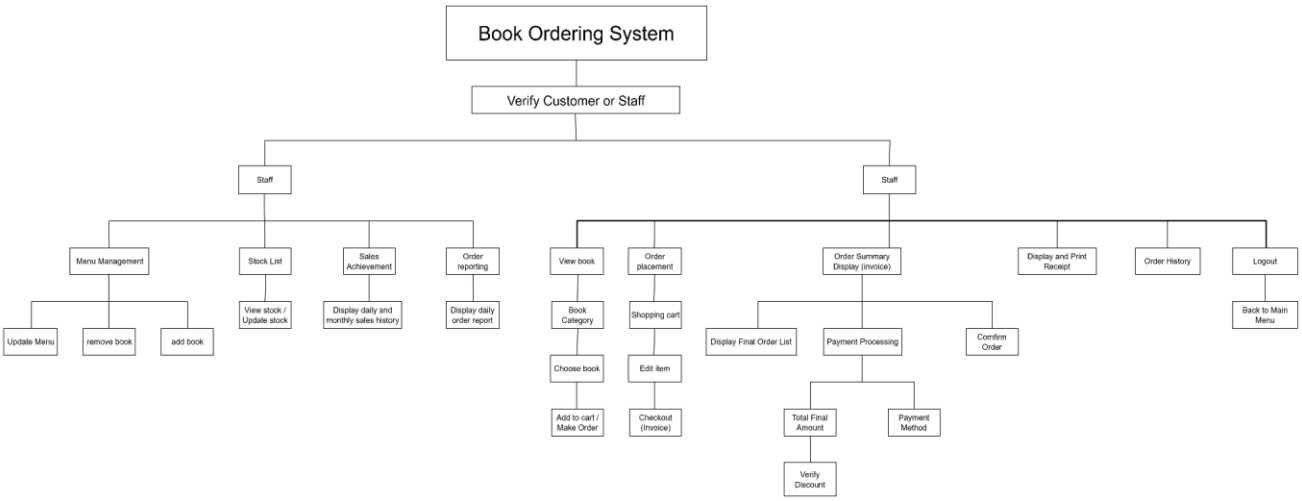
Variable	Data Tyoe	Purpose
bookcatalogue	**bookcatalogue[MAX_BOOKS]**	**Store all available books info**
customer_info	**customer_info[MAX_CUSTOMER]**	**Store all registered customer info**
order_detail	**vector<order>**	**store all order details**
daily_sales	**sales[1000]**	**store daily sales**
monthly_sales	**sales[1000]**	**store monthly sales**
total_sold	**int[MAX_BOOKS]**	**store how many copies of each book have been sold**
bookCount	**int**	**store total number of books**
customerCount	**int**	**store total number of customers**
dailyCount	**int**	**store number of daily sales records**
monthlyCount	**int**	**store number of monthly sales records**
key	**char**	**wait for user to press enter key to return**
username	**string**	**store the username of customer**
password	**string**	**store the password of customer**
choice	**int**	**store user selection**
found	**bool**	**check if an item exists**
id	**int**	**store book id**

i	**int**	**loop counters**
newBook	**book_catalogue**	**store new book info**
j	**int**	**loop counters**
k	**int**	**loop counters**
date	**string**	**store date entered by user/ current date**
month	**string**	**store month entered by user/current month**
total_revenue	**double**	**store total sales revenue**
total_quantity	**int**	**store total sales quantity**
cart	**CartItem[MAX_CART_ITEMS]**	**store the items in shopping cart**
cartCount	**int**	**store the number of items in shopping cart**
order_id	**int**	**store the order id**
loginSuccess	**bool**	**check if the login action is successful**
newCustomer	**customer**	**store new customer info**
customerChoice	**int**	**store customer selection**
already_quantity	**int**	**store the quantity of items that have been added into shopping cart**
quantity	**int**	**store the quantity of books ordered**
continueOrder	**char**	**store the customer choice to continue ordering**
orderNum	**int**	**store the number of order**
confirm	**char**	**store the customer choice to remove an item in cart**
itemTotal	**int**	**store the total cost of a cart item**
confirmCheckout	**char**	**store the customer choice to continue checkouting**
typePaymentMethod	**string**	**stores selected payment method name**
total	**double**	**store the items in cart total price**
subtotal	**double**	**store the total price of items ordered before tax**

paymentMethod	**char**	**store the selected payment method code**
discount	**double**	**store the discount amount applied**
serviceTax	**discount**	**store the amount of service tax needed to pay**
SST	**double**	**store the amount of sst need to pay**
totalPayable	**double**	**store the total amount needed to pay by customer**
paymentAmount	**double**	**store the payment pay by customer**
change	**double**	**store the amount of changes**
Total	**double**	**store the total amount of an item**

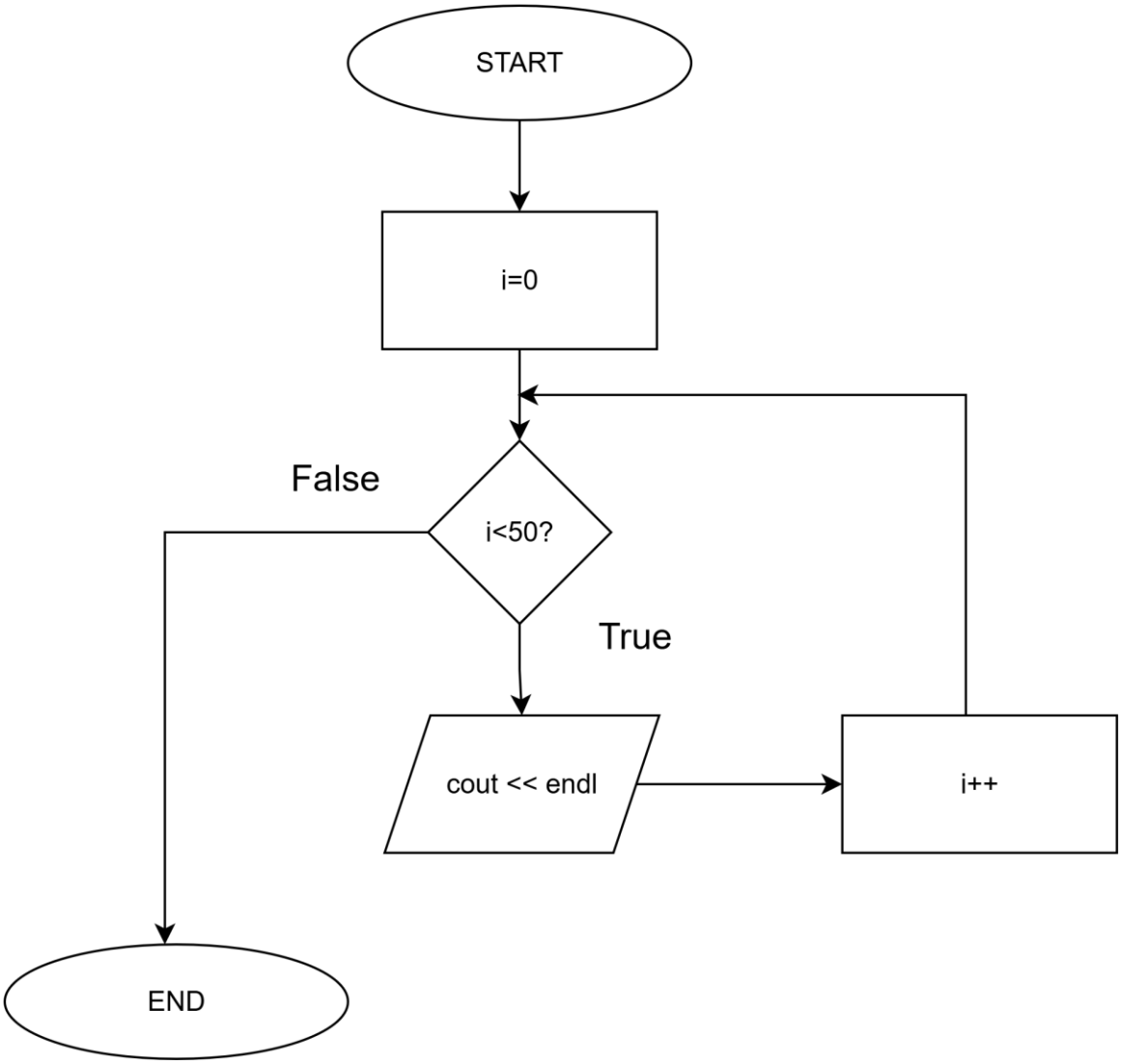
3.0 OVERALL PROGRAM DESIGN

3.1 Structure Chart

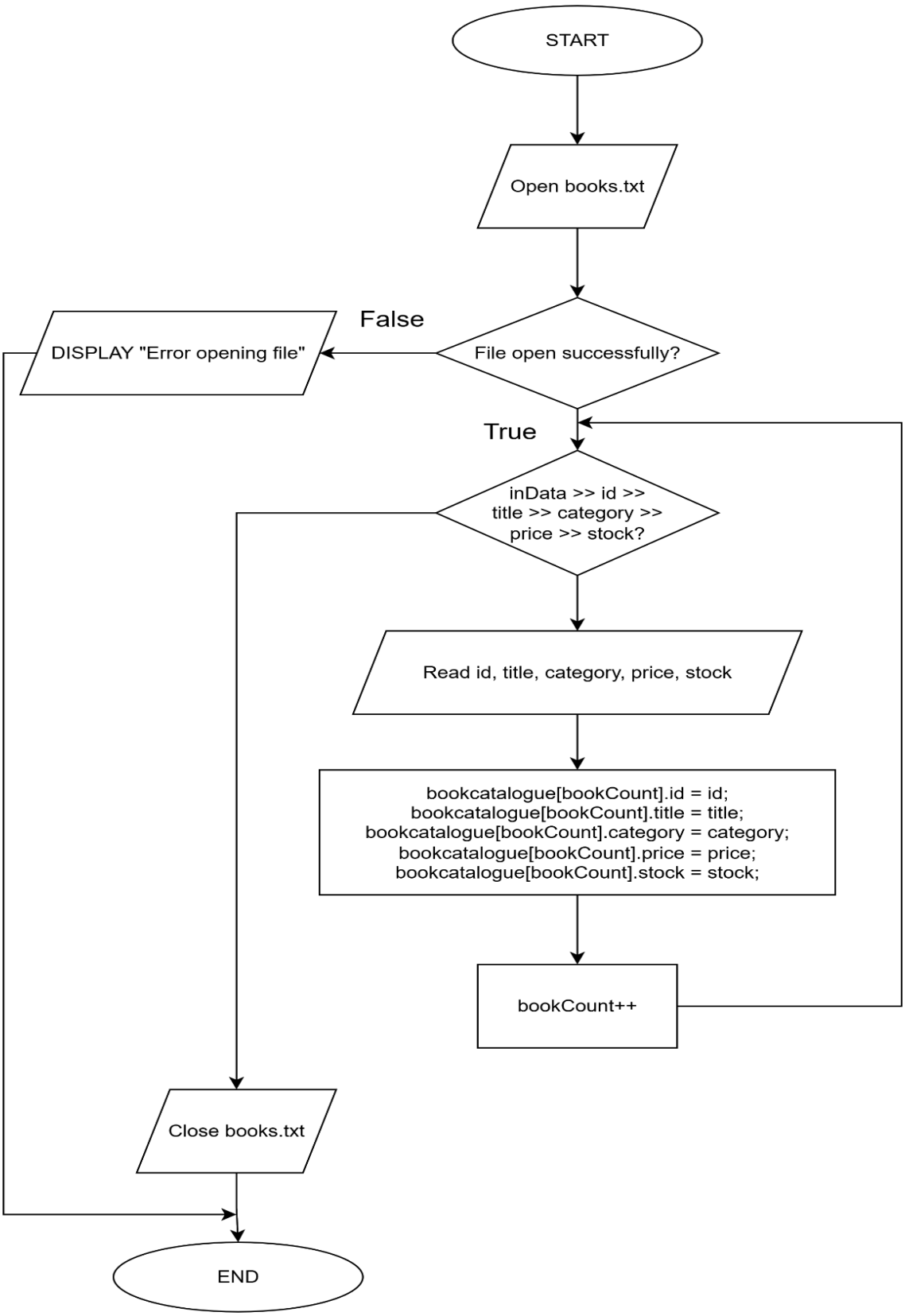


3.2 Flowchart

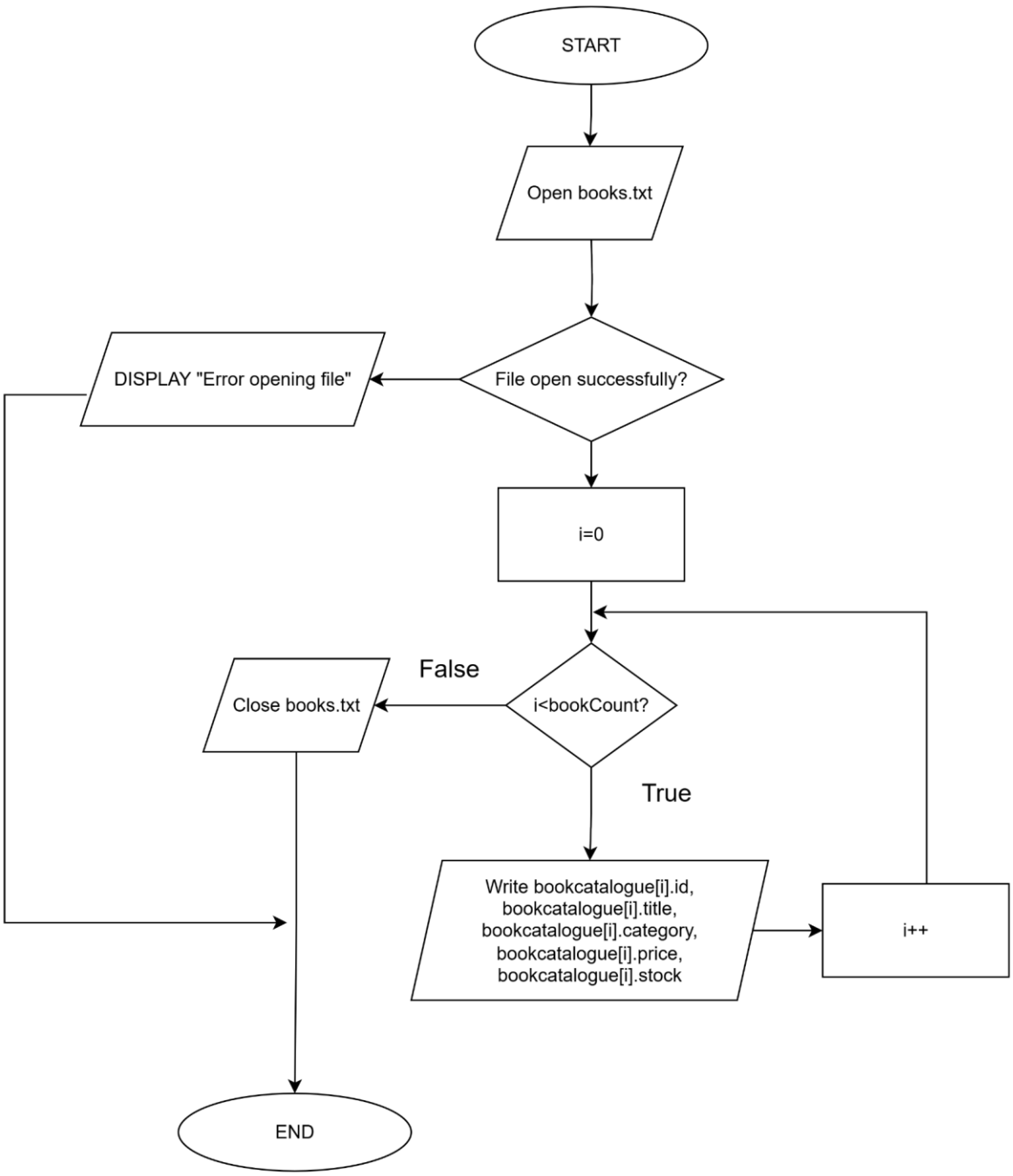
1. **void clearScreen();**



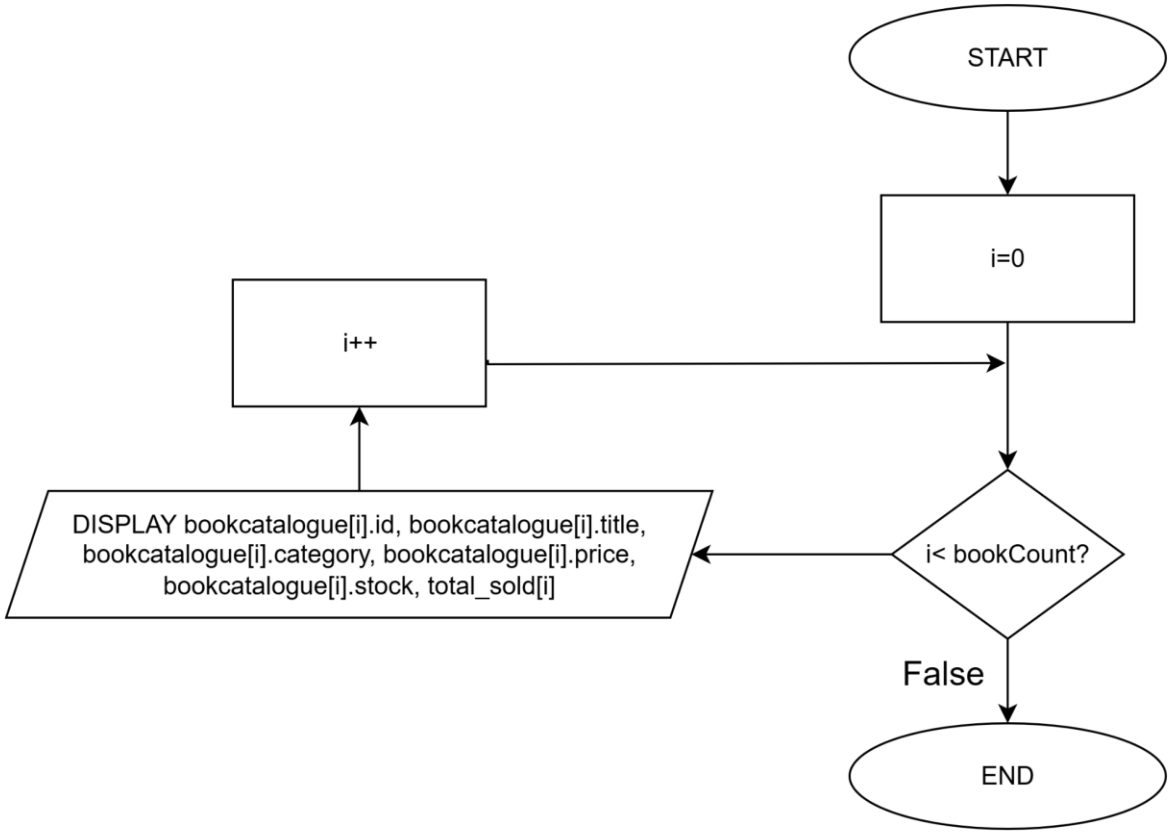
2. void loadBook();



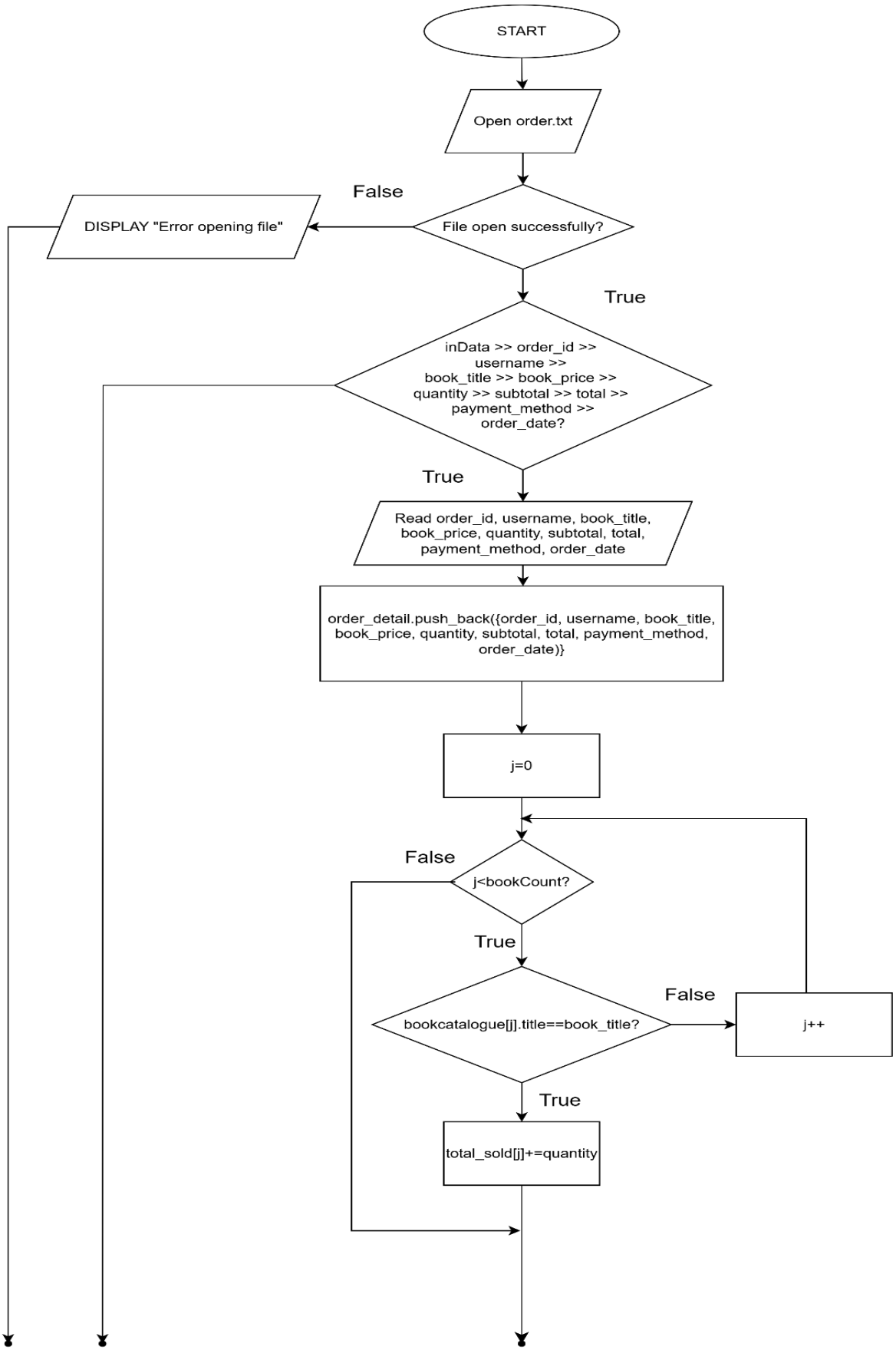
3. **void saveBook();**

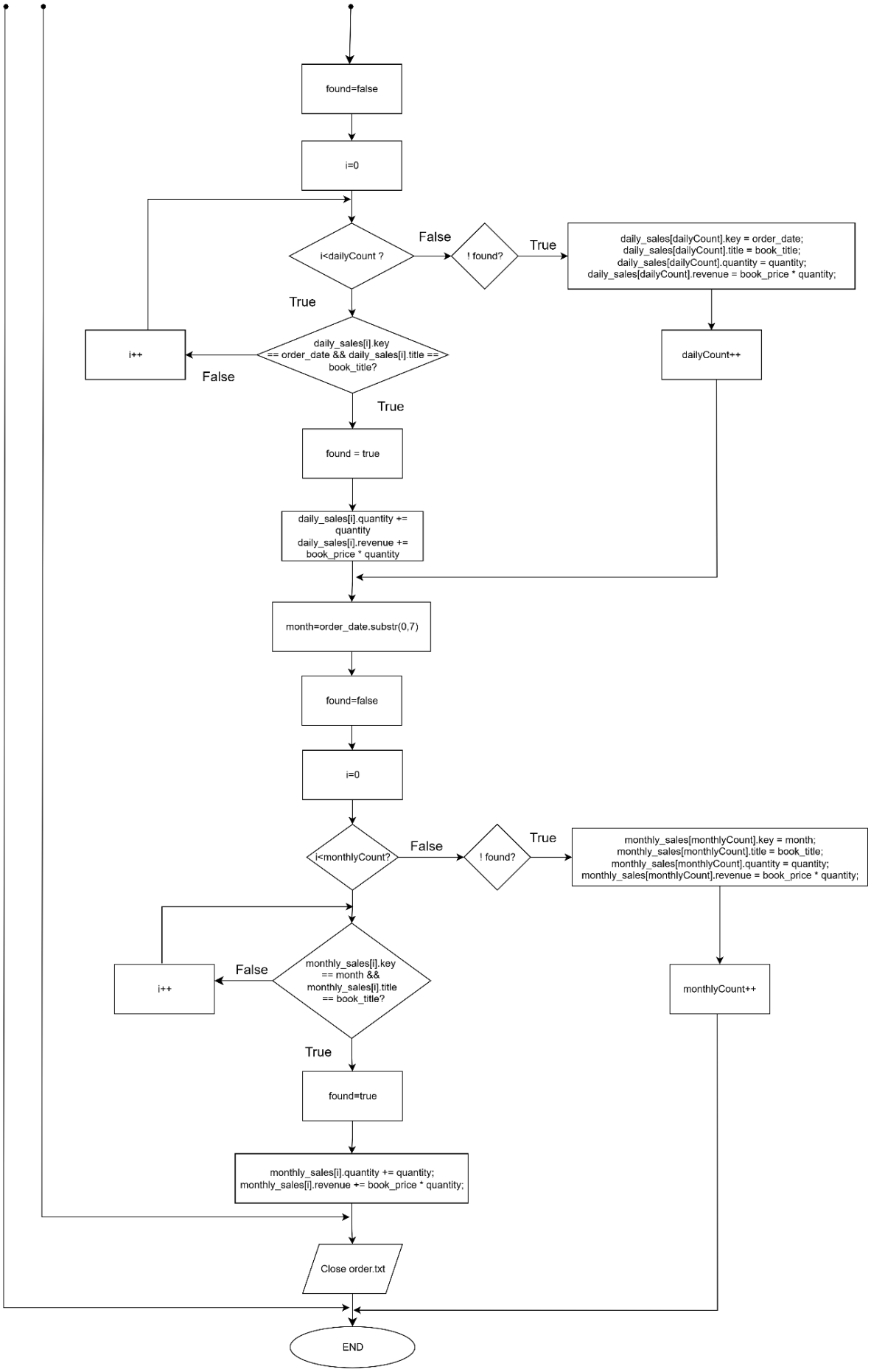


4. **void displayBook();**

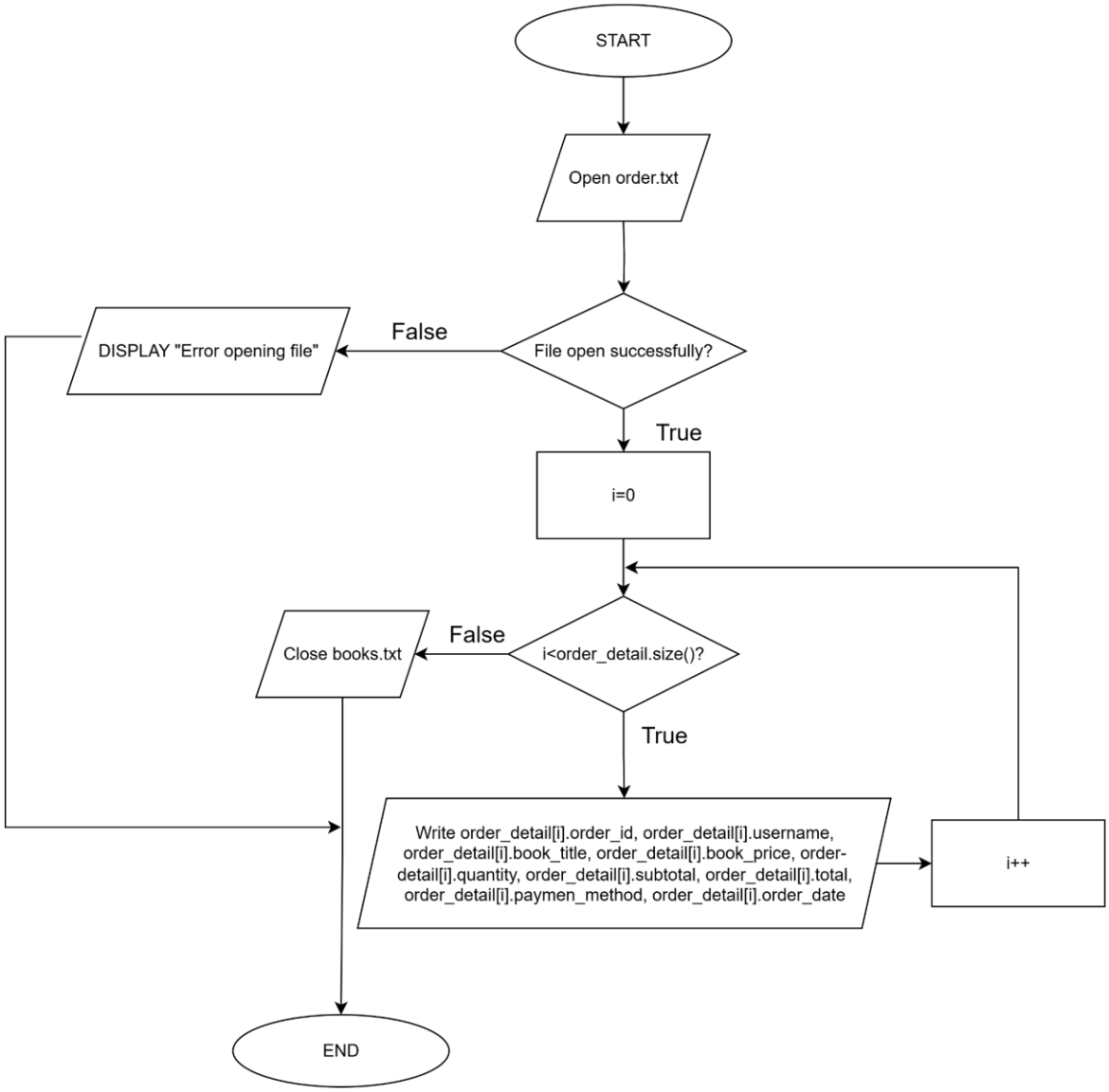


5. void loadOrder();

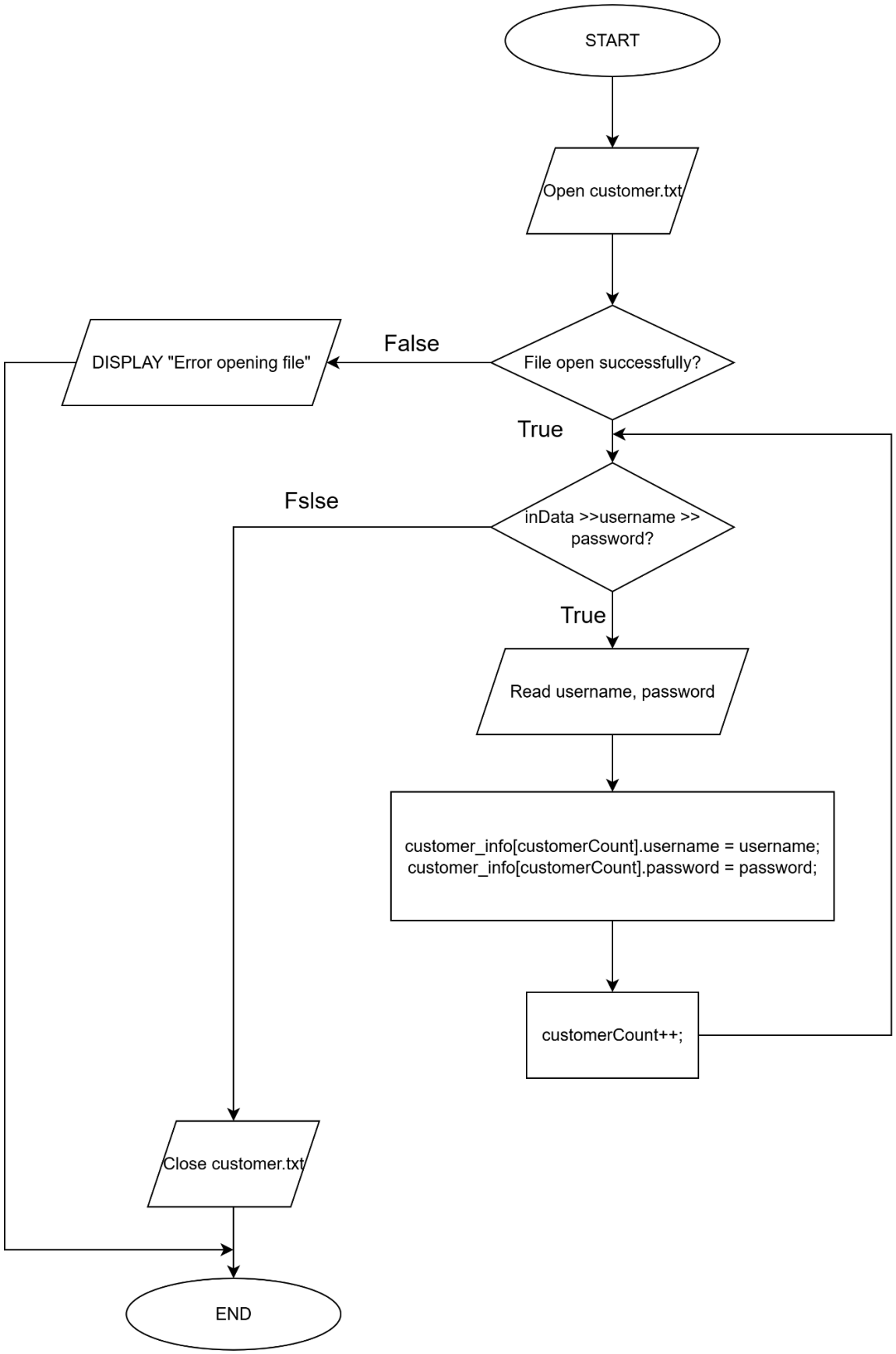



found=false
i=0
i<dailyCount ?
False
! found?
True
daily_sales[dailyCount].key = order_date;
daily_sales[dailyCount].title = book_title;
daily_sales[dailyCount].quantity = quantity;
daily_sales[dailyCount].revenue = book_price * quantity;
True
daily_sales[i].key
== order_date && daily_sales[i].title ==
book_title?
i++
False
dailyCount++
True
found = true
daily_sales[i].quantity +=
quantity
daily_sales[i].revenue +=
book_price * quantity
month=order_date.substr(0,7)
found=false
i=0
i<monthlyCount?
False
! found?
True
monthly_sales[monthlyCount].key = month;
monthly_sales[monthlyCount].title = book_title;
monthly_sales[monthlyCount].quantity = quantity;
monthly_sales[monthlyCount].revenue = book_price * quantity;
monthly_sales[i].key
== month &&
monthly_sales[i].title
== book_title?
False
i++
monthlyCount++
True
found=true
monthly_sales[i].quantity += quantity;
monthly_sales[i].revenue += book_price * quantity;
Close order.txt
END

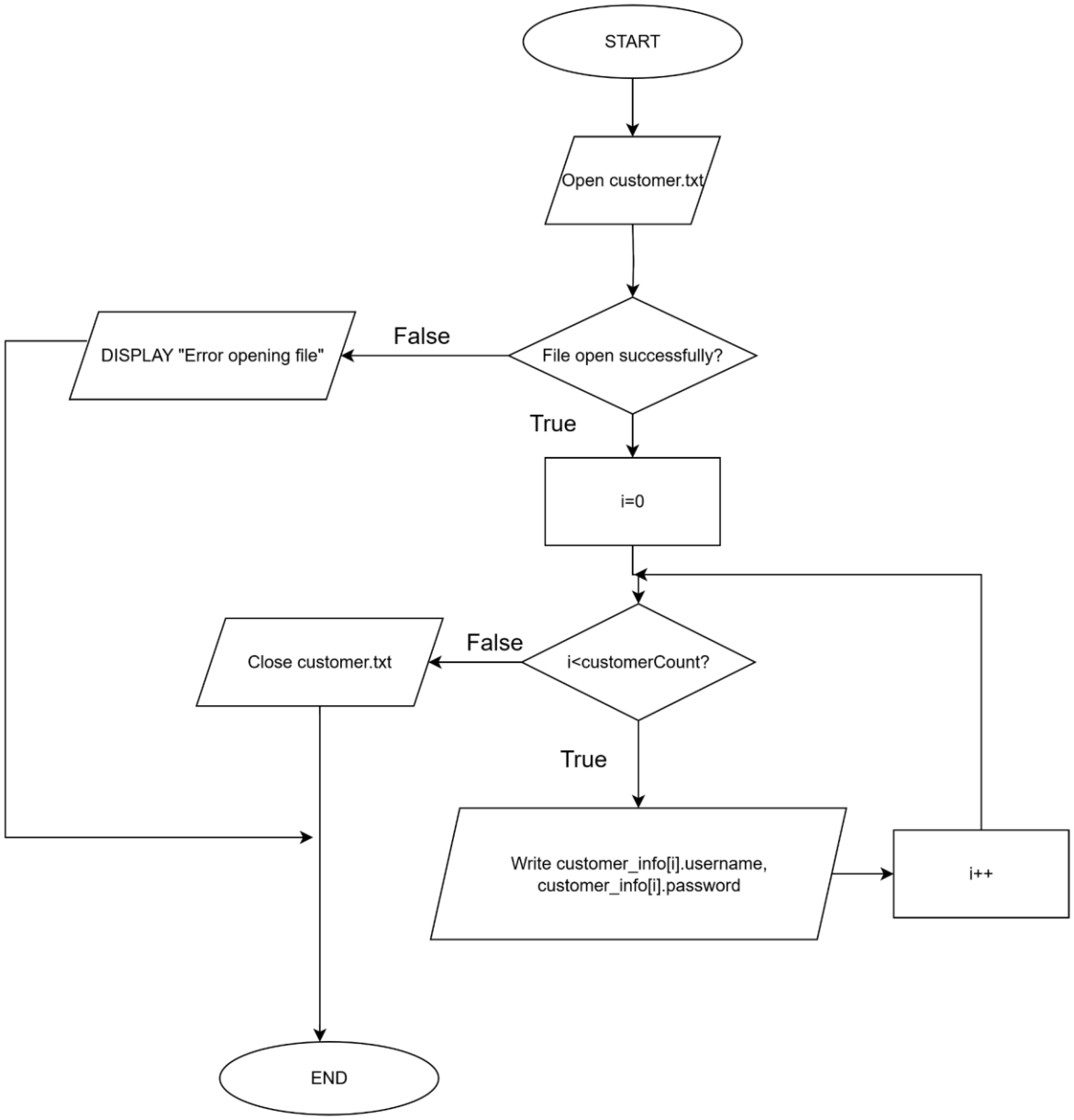
6. void saveOrder();



7. **void loadCustomer();**

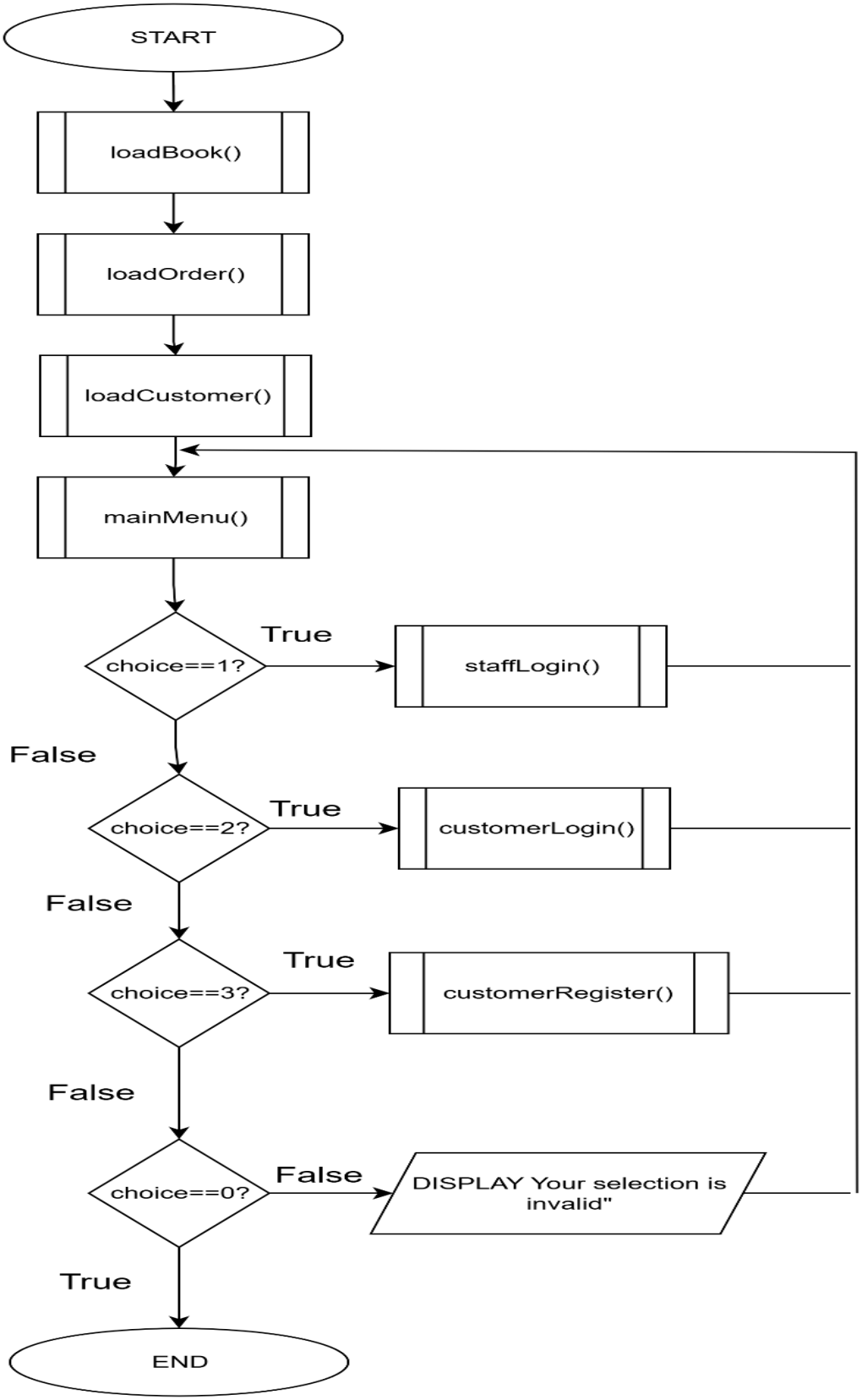


8. **void saveCustomer();**

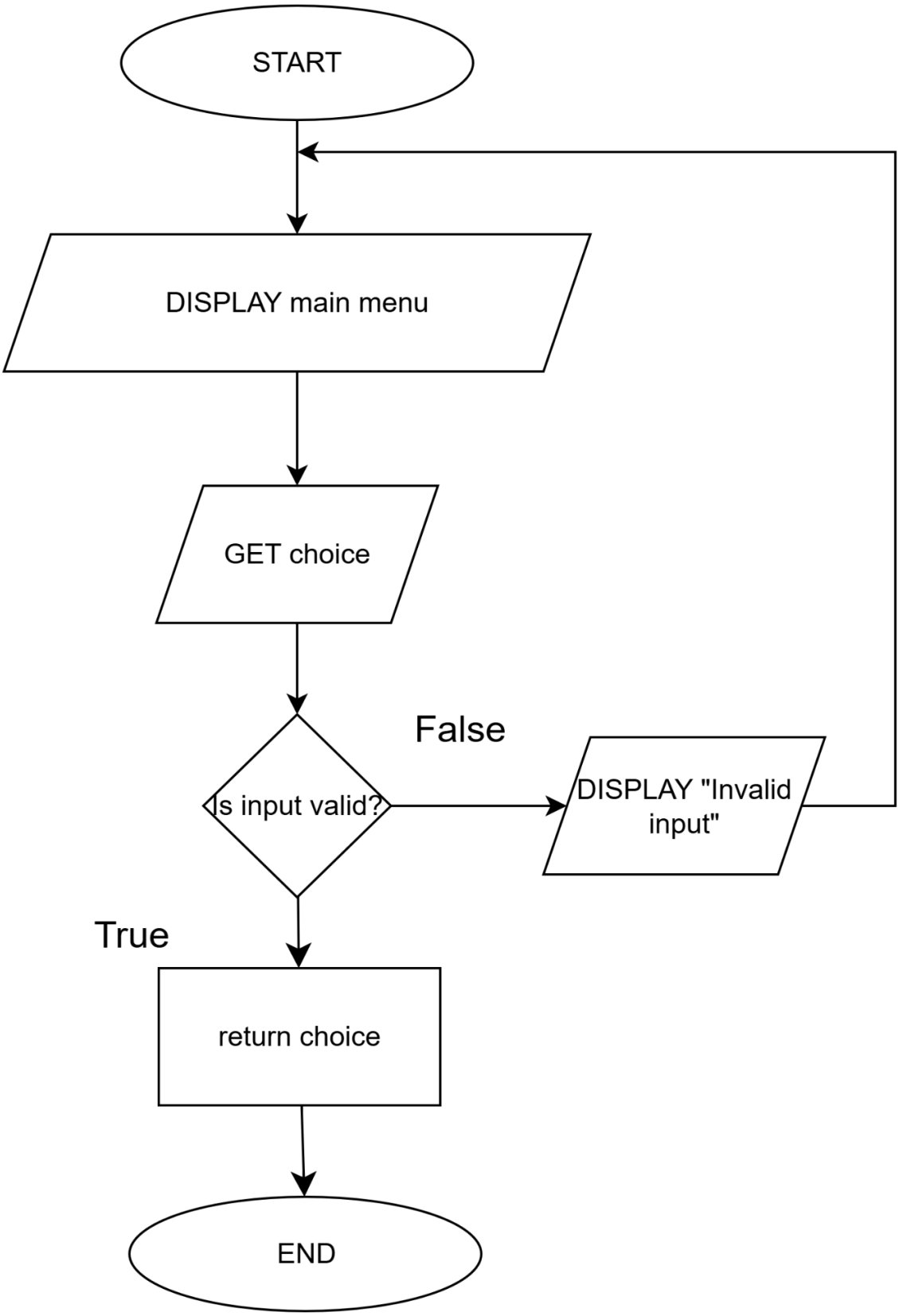


3.2.1 Main Flowchart

1. **int main();**

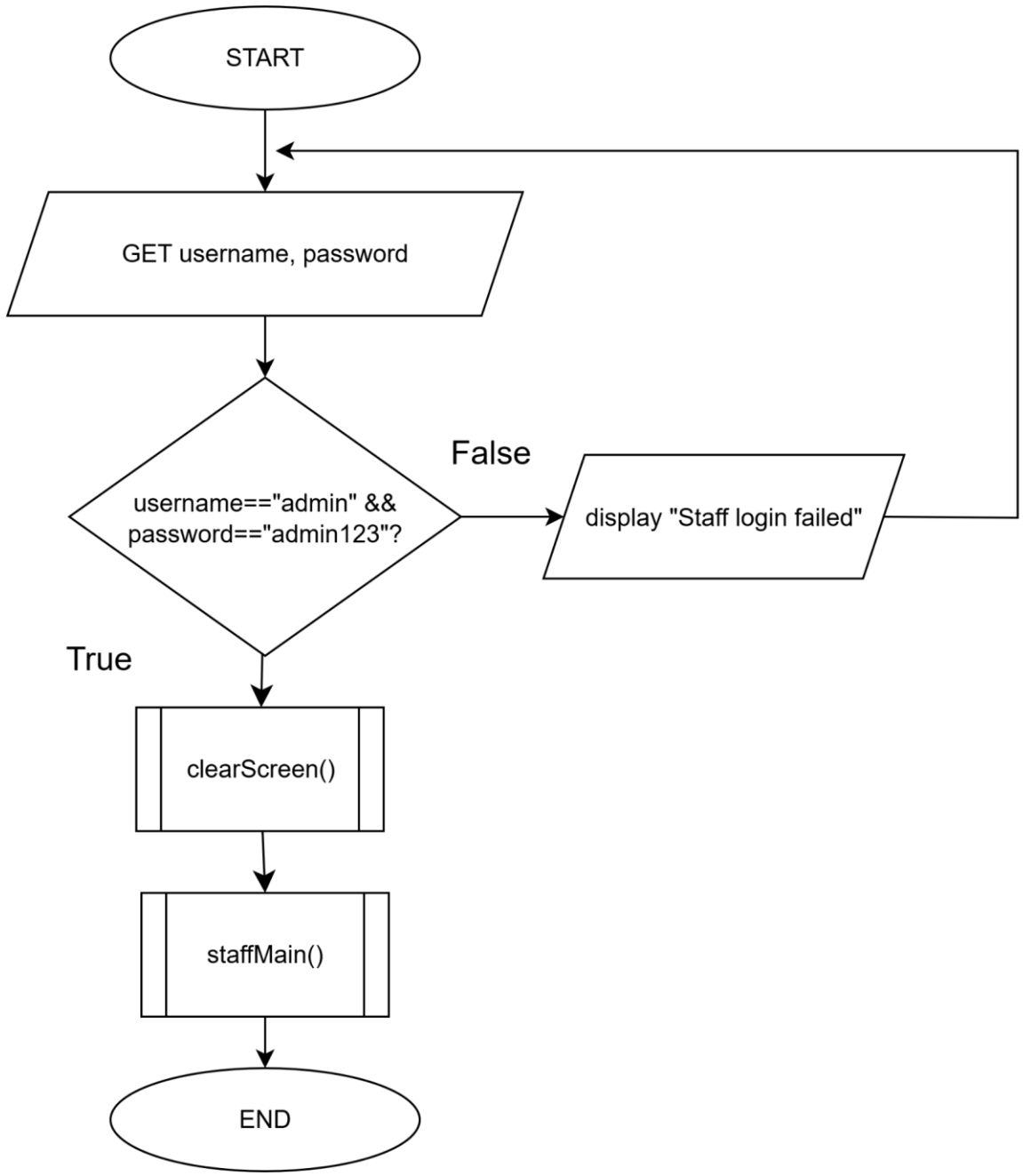


2. **int mainMenu();**

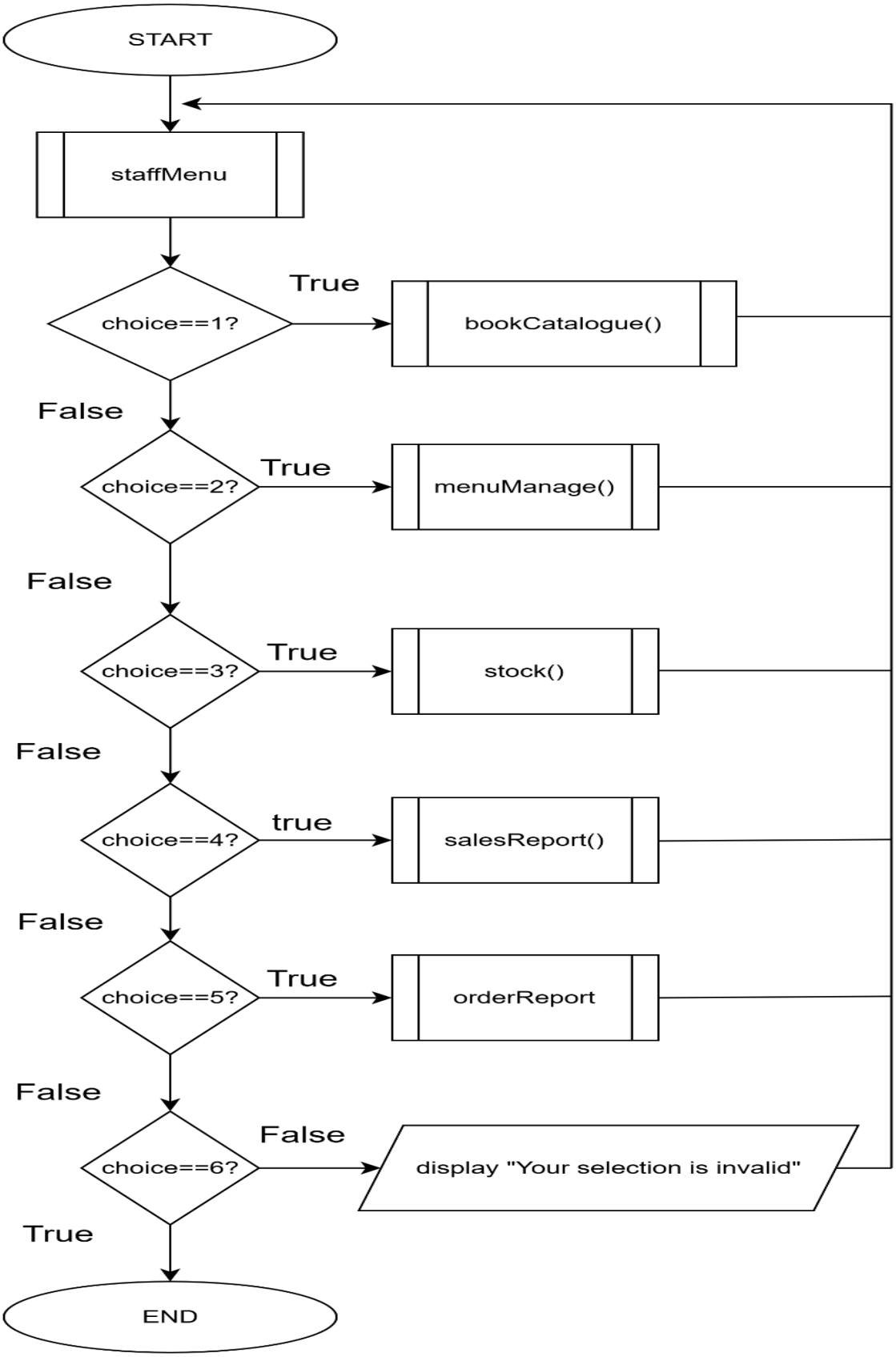


3.2.2 Staff Flowchart

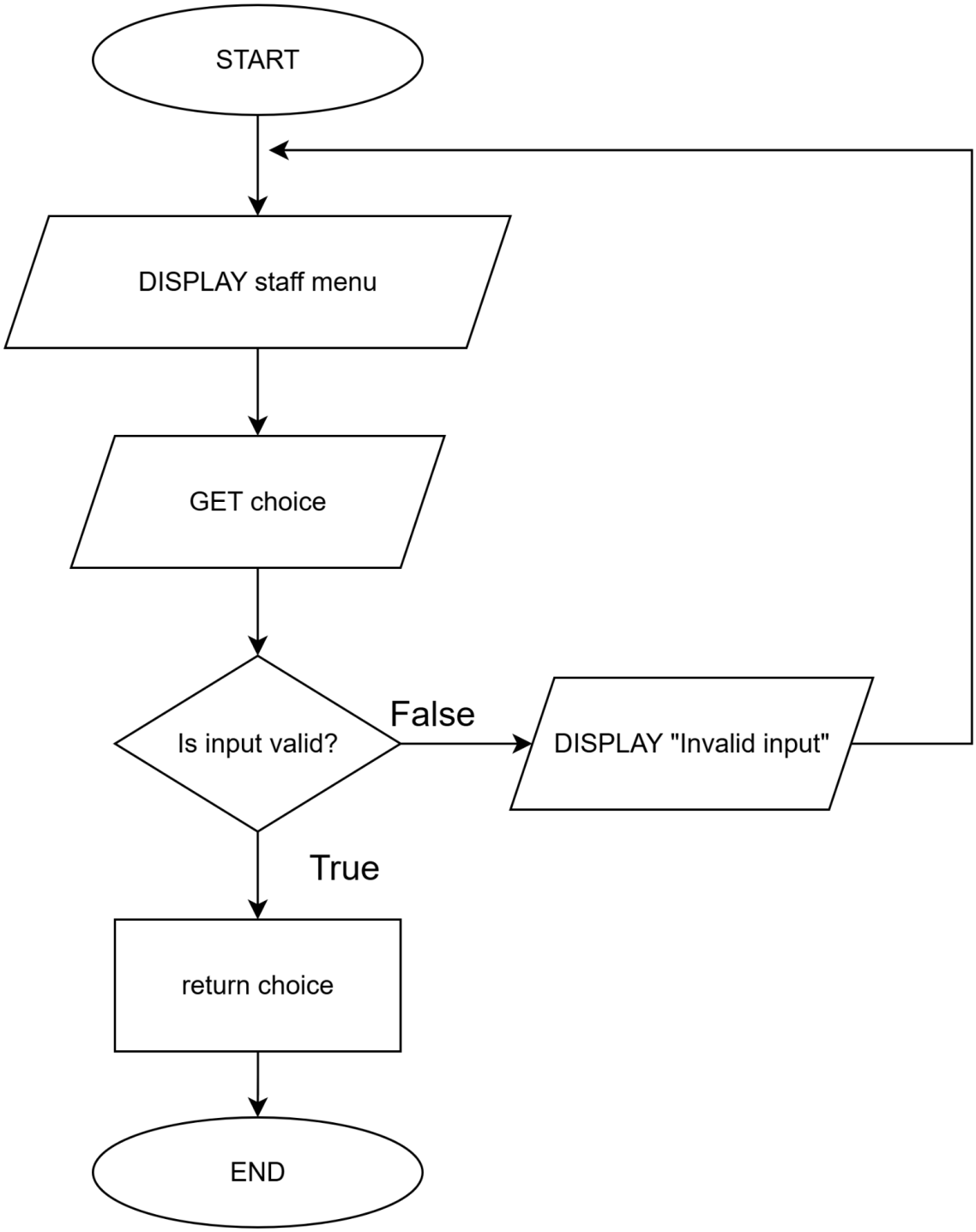
1. **void staffLogin();**



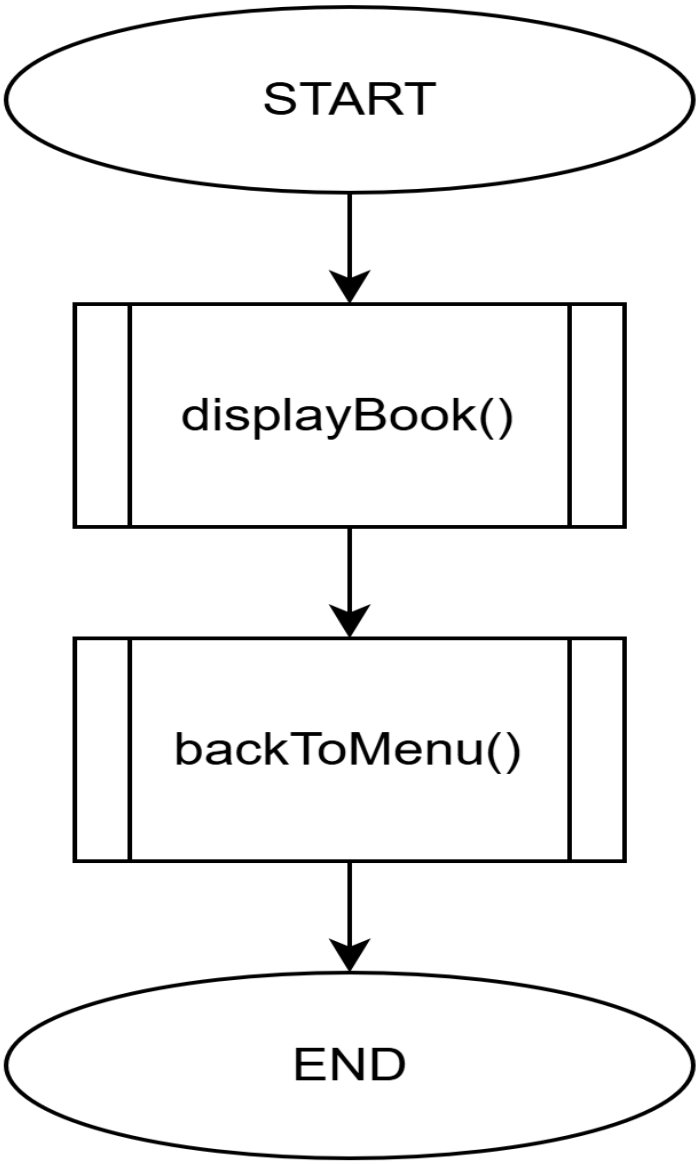
2. **void staffMain();**



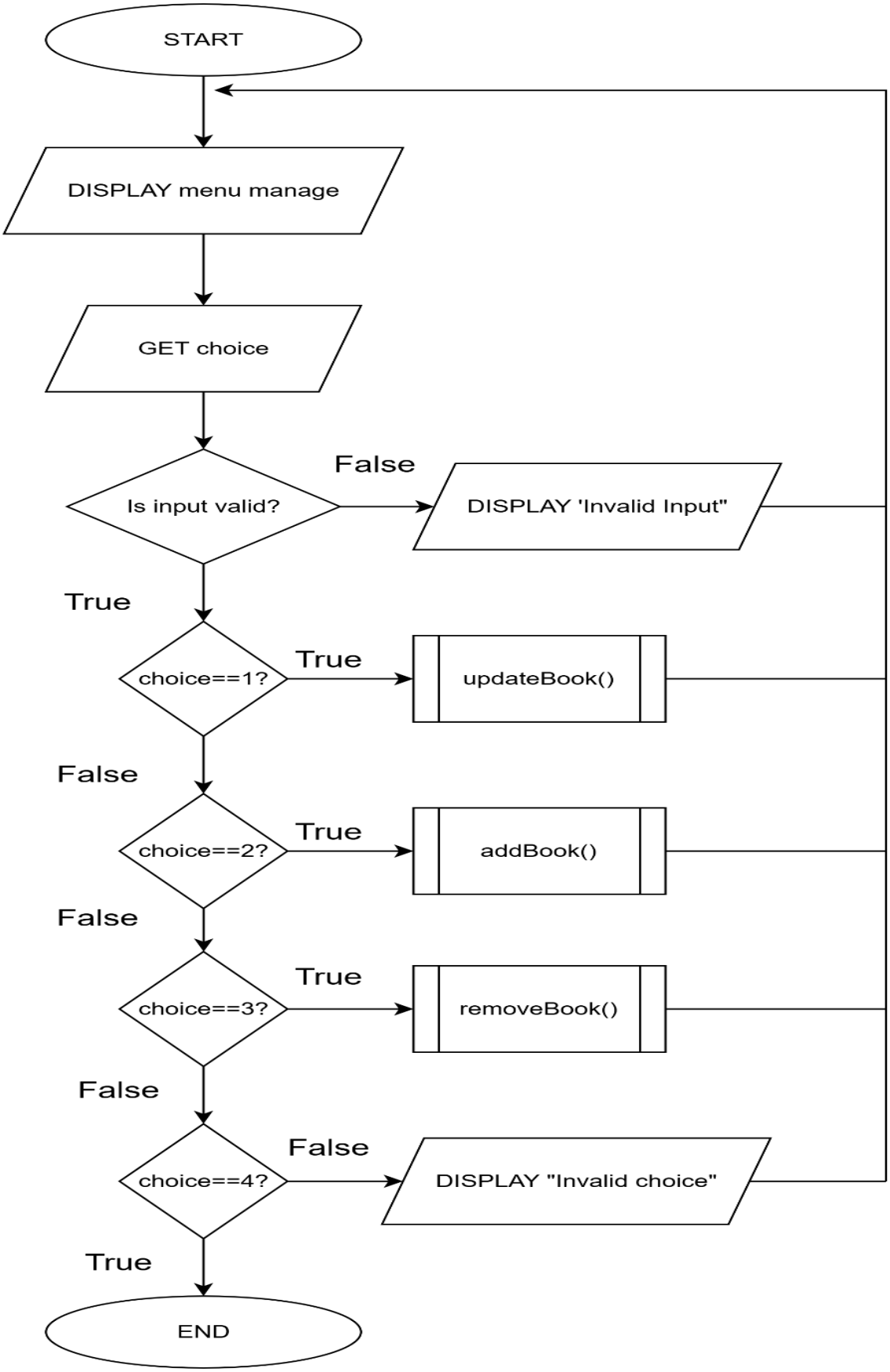
3. int staffMenu();



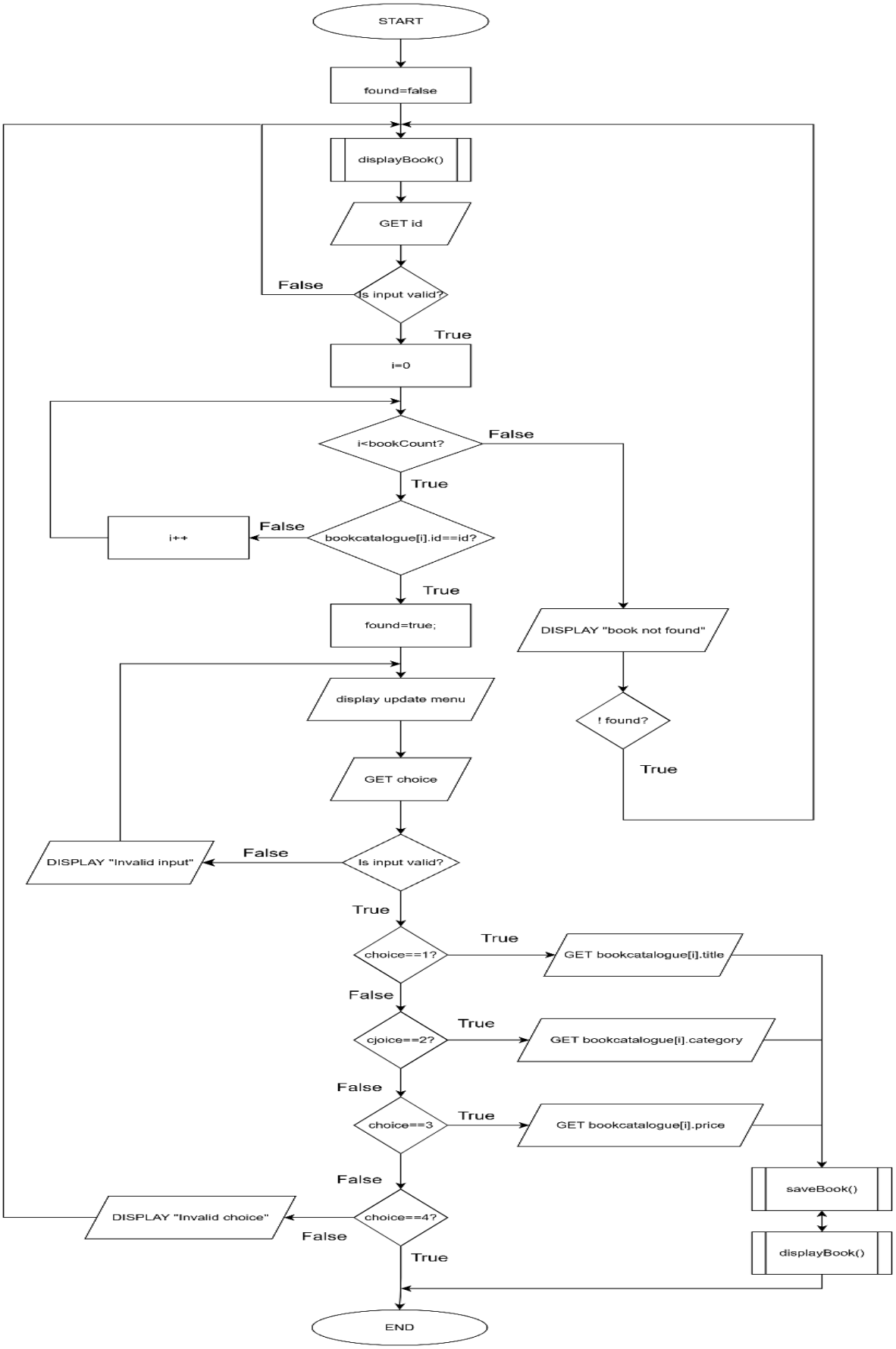
4. **void bookcatalogue();**



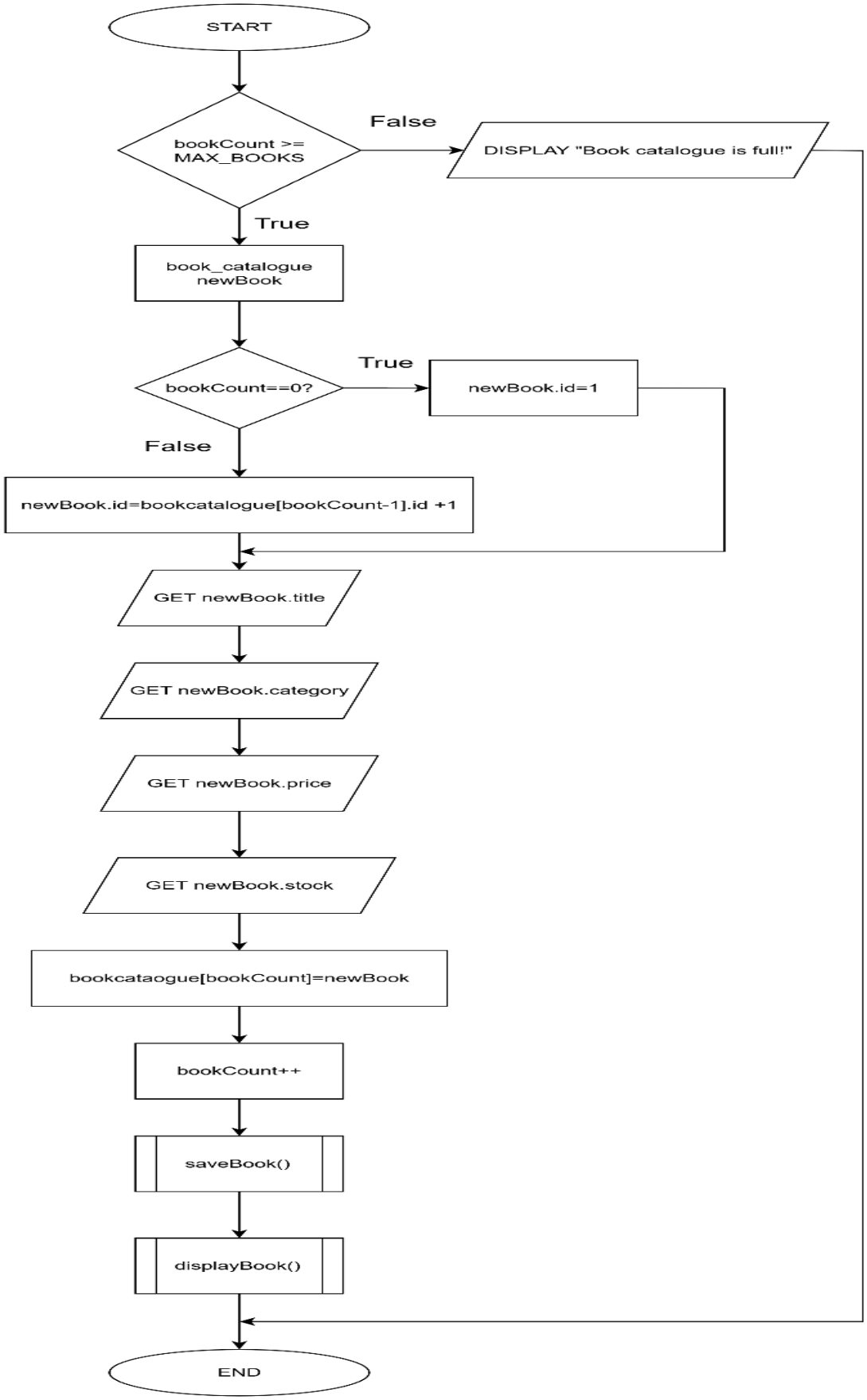
5. **void menuManage();**



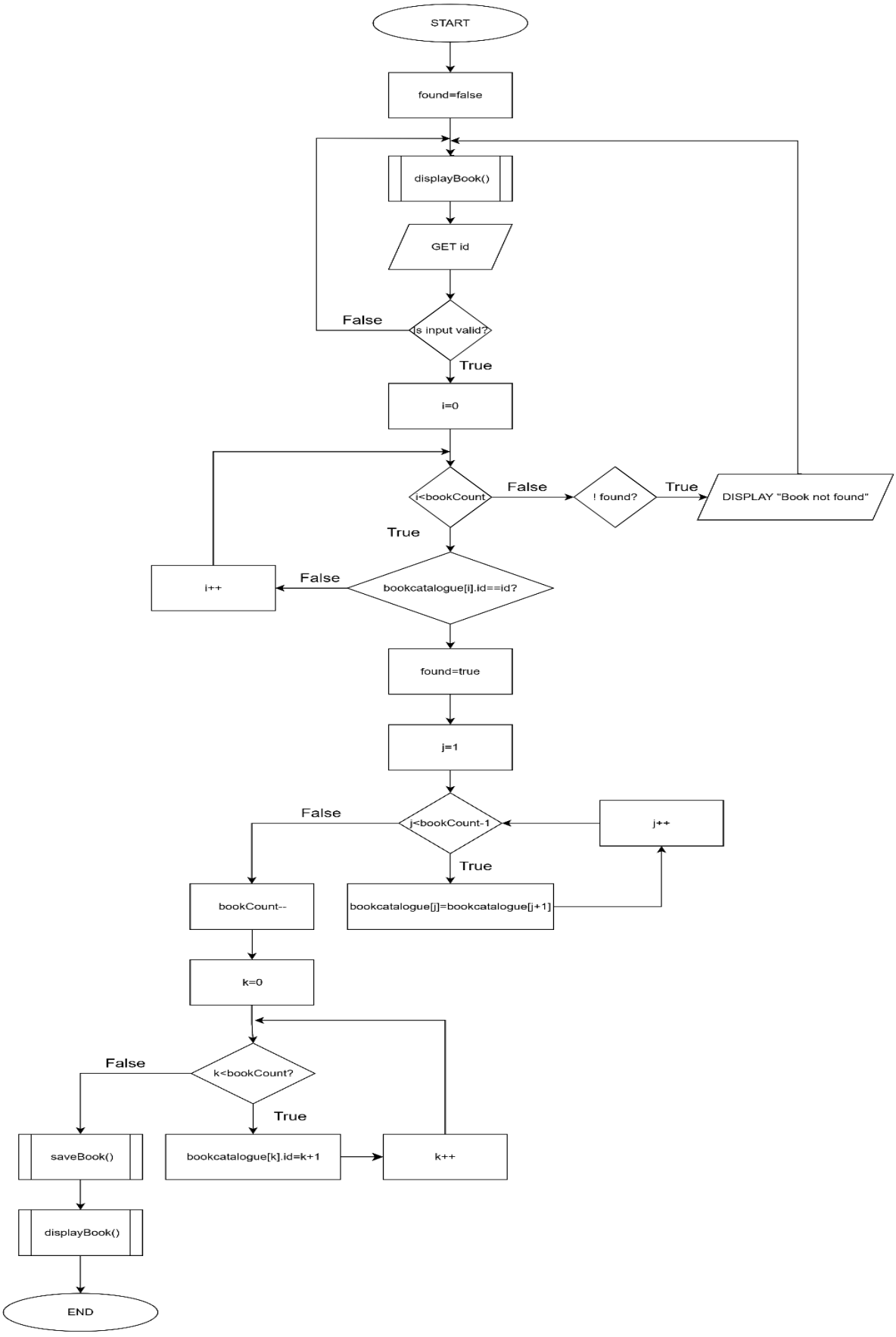
6. void updateBook();



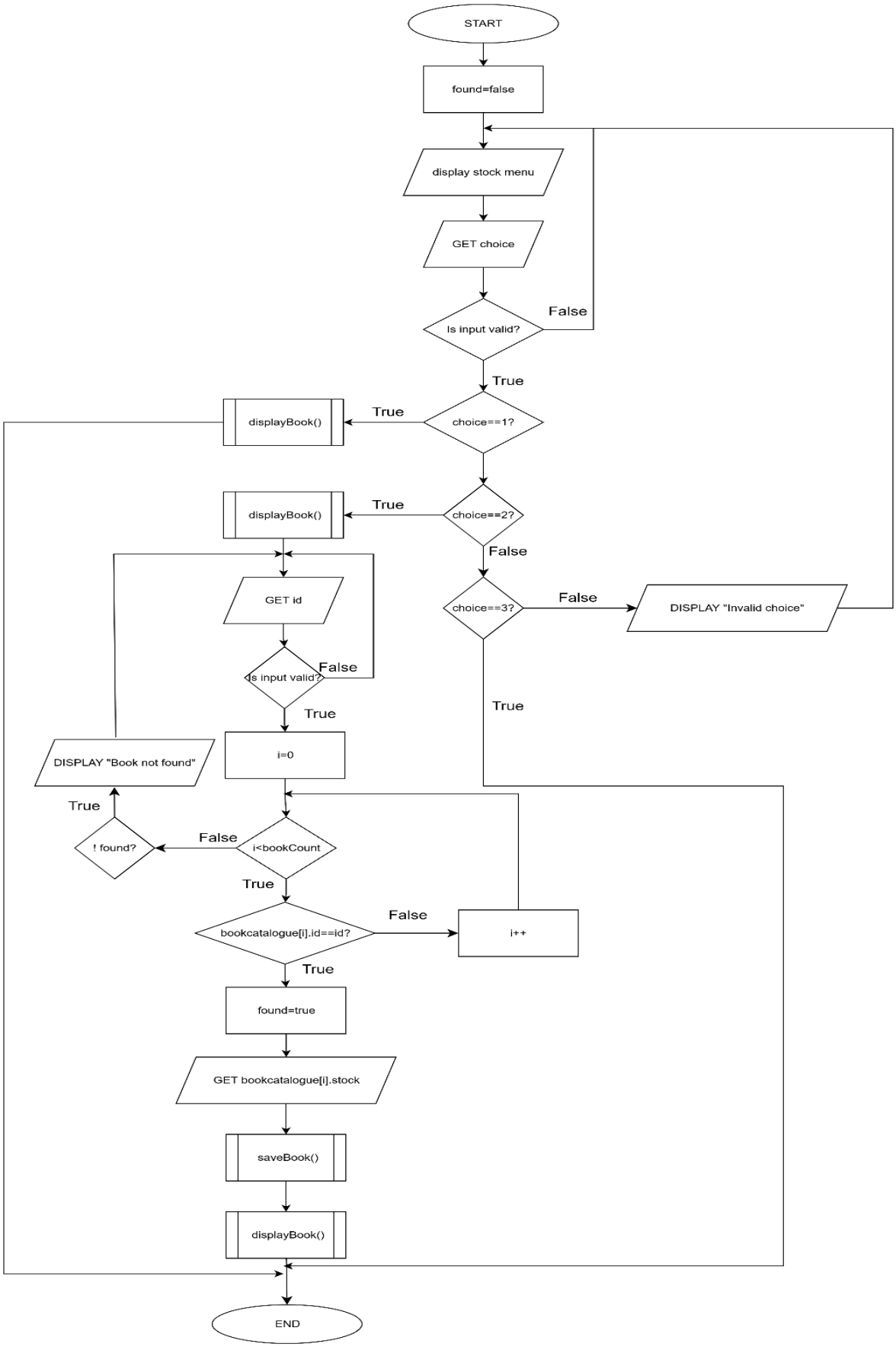
7. void addBook();



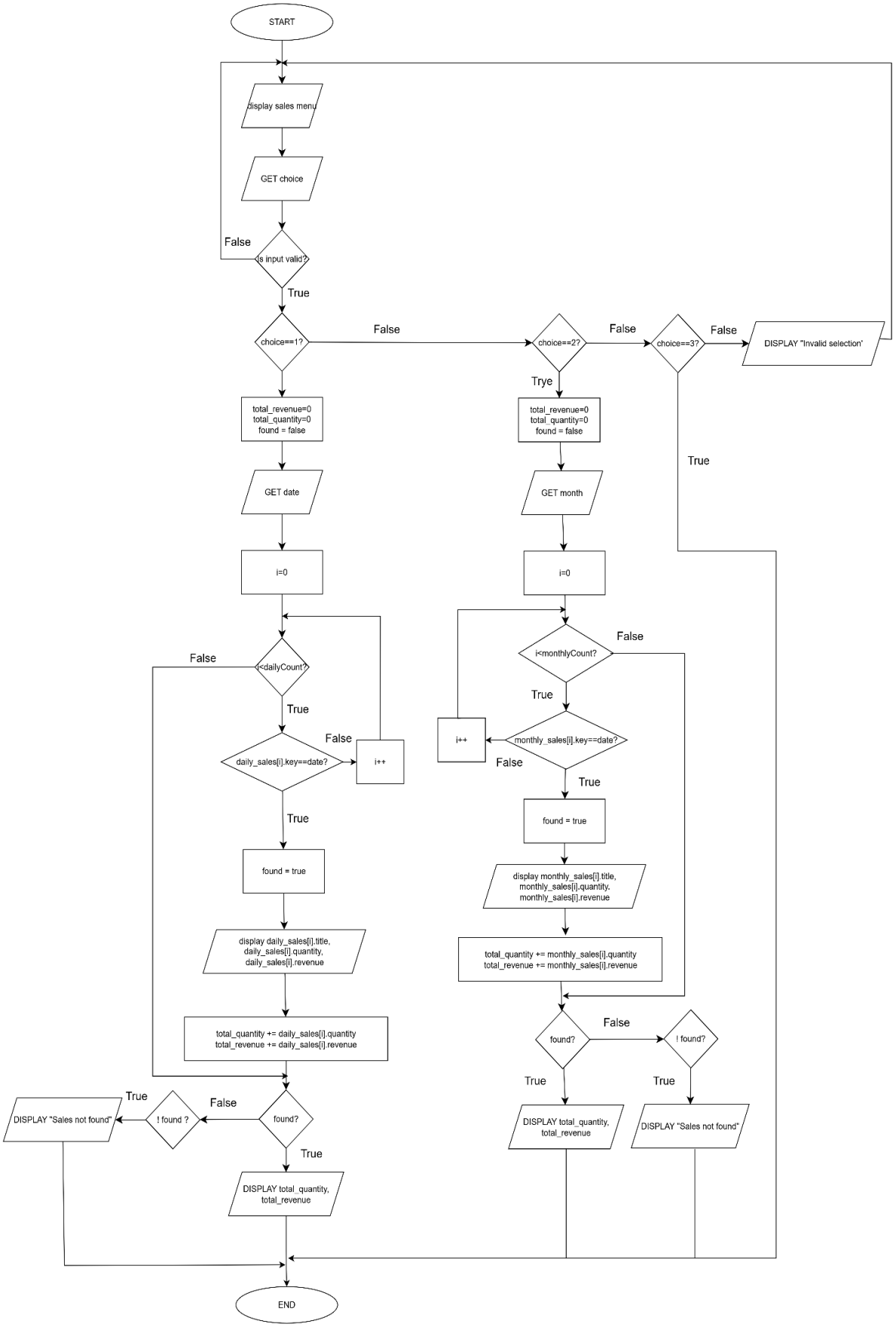
8. void removeBook();


START
found=false
displayBook()
GET id
False
Is input valid?
True
i=0
i<bookCount
False
! found?
True
DISPLAY "Book not found"
True
i++
False
bookcatalogue[i].id==id?
found=true
j=1
False
j<bookCount-1
j++
True
bookCount--
bookcatalogue[j]=bookcatalogue[j+1]
k=0
False
k<bookCount?
True
saveBook()
bookcatalogue[k].id=k+1
k++
displayBook()
END

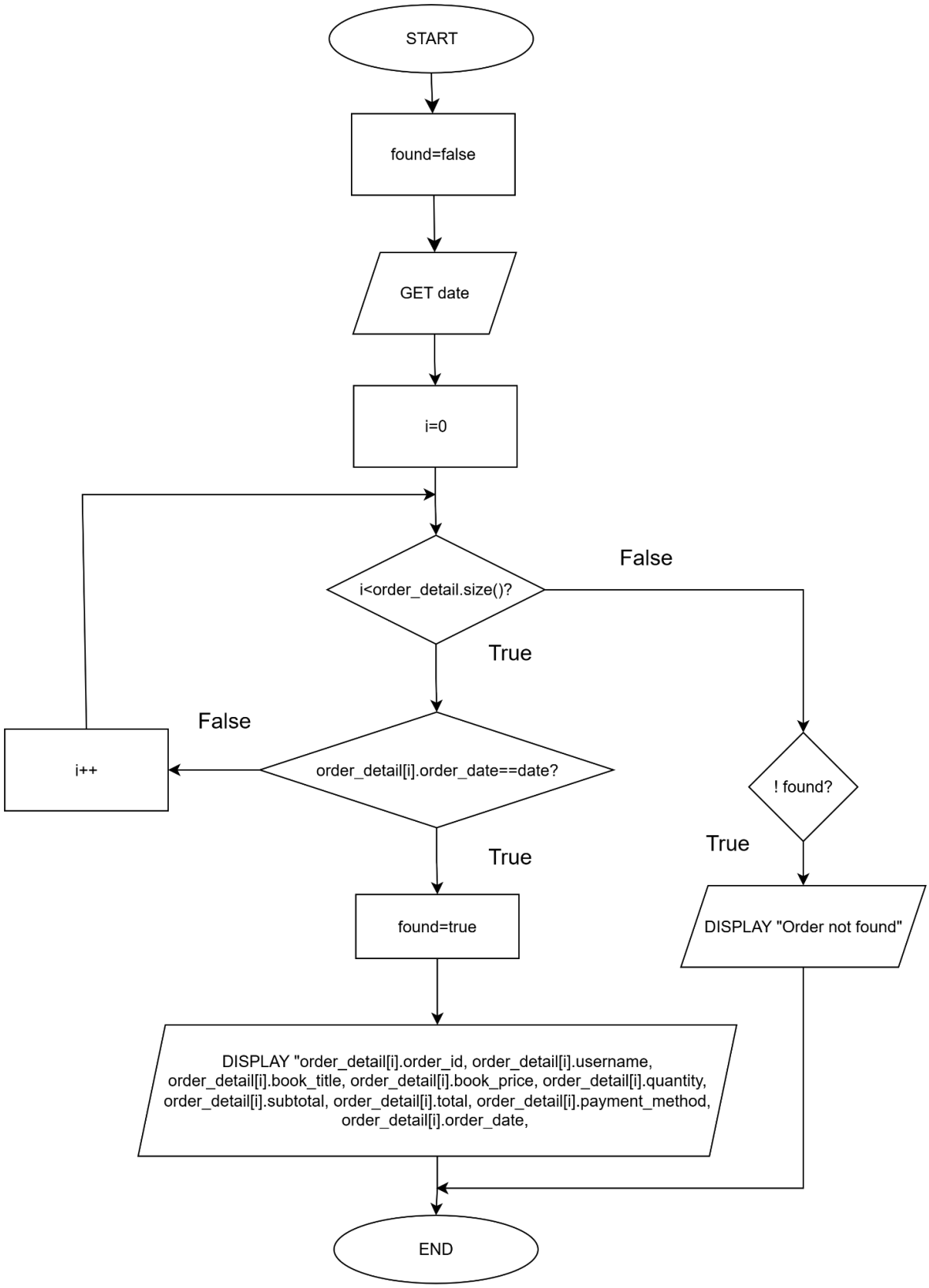
9. void stock();



10. void salesReport();

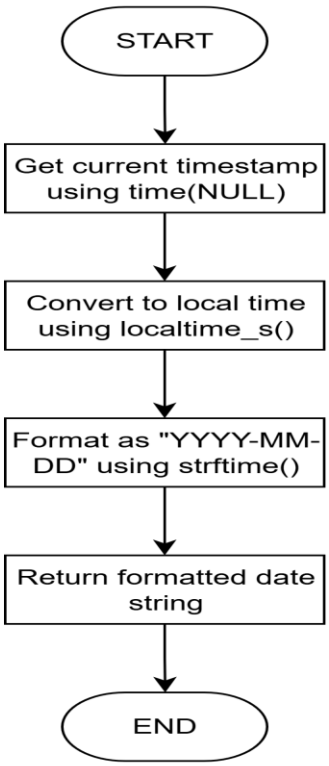


11. void orderReport();

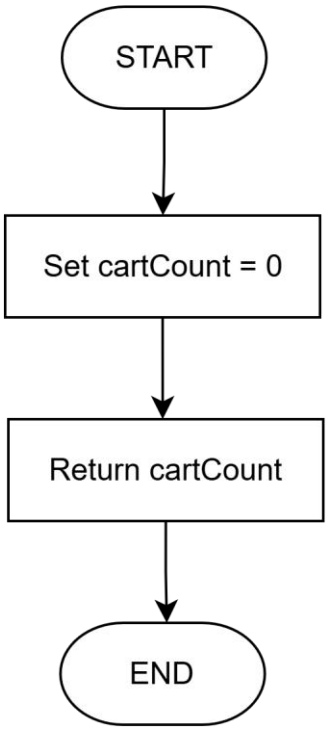


3.2.3 Customer Flowchart

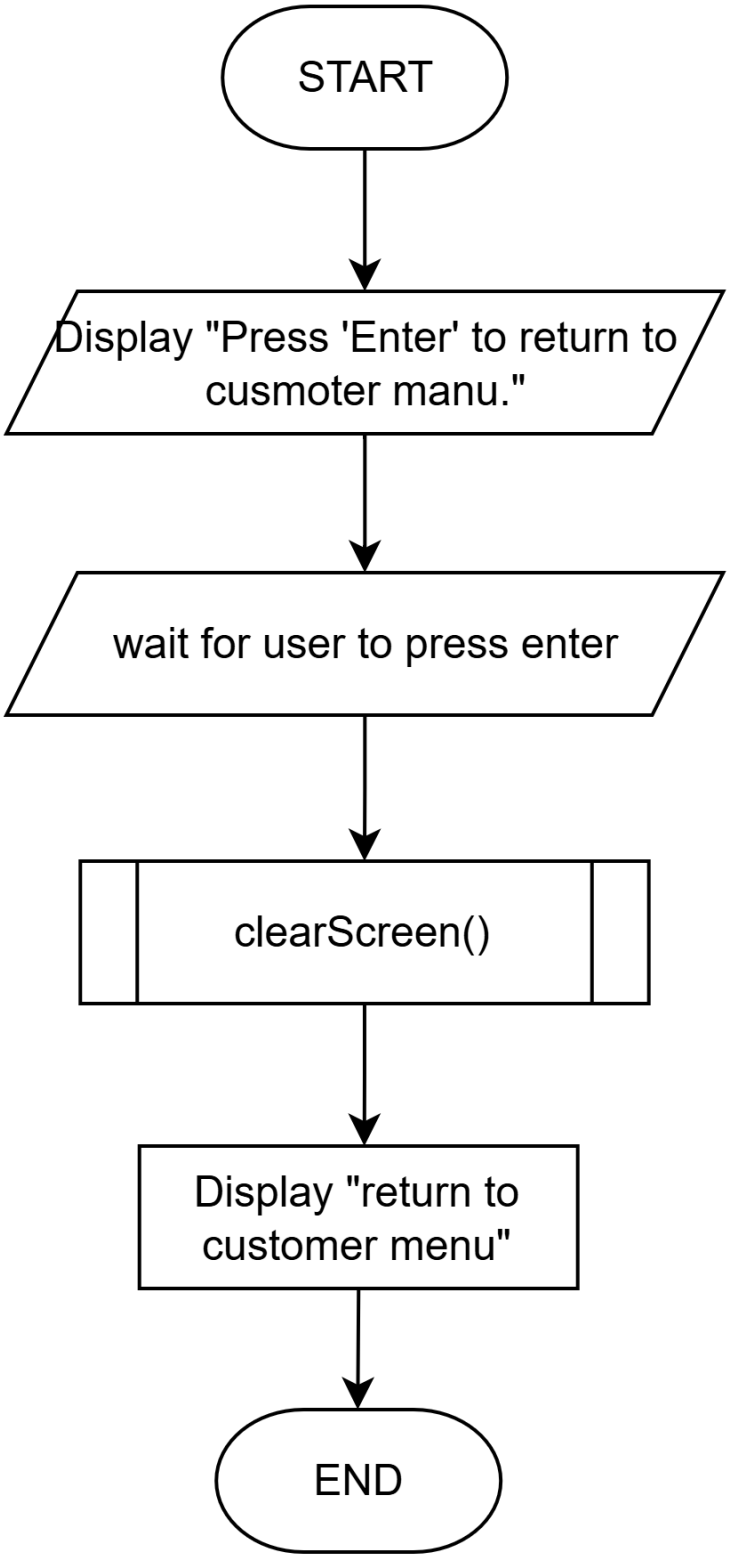
1. **string getCurrentDate( )**



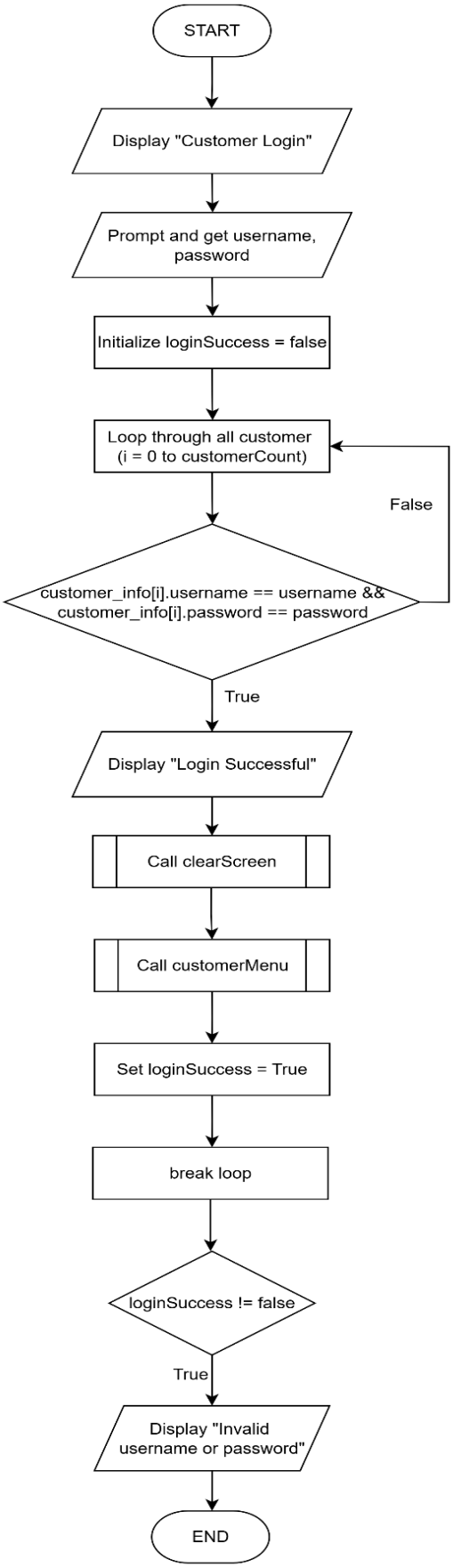
2. **int clearShoppingCart( )**



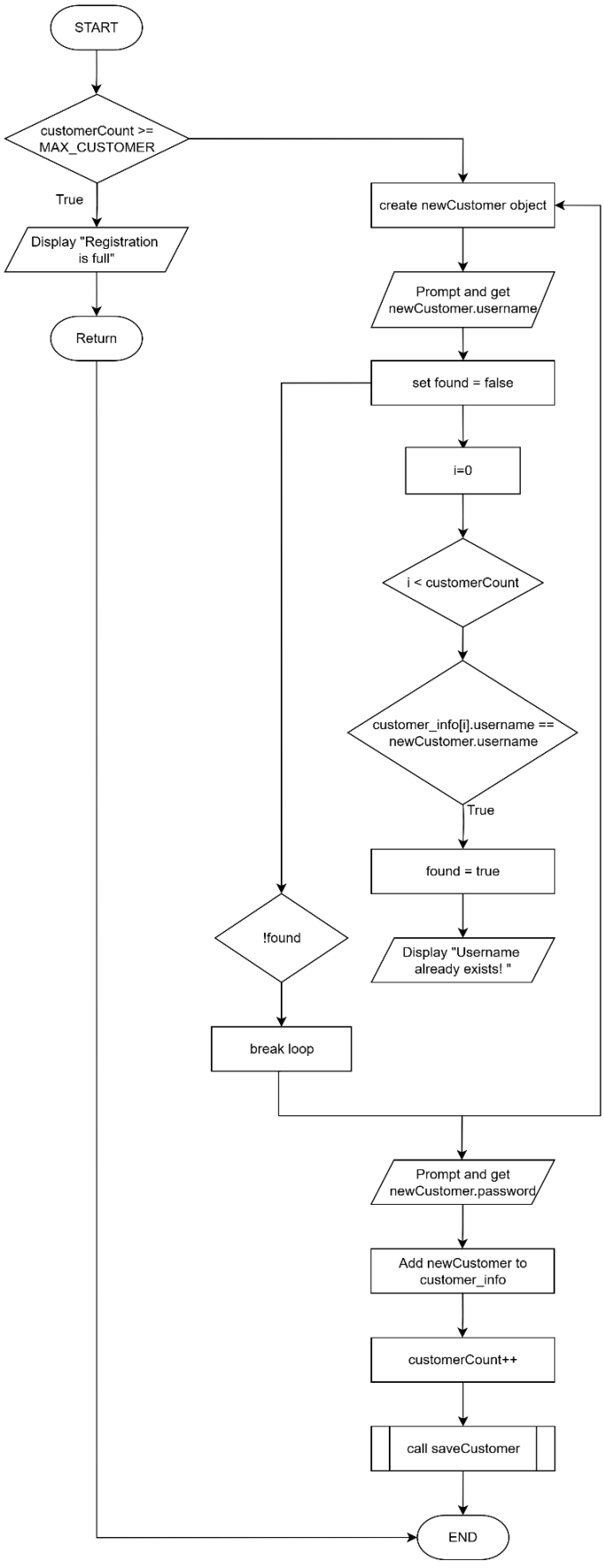
3. **Void backToMenuPage**



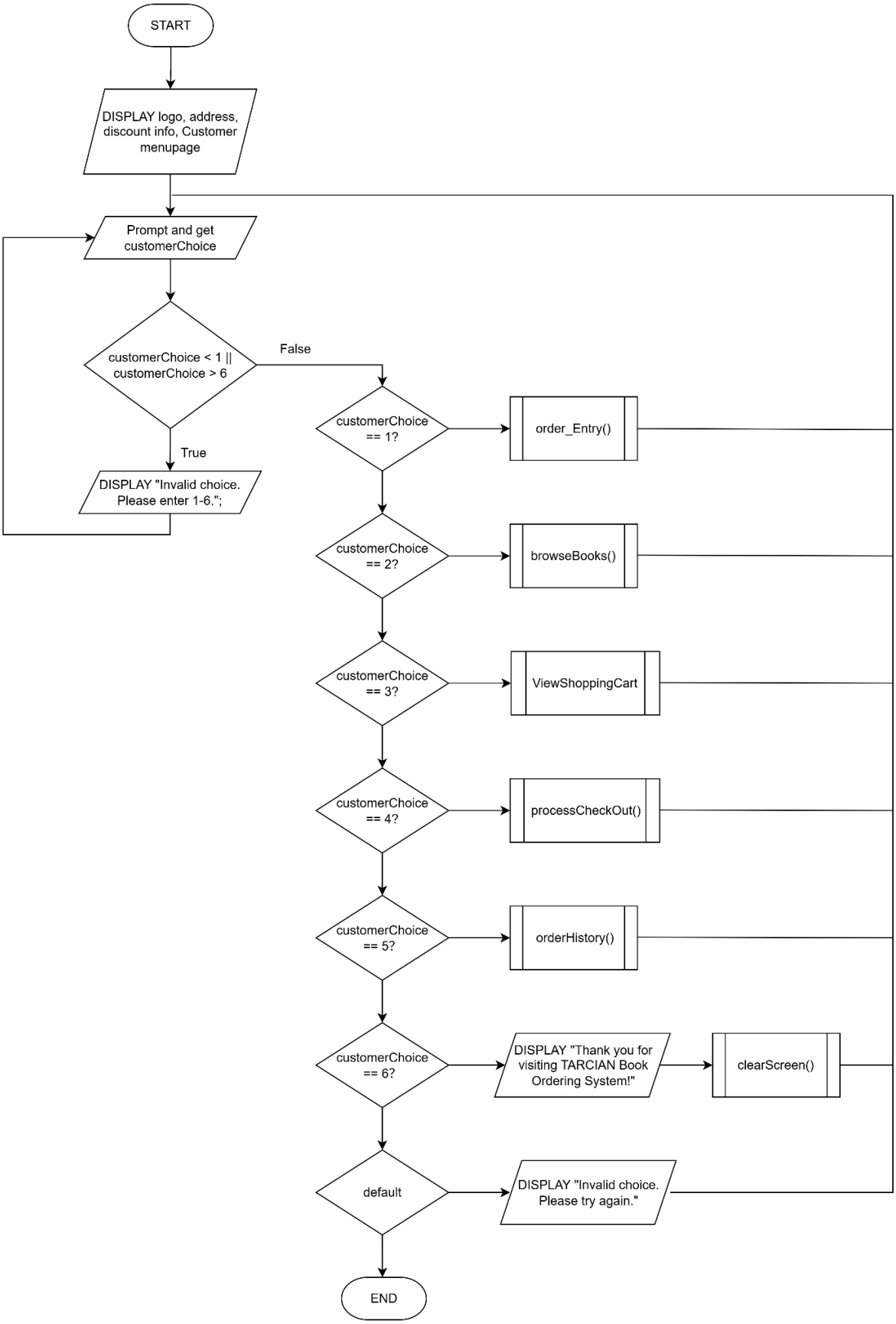
4. **Void customerLogin( )**



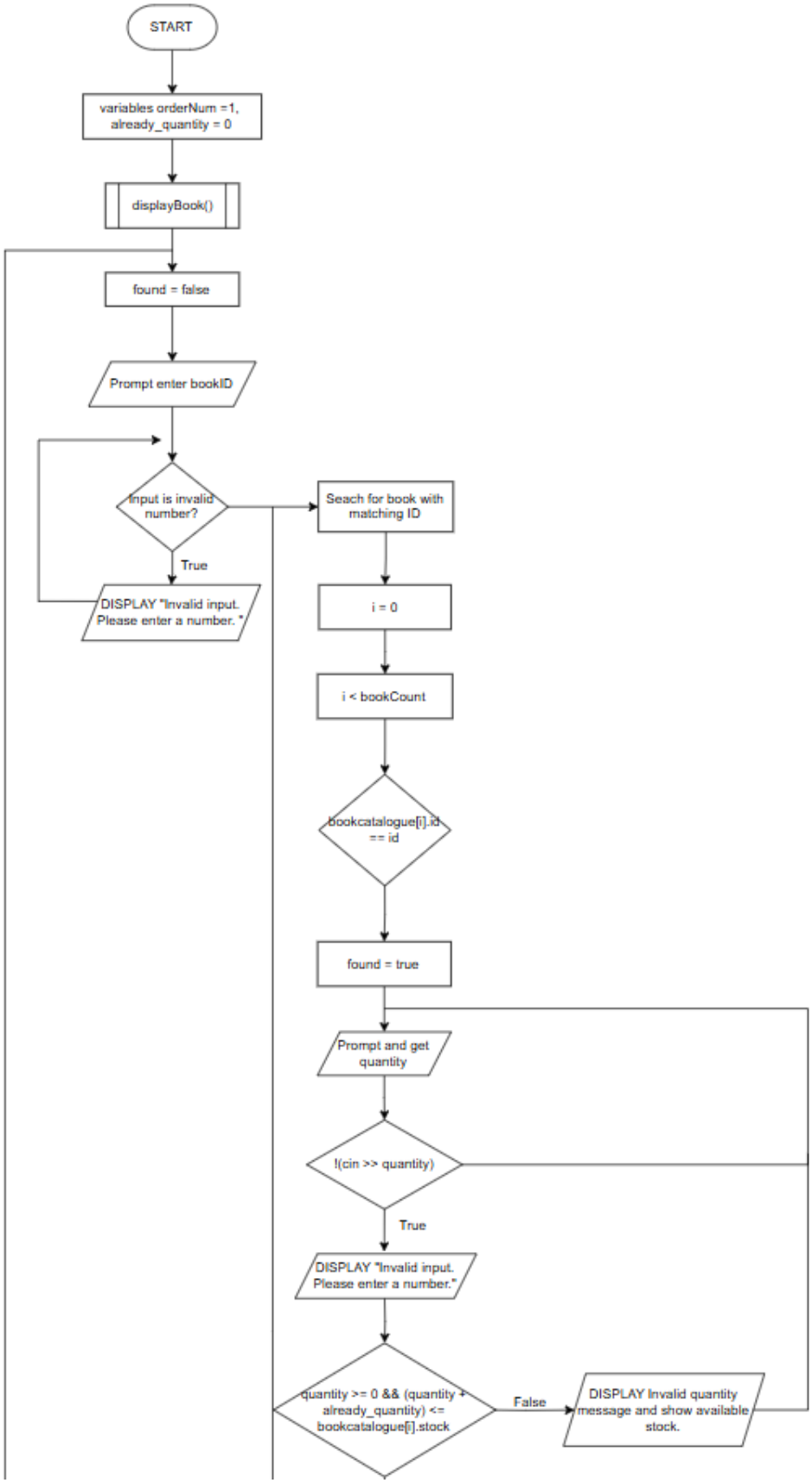
5. Void customerRegister( )

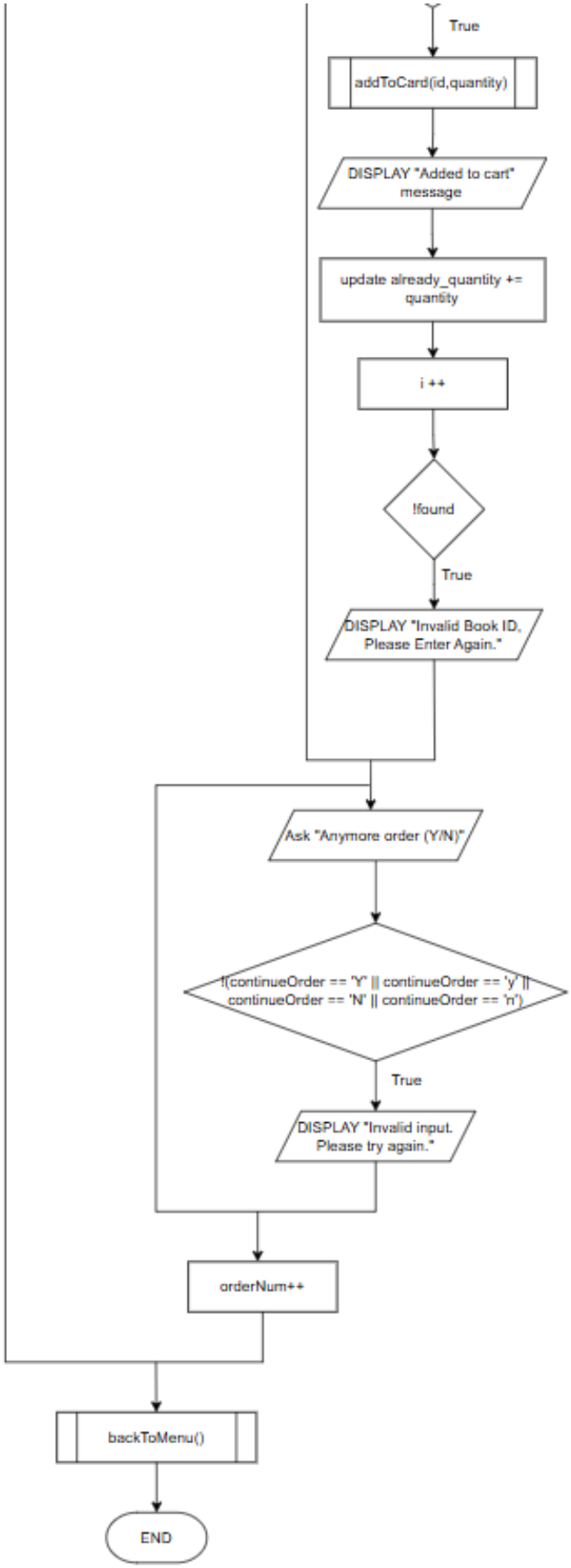


6. Void customerMenu( )

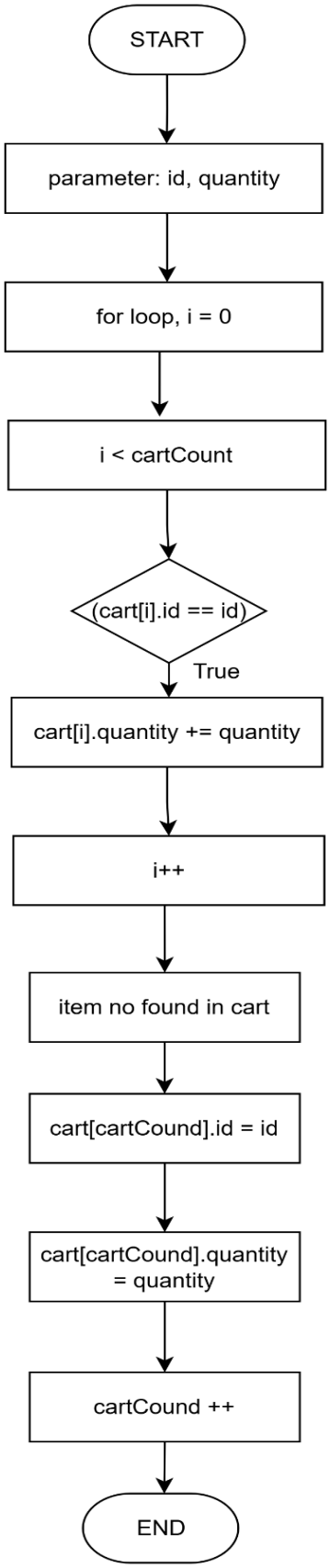


7. Void order_Entry( )

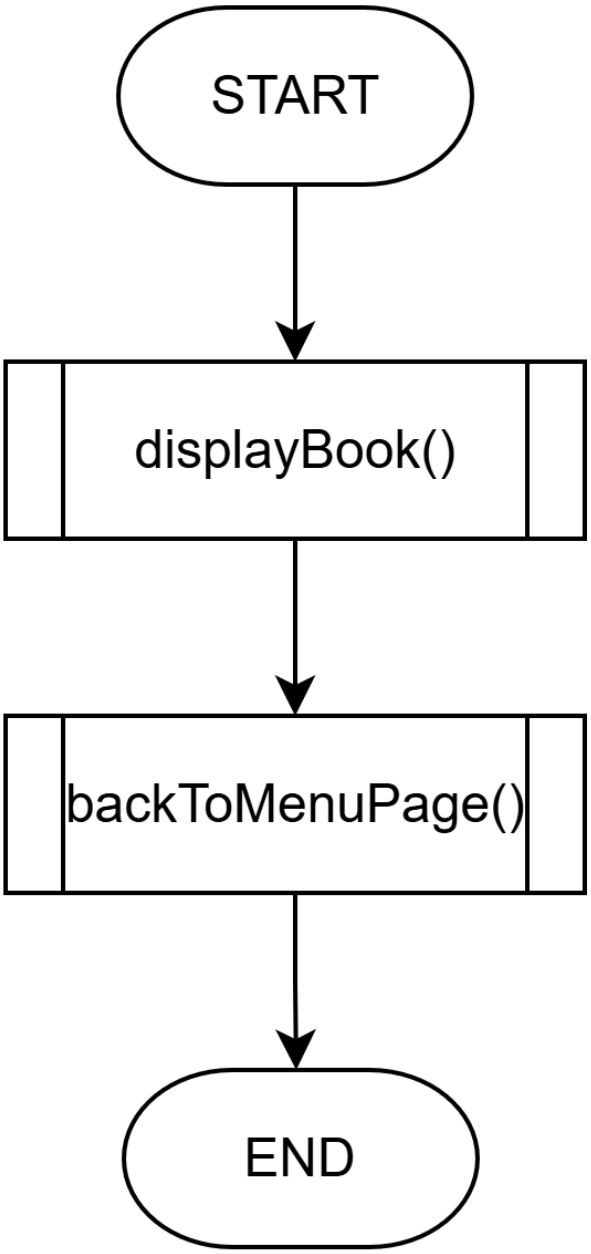

START
variables orderNum =1, already_quantity = 0
displayBook()
found = false
Prompt enter bookID
Input is invalid number?
True
DISPLAY "Invalid input. Please enter a number. "
Seach for book with matching ID
i = 0
i < bookCount
bookcatalogue[i].id == id
found = true
Prompt and get quantity
!(cin >> quantity)
True
DISPLAY "Invalid input. Please enter a number."
quantity >= 0 && (quantity + already_quantity) <= bookcatalogue[i].stock
False
DISPLAY Invalid quantity message and show available stock.


True
addToCard(id,quantity)
DISPLAY "Added to cart" message
update already_quantity += quantity
i ++
!found
True
DISPLAY "Invalid Book ID, Please Enter Again."
Ask "Anymore order (Y/N)"
!(continueOrder == 'Y' || continueOrder == 'y' || continueOrder == 'N' || continueOrder == 'n')
True
DISPLAY "Invalid input. Please try again."
orderNum++
backToMenu()
END

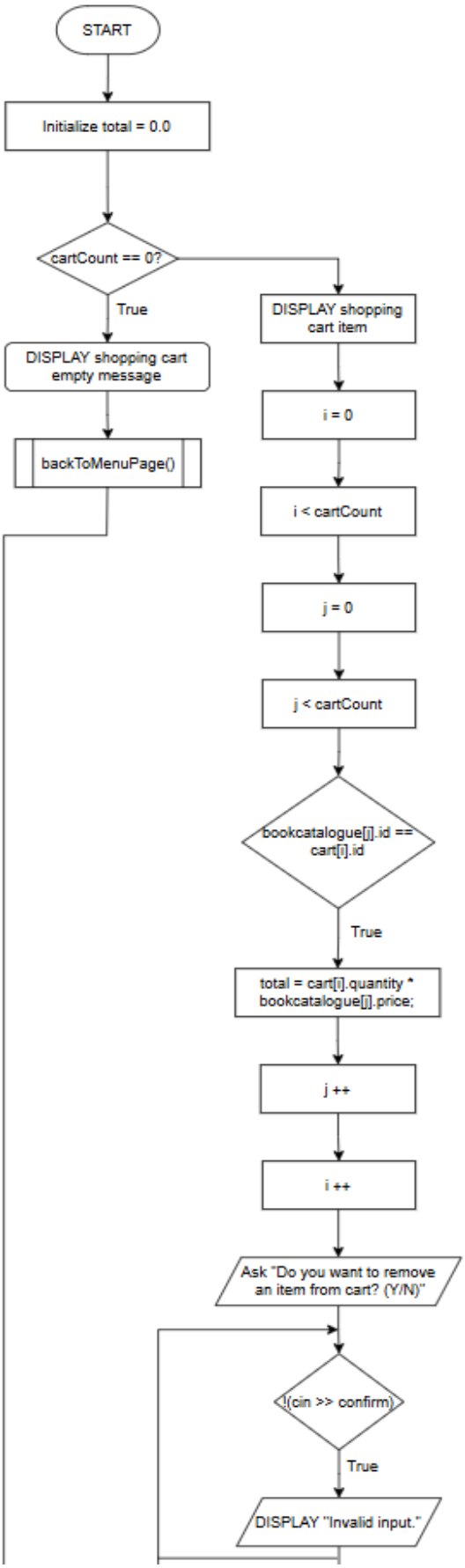
8. Void addToCart(int, int)

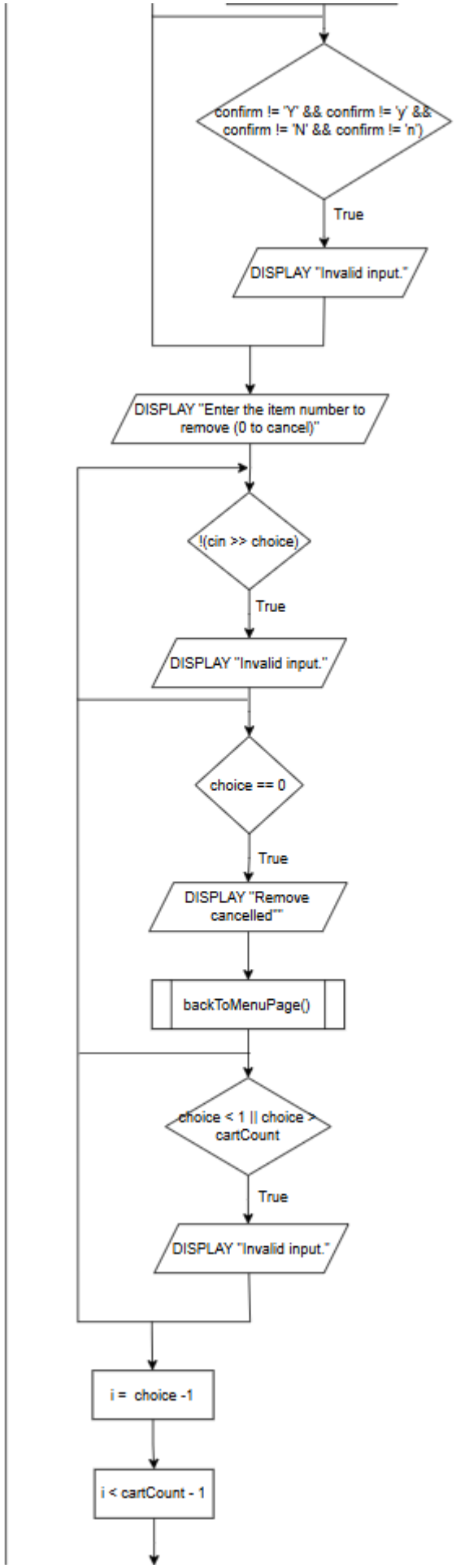


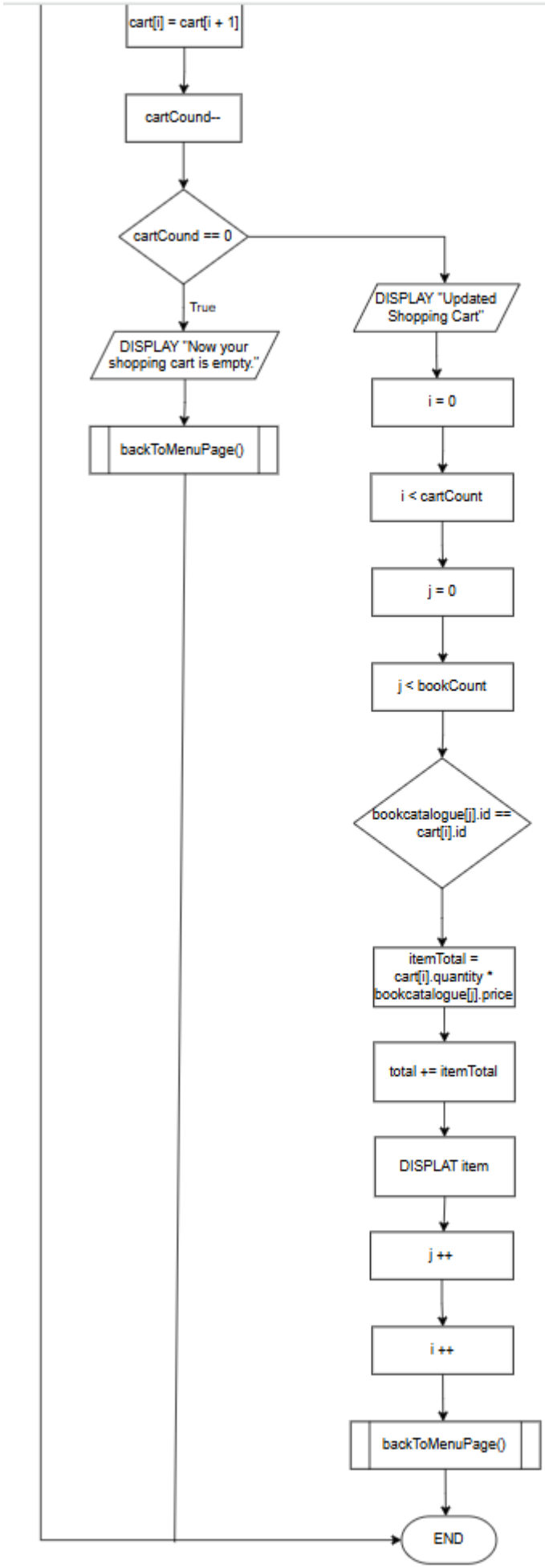
9. Void browseBooks( )



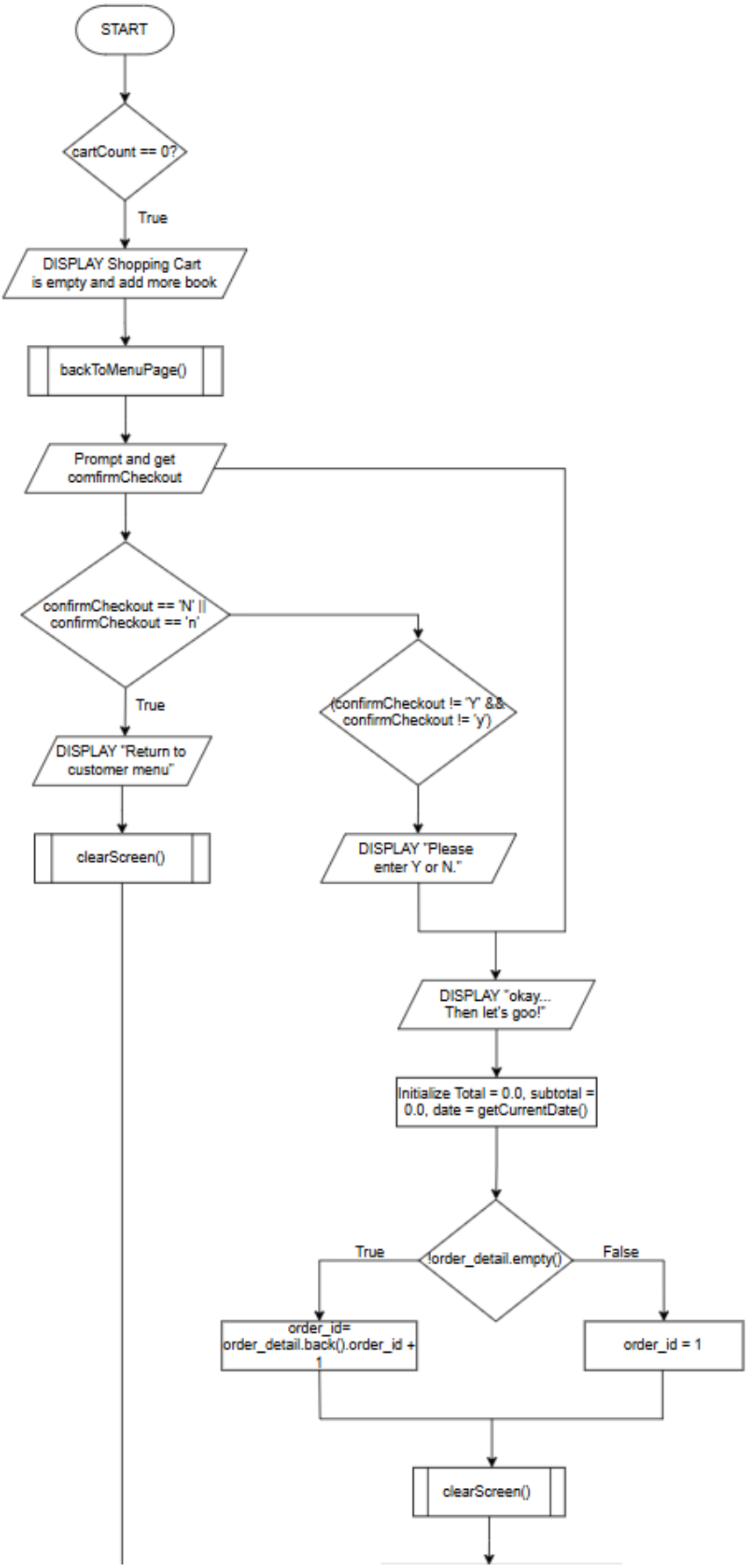
10. Void ViewShoppingCart

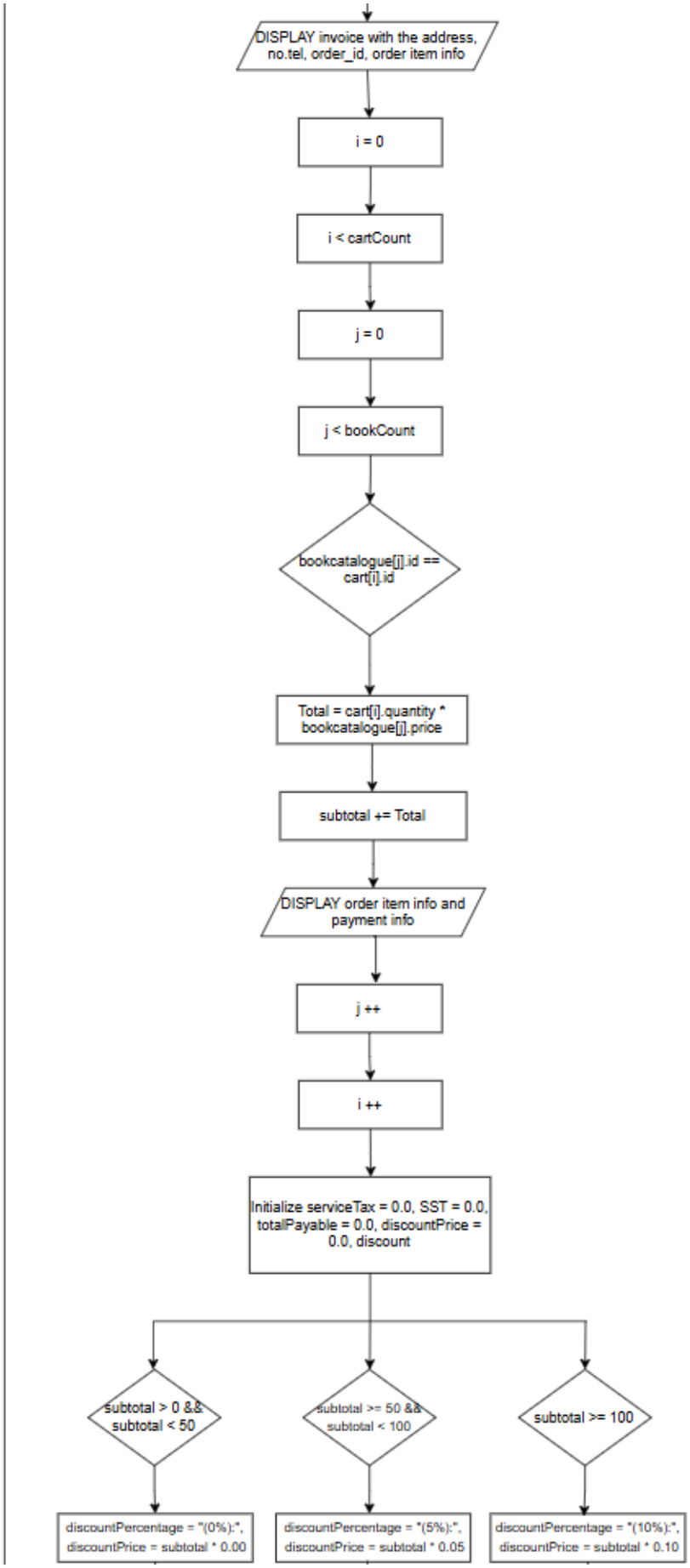

START
Initialize total = 0.0
cartCount == 0?
True
DISPLAY shopping cart empty message
backToMenuPage()
DISPLAY shopping cart item
i = 0
i < cartCount
j = 0
j < cartCount
bookcatalogue[j].id == cart[i].id
True
total = cart[i].quantity * bookcatalogue[j].price;
j ++
i ++
Ask "Do you want to remove an item from cart? (Y/N)"
!(cin >> confirm)
True
DISPLAY "Invalid input."

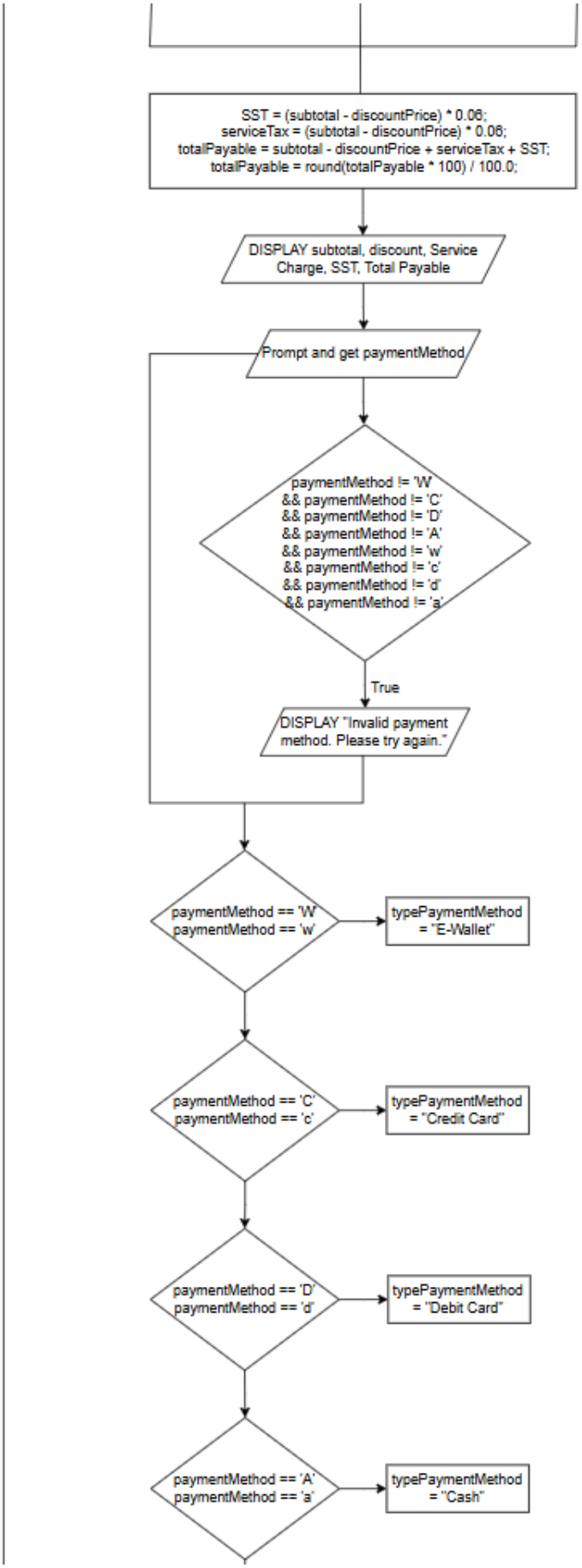

confirm != 'Y' && confirm != 'y' && confirm != 'N' && confirm != 'n')
True
DISPLAY "Invalid input."
DISPLAY "Enter the item number to remove (0 to cancel)"
!(cin >> choice)
True
DISPLAY "Invalid input."
choice == 0
True
DISPLAY "Remove cancelled""
backToMenuPage()
choice < 1 || choice > cartCount
True
DISPLAY "Invalid input."
i = choice -1
i < cartCount - 1

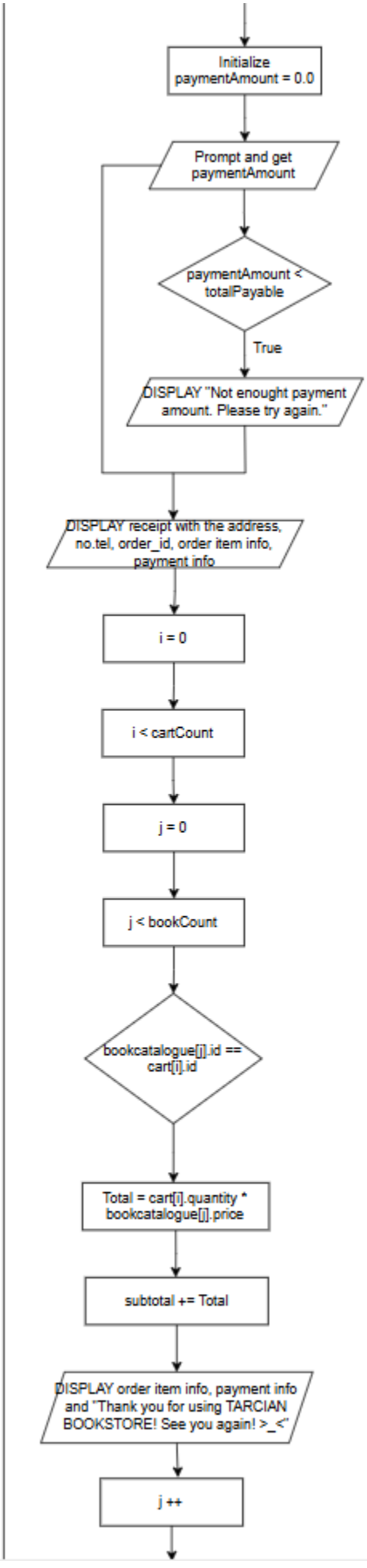

cart[i] = cart[i + 1]
cartCound--
cartCound == 0
True
DISPLAY "Now your shopping cart is empty."
backToMenuPage()
DISPLAY "Updated Shopping Cart"
i = 0
i < cartCount
j = 0
j < bookCount
bookcatalogue[j].id == cart[i].id
itemTotal = cart[i].quantity * bookcatalogue[j].price
total += itemTotal
DISPLAT item
j ++
i ++
backToMenuPage()
END

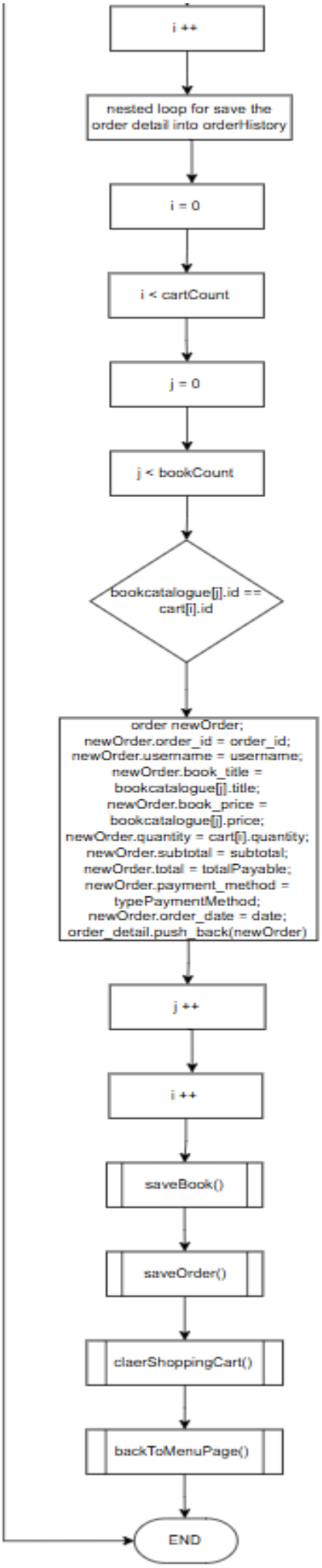
11. Void processCheckOut( )



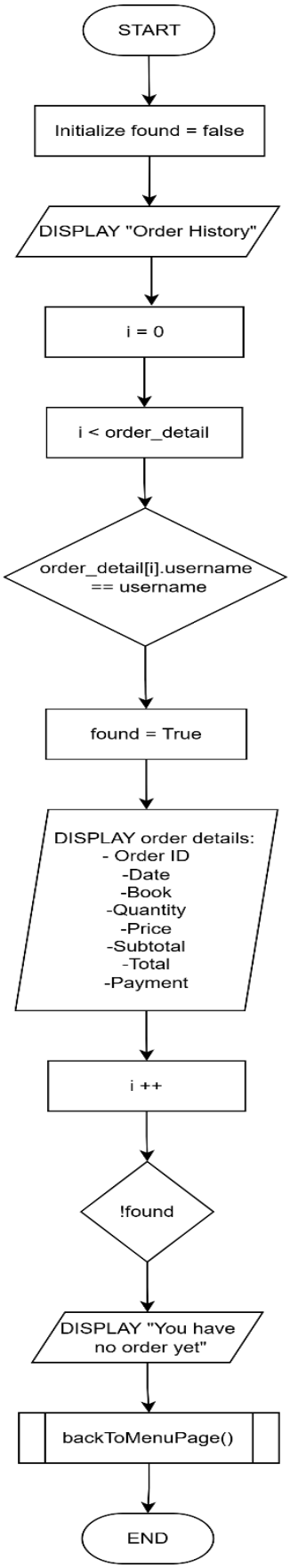

DISPLAY invoice with the address, no.tel, order_id, order item info
i = 0
i < cartCount
j = 0
j < bookCount
bookcatalogue[j].id == cart[i].id
Total = cart[i].quantity * bookcatalogue[j].price
subtotal += Total
DISPLAY order item info and payment info
j ++
i ++
Initialize serviceTax = 0.0, SST = 0.0, totalPayable = 0.0, discountPrice = 0.0, discount
subtotal > 0 && subtotal < 50
subtotal >= 50 && subtotal < 100
subtotal >= 100
discountPercentage = "(0%):", discountPrice = subtotal * 0.00
discountPercentage = "(5%):", discountPrice = subtotal * 0.05
discountPercentage = "(10%):", discountPrice = subtotal * 0.10


SST = (subtotal - discountPrice) * 0.06;
serviceTax = (subtotal - discountPrice) * 0.06;
totalPayable = subtotal - discountPrice + serviceTax + SST;
totalPayable = round(totalPayable * 100) / 100.0;
DISPLAY subtotal, discount, Service Charge, SST, Total Payable
Prompt and get paymentMethod
paymentMethod != 'W' && paymentMethod != 'C' && paymentMethod != 'D' && paymentMethod != 'A' && paymentMethod != 'w' && paymentMethod != 'c' && paymentMethod != 'd' && paymentMethod != 'a'
True
DISPLAY "Invalid payment method. Please try again."
paymentMethod == 'W' paymentMethod == 'w'
typePaymentMethod = "E-Wallet"
paymentMethod == 'C' paymentMethod == 'c'
typePaymentMethod = "Credit Card"
paymentMethod == 'D' paymentMethod == 'd'
typePaymentMethod = "Debit Card"
paymentMethod == 'A' paymentMethod == 'a'
typePaymentMethod = "Cash"


Initialize paymentAmount = 0.0
Prompt and get paymentAmount
paymentAmount < totalPayable
True
DISPLAY "Not enought payment amount. Please try again."
DISPLAY receipt with the address, no.tel, order_id, order item info, payment info
i = 0
i < cartCount
j = 0
j < bookCount
bookcatalogue[j].id == cart[i].id
Total = cart[i].quantity * bookcatalogue[j].price
subtotal += Total
DISPLAY order item info, payment info and "Thank you for using TARCIAN BOOKSTORE! See you again! >_<"
j ++


i ++
nested loop for save the order detail into orderHistory
i = 0
i < cartCount
j = 0
j < bookCount
bookcatalogue[j].id == cart[i].id
order newOrder;
newOrder.order_id = order_id;
newOrder.username = username;
newOrder.book_title = bookcatalogue[j].title;
newOrder.book_price = bookcatalogue[j].price;
newOrder.quantity = cart[i].quantity;
newOrder.subtotal = subtotal;
newOrder.total = totalPayable;
newOrder.payment_method = typePaymentMethod;
newOrder.order_date = date;
order_detail.push_back(newOrder)
j ++
i ++
saveBook()
saveOrder()
claerShoppingCart()
backToMenuPage()
END

12. Void orderHistory( )



4.0 ADDITIONAL FUNCTIONALITIES

4.1 Order History

```
Dear customer. Welcome to TARCIAN Book Ordering System!

1. Order Entry
2. Browse all Books
3. View Shopping Cart
4. Process to Checkout
5. View Order History
6. Logout

Choice: 5
```

```
==================== ORDER HISTORY ====================

Order ID: 17
Date    : 2025-09-09
Book    : The Last Kingdom
Quantity: 1
Price   : RM 34.90
Subtotal: RM 34.90
Total   : RM 40.48
Payment : Cash



Press 'Enter' to return to customer menu.....
```

- **Allow customer to view their order history**
- **So, customer can retrieve the items they have purchased**

4.2 Staff / Customer Login and Registration

```
==== TARCIAN BOOKSTORE ORDERING SYSTEM ====

1. Staff Login
2. Customer Login
3. Customer Register
0. Exit

Enter your choice: 1

[Staff Login]
Username: admin
Password: admin123
```

- **staff can login into system using valid username and password**

```
==== TARCIAN BOOKSTORE ORDERING SYSTEM ====

1. Staff Login
2. Customer Login
3. Customer Register
0. Exit

Enter your choice: 2

========== Customer Login ==========
Enter username: nicolechia
Enter password: 0123

Customer login successful
```

- customer can login into system using registered account

```
==== TARCIAN BOOKSTORE ORDERING SYSTEM ====

1. Staff Login
2. Customer Login
3. Customer Register
0. Exit

Enter your choice: 3

========== CUSTOMER REGISTRATION ==========
Enter your username: peien
Enter your password: 0513
```

- **customer can create new account**

4.3 Discount

```
 ------------------------------------
|                                    |
| Special OFFER !!!                  |
| Spend RM50 to get a 5% discount!!  |
| Spend RM100 to get a 10% discount!! |
|____________________________________|
```

- **provide discount to customer to increase user experience**

4.4 Edit Book Catalogue

- **Update Book info**

```
==== Manage Book Catalogue ====

1. Update Book Info
2. Add New Book
3. Remove Book
4. Return

Enter your choice: 1

================================== BOOK CATALOGUE ==================================

ID      TITLE                                CATEGORY        PRICE     STOCK    SOLD
------------------------------------------------------------------------------------
1       Book A                               Fiction         29.50      0         7
2       The Last Kingdom                     Fiction         35.00      15        0
3       My Cat Is Secretly an Alien          Fiction         28.50      9         0
4       The Girl Who Spoke to Shadow         Fiction         32.90      5         1
5       How to Marry a Dragon                Fiction         26.40      1         4
6       The Science of Learning              Non-fiction     40.00      7         0
7       How to Lie with Statistics           Non-fiction     33.30      5         1
8       Understanding Human Behavior         Non-fiction     38.80      9         0
9       Why need to go to SCHOOL             Non-fiction     25.00      0         3
10      Adventures in Alphabet Land          Children        21.00      8         0
11      Why Monsters Need Do Homework        Children        18.90      10        0
12      The Magical Pencil                   Children        20.50      8         0
13      How to Train Your Teddy Bear         Children        19.99      6         0
14      A Thousand Autumns                   Novel           36.60      17        3
15      I Fell in Love with a Sandwich       Novel           27.70      7         0
16      The Silence Between Stars            Novel           34.20      6         0
17      My Ex Turned Into a Vampire          Novel           29.90      4         0
18      C++ Programming Handbook             Reference       50.00      8         2
19      My Code Works Only at Midnight       Reference       24.40      5         0
20      Advanced Algorithms                  Reference       48.80      0         6
21      C++ Projects for Lazy Students       Reference       31.10      6         1

Enter book id to update book info:
```

- **Add new book**

```
Enter your choice: 2
Enter book title: Book C
Enter book category: Non-Fiction
Enter book price: 20.00
Enter book stock: 15

Book added successfully!

============================== CURRENT BOOK CATALOGUE ==============================

ID      TITLE                                CATEGORY        PRICE     STOCK    SOLD
--------------------------------------------------------------------------------
1       Book B                               Fiction         29.50      0         7
2       The Last Kingdom                     Fiction         35.00      15        0
3       My Cat Is Secretly an Alien          Fiction         28.50      9         0
4       The Girl Who Spoke to Shadow         Fiction         32.90      5         1
5       How to Marry a Dragon                Fiction         26.40      1         4
6       The Science of Learning              Non-fiction     40.00      7         0
7       How to Lie with Statistics           Non-fiction     33.30      5         1
8       Understanding Human Behavior         Non-fiction     38.80      9         0
9       Why need to go to SCHOOL             Non-fiction     25.00      0         3
10      Adventures in Alphabet Land          Children        21.00      8         0
11      Why Monsters Need Do Homework        Children        18.90      10        0
12      The Magical Pencil                   Children        20.50      8         0
13      How to Train Your Teddy Bear         Children        19.99      6         0
14      A Thousand Autumns                   Novel           36.60      17        3
15      I Fell in Love with a Sandwich       Novel           27.70      7         0
16      The Silence Between Stars            Novel           34.20      6         0
17      My Ex Turned Into a Vampire          Novel           29.90      4         0
18      C++ Programming Handbook             Reference       50.00      8         2
19      My Code Works Only at Midnight       Reference       24.40      5         0
20      Advanced Algorithms                  Reference       48.80      0         6
21      C++ Projects for Lazy Students       Reference       31.10      6         1
22      Book C                               Non-Fiction     20.00      15        0


Press 'Enter' to return to menu.....
```

- **Remove book**

```
==== Manage Book Catalogue ====

1. Update Book Info
2. Add New Book
3. Remove Book
4. Return

Enter your choice: 3

================================== BOOK CATALOGUE ==================================

ID      TITLE                                CATEGORY        PRICE     STOCK    SOLD
------------------------------------------------------------------------------------
1       Book B                               Fiction         29.50      0         7
2       The Last Kingdom                     Fiction         35.00      15        0
3       My Cat Is Secretly an Alien          Fiction         28.50      9         0
4       The Girl Who Spoke to Shadow         Fiction         32.90      5         1
5       How to Marry a Dragon                Fiction         26.40      1         4
6       The Science of Learning              Non-fiction     40.00      7         0
7       How to Lie with Statistics           Non-fiction     33.30      5         1
8       Understanding Human Behavior         Non-fiction     38.80      9         0
9       Why need to go to SCHOOL             Non-fiction     25.00      0         3
10      Adventures in Alphabet Land          Children        21.00      8         0
11      Why Monsters Need Do Homework        Children        18.90      10        0
12      The Magical Pencil                   Children        20.50      8         0
13      How to Train Your Teddy Bear         Children        19.99      6         0
14      A Thousand Autumns                   Novel           36.60      17        3
15      I Fell in Love with a Sandwich       Novel           27.70      7         0
16      The Silence Between Stars            Novel           34.20      6         0
17      My Ex Turned Into a Vampire          Novel           29.90      4         0
18      C++ Programming Handbook             Reference       50.00      8         2
19      My Code Works Only at Midnight       Reference       24.40      5         0
20      Advanced Algorithms                  Reference       48.80      0         6
21      C++ Projects for Lazy Students       Reference       31.10      6         1
22      Book C                               Non-Fiction     20.00      15        0

Enter book id to remove:
```

5.0 PROGRAM TESTING & OUTPUTS

5.1 Test Data for 3 Times Program Testing for Whole Program

5.1.1 Normal Process and Expected Output (For Customer Order and Purchase)

Test Data 1:

Step	Test Data Input	Expected Output
1	Main Menu: Enter 2	System displays "Customer Login" page
2	Username: nicolechia Password: 0123	System displays "Customer login successful" and goes to customer menu
3	Customer Menu: Enter 1	System displays "BOOK CATALOGUE" and order entry options
4	Book ID: Enter 1	System accepts input and displays "Enter quantity: "
5	Quantity: Enter 2	System accepts input, adds to cart, and displays "Added to cart: 2 copy/copies of [Book Title]"
6	Anymore Order: Enter n	System displays "Press 'Enter' to return to customer menu"
7	Press Enter	System returns to customer menu
8	Customer Menu: Enter 2	System displays "BOOK CATALOGUE" and displays "Press 'Enter' to return to customer menu"
9	Press Enter	System returns to customer menu
10	Customer Menu: Enter 3	System displays shopping cart with items
11	Remove item: Enter N	System displays "Okay, nothing removed" and returns to menu
12	Customer Menu: Enter 4	System displays checkout confirmation
13	Confirm checkout: Enter Y	System displays invoice with order details
14	Payment method: Enter W	System accepts e-Wallet and prompts for payment amount
15	Payment Amount: Enter 100	System displays receipt with order details and return to customer menu
16	Customer Menu: Enter 5	System displays order history and Press ‘Enter’ to return to customer menu”
17	Press Enter	System returns to customer menu
18	Customer Menu: Enter 6	System displays Main Menu
19	Main Menu: 0	System exit the program

Program Testing 1:

A. At the first, the system requires users to identify their role whether they are customer or staff. In this case, the user chooses "2" as their role and the system will display the customer login page for the user. The system prompts the user to enter username and password for authentication. The user enters username "nicolechia" and password "0123". After validation, the system displays "Customer login successful" and redirects to the customer menu.

```
==== TARCIAN BOOKSTORE ORDERING SYSTEM ====

1. Staff Login
2. Customer Login
3. Customer Register
0. Exit

Enter your choice: 2

========== Customer Login ==========
Enter username: nicolechia
Enter password: 0123

Customer login successful
```

B. There are 6 options on the customer menu with TARCIAN BOOKSTORE ASCII banner and special discount offers displayed. After selecting option '1' for "Order Entry", the system will display the complete book catalogue with all available books showing ID, title, category, price, stock and sold information for customer viewing.

```
 _____ _    ____  ____ ___   _    _  _
|_   _|/ \  |  _ \ / ___|_ _| / \  | \ | |
  | | / _ \ | |_) | |    | | / _ \ |  \| |
  | |/ ___ \|  _ <| |___ | |/ ___ \| |\  |
 _|_/_/ __\_\_|_\_\\____|___/_/___\_\_| \_|_  _____
| __ ) / _ \ / _ \| |/ / ___|_   _/ _ \|  _ \| ____|
|  _ \| | | | | | | ' /\___ \ | || | | | |_) |  _|
| |_) | |_| | |_| | . \ ___) || || |_| |  _ <| |___
|____/ \___/ \___/|_|\_\____/ |_| \___/|_| \_\_____|

==================================== Customer Menu ====================================

                                TARCIAN BOOKSTORE
                            No. K666, Block K TARUMT,
                          Jalan Genting Klang, Setapak, 53000
                                  Kuala Lumpur.


 ______________________________________
|                                      |
| Special OFFER !!!                    |
| Spend RM50 to get a 5% discount!!    |
| Spend RM100 to get a 10% discount!!  |
|______________________________________|

Dear customer. Welcome to TARCIAN Book Ordering System!

1. Order Entry
2. Browse all Books
3. View Shopping Cart
4. Process to Checkout
5. View Order History
6. Logout

Choice: 1

You have selected option 1.
```

C. The system shows "Order 1" and prompts the user to "Enter the book ID to add to shopping cart". The user enters book ID "1" and the system validates the input, then displays "Enter quantity:" for the selected book. The user enters "n" to finish the ordering process. The system displays "Press 'Enter' to return to customer menu" message and waits for user input to continue.

```
================================== BOOK CATALOGUE ==================================

ID      TITLE                                CATEGORY        PRICE     STOCK    SOLD
-----------------------------------------------------------------------------------
1       Book A                               Fiction         29.50     10       8
2       The Last Kingdom                     Fiction         35.00     17       11
3       My Cat Is Secretly an Alien          Fiction         28.50     29       0
4       The Girl Who Spoke to Shadow         Fiction         32.90     15       1
5       How to Marry a Dragon                Fiction         26.40     15       9
6       The Science of Learning              Non-fiction     40.00     7        7
7       How to Lie with Statistics           Non-fiction     33.30     5        1
8       Understanding Human Behavior         Non-fiction     38.80     9        0
9       Why need to go to SCHOOL             Non-fiction     25.00     25       3
10      Adventures in Alphabet Land          Children        21.00     8        0
11      Why Monsters Need Do Homework        Children        18.90     10       0
12      The Magical Pencil                   Children        20.50     8        0
13      How to Train Your Teddy Bear         Children        19.99     10       0
14      A Thousand Autumns                   Novel           36.60     17       3
15      I Fell in Love with a Sandwich       Novel           27.70     7        0
16      The Silence Between Stars            Novel           34.20     26       0
17      My Ex Turned Into a Vampire          Novel           29.90     9        0
18      C++ Programming Handbook             Reference       50.00     8        2
19      My Code Works Only at Midnight       Reference       24.40     15       0
20      Advanced Algorithms                  Reference       48.80     11       6
21      C++ Projects for Lazy Students       Reference       31.10     21       1
22      Book C                               Reference       12.00     15       0

Order 1
Enter the book ID to add to shopping cart: 1
Enter quantity: 2
Added to cart: 2 copy/copies of "Book A"

Anymore order (Y/N)? n


Press 'Enter' to return to customer menu.....
```

D. The user presses Enter key and the system clears the screen, then returns to the customer menu displaying all 6 available options again for further selection. The user selects option '3' for "View Shopping Cart" and the system displays the shopping cart contents in a formatted table showing book ID, book title, category, unit price, stock and sold information. Then press ‘Enter’ will return to the customer page.

```
Dear customer. Welcome to TARCIAN Book Ordering System!

1. Order Entry
2. Browse all Books
3. View Shopping Cart
4. Process to Checkout
5. View Order History
6. Logout

Choice: 2

You have selected option 2.


================================== BOOK CATALOGUE ==================================

ID      TITLE                                CATEGORY        PRICE     STOCK    SOLD
------------------------------------------------------------------------------------
1       Book A                               Fiction         29.50     10         8
2       The Last Kingdom                     Fiction         35.00     17        11
3       My Cat Is Secretly an Alien          Fiction         28.50     29         0
4       The Girl Who Spoke to Shadow         Fiction         32.90     15         1
5       How to Marry a Dragon                Fiction         26.40     15         9
6       The Science of Learning              Non-fiction     40.00     7          7
7       How to Lie with Statistics           Non-fiction     33.30     5          1
8       Understanding Human Behavior         Non-fiction     38.80     9          0
9       Why need to go to SCHOOL             Non-fiction     25.00     25         3
10      Adventures in Alphabet Land          Children        21.00     8          0
11      Why Monsters Need Do Homework        Children        18.90     10         0
12      The Magical Pencil                   Children        20.50     8          0
13      How to Train Your Teddy Bear         Children        19.99     10         0
14      A Thousand Autumns                   Novel           36.60     17         3
15      I Fell in Love with a Sandwich       Novel           27.70     7          0
16      The Silence Between Stars            Novel           34.20     26         0
17      My Ex Turned Into a Vampire          Novel           29.90     9          0
18      C++ Programming Handbook             Reference       50.00     8          2
19      My Code Works Only at Midnight       Reference       24.40     15         0
20      Advanced Algorithms                  Reference       48.80     11         6
21      C++ Projects for Lazy Students       Reference       31.10     21         1
22      Book C                               Reference       12.00     15         0


Press 'Enter' to return to customer menu.....
```

E. The user selects option '3' for "View Shopping Cart" and the system displays the shopping cart contents in a formatted table showing Number of book, book title, quantity, unit price, and total price for each item in the cart. The system asks "Do you want to remove an item from cart? (Y/N):" and the user enters "n". The system displays "Okay, nothing removed" and then shows "Press 'Enter' to return to customer menu" for navigation back.

```
Dear customer. Welcome to TARCIAN Book Ordering System!

1. Order Entry
2. Browse all Books
3. View Shopping Cart
4. Process to Checkout
5. View Order History
6. Logout

Choice: 3

You have selected option 3.


========================== Hi I'm your Shopping Cart ==========================

No.  Title                                Qty       Price (RM)     Total (RM)
-------------------------------------------------------------------------------
1    Book A                                2         29.50          59.00

Do you want to remove an item from cart? (Y/N): n
Okay, nothing removed.


Press 'Enter' to return to customer menu.....
```

F. The user selects option '4' for "Process to Checkout" and the system asks "Are you confirm to process checkout (Y/N):". The user enters "Y" and the system displays "Okay... Then let's goo!" confirmation message.

```
Dear customer. Welcome to TARCIAN Book Ordering System!

1. Order Entry
2. Browse all Books
3. View Shopping Cart
4. Process to Checkout
5. View Order History
6. Logout

Choice: 4

You have selected option 4.

Are you confirm to process checkout (Y/N): y
Okay... Then let's goo!
```

G. The system clears the screen and generates a detailed invoice showing information including address and contact details, invoice number, current date, and itemized list of purchased books with individual calculations. The system automatically calculates subtotal, applies discount if applicable (5% for RM50+, 10% for RM100+), adds service charge (6%) and SST tax (6%), then displays the total payable amount. The system prompts for payment method selection with options "(W = e-Wallet, C = Credit card, D = Debit card, A = Cash)". The user enters "w" for e-Wallet payment method and the system accepts the selection. Then the system prompts "Please enter your payment amount (Eg.Rm 120.80): RM" and the user enters "100.00" as payment amount.

```
Invoice:

                                TARCIAN BOOKSTORE
                            No. K666, Block K TARUMT,
                        Jalan Genting Klang, Setapak, 53000
                                Kuala Lumpur.
                              Tel : 03 - 1314520

Invoice No. INV: 20                                                  Date: 2025-09-09
-------------------------------------------------------------------------------------
ID   Title                              Qty       Price (RM)     Total (RM)
-------------------------------------------------------------------------------------
1    Book A                              2         29.50          59.00

Subtotal :                                                        RM59.00
Discount (5%) :                                                 - RM 2.95
Service Charge (6%) :                                             RM 3.36
SST Tax (6%) :                                                    RM 3.36

Total Payable:                                                    RM62.78


(W = e-Wallet, C = Credit card, D = Debit card, A = Cash)
Select payment method: w

Please enter you payment amount (Eg.Rm 120.80): RM 100
```

H. The system validates that the payment amount is sufficient, processes the payment, and generates a complete receipt showing all transaction details including change amount. The system then updates book stock, saves order details to file, clears the shopping cart, and returns to customer menu for additional operations.

```
Receipt:

                                TARCIAN BOOKSTORE
                            No. K666, Block K TARUMT,
                          Jalan Genting Klang, Setapak, 53000
                                Kuala Lumpur.
                              Tel : 03 - 1314520

Receipt No.RCP 23                                                  Date: 2025-09-09
-------------------------------------------------------------------------------
ID   Title                              Qty       Price (RM)     Total (RM)
-------------------------------------------------------------------------------
1    Book A                              2         29.50          59.00
                                                                 ----------------
Subtotal :                                                         RM59.00
Discount (5%) :                                                  - RM 2.95
Service Charge (6%) :                                              RM 3.36
SST Tax (6%) :                                                     RM 3.36

Total Payable:                                                     RM62.78
                                                                 ----------------


Payment Method:                                                    E-Wallet
Payment Amount:                                                    RM100.00
Change:                                                            RM37.22

Thank you for using TARCIAN BOOKSTORE! See you next time! >_<


Press 'Enter' to return to customer menu.....
```

I. The user selects option '5' for "View Order History" to access their previous purchase records. The system displays "ORDER HISTORY" header and searches for all orders associated with the logged-in customer. The system displays complete order history showing all previous purchases made by the current customer. Besides, the system shows Order ID, Date, Book title, Quantity, Price per unit, Subtotal amount, and Payment Method used for the transaction.

```
Dear customer. Welcome to TARCIAN Book Ordering System!

1. Order Entry
2. Browse all Books
3. View Shopping Cart
4. Process to Checkout
5. View Order History
6. Logout

Choice: 5

You have selected option 5.


==================== ORDER HISTORY ====================
```

```
Order ID: 23
Date    : 2025-09-09
Book    : Book A
Quantity: 2
Price   : RM 29.50
Subtotal: RM 59.00
Total   : RM 62.78
Payment Method : E-Wallet



Press 'Enter' to return to customer menu.....
```

J. From the customer menu, the user selects option '6' for "Logout" to exit the customer interface. The system displays "Thank you for visiting TARCIAN Book Ordering System!" appreciation message and clears the screen.

```
Dear customer. Welcome to TARCIAN Book Ordering System!

1. Order Entry
2. Browse all Books
3. View Shopping Cart
4. Process to Checkout
5. View Order History
6. Logout

Choice: 6

You have selected option 6.

Thank you for visiting TARCIAN Book Ordering System!
```

K. At the main menu, the user selects option "0" to exit the entire system. The system displays "Exiting system..." message and terminates the program execution completely.

```
==== TARCIAN BOOKSTORE ORDERING SYSTEM ====

1. Staff Login
2. Customer Login
3. Customer Register
0. Exit

Enter your choice: 0

Exiting system...
```

5.1.2 Test Data For Staff and Management

Main page

```
==== TARCIAN BOOKSTORE ORDERING SYSTEM ====

1. Staff Login
2. Customer Login
3. Customer Register
0. Exit

Enter your choice:
```

```
Enter your choice: 5

Your selection is invalid.

==== TARCIAN BOOKSTORE ORDERING SYSTEM ====

1. Staff Login
2. Customer Login
3. Customer Register
0. Exit

Enter your choice:
```

```
Enter your choice: pe

Invalid input. Please enter a number.

==== TARCIAN BOOKSTORE ORDERING SYSTEM ====

1. Staff Login
2. Customer Login
3. Customer Register
0. Exit

Enter your choice:
```

- if the user accidentally enters the wrong input, the program will prompt an error message and the user is asked to enter the input again.

Staff Login

```
==== TARCIAN BOOKSTORE ORDERING SYSTEM ====

1. Staff Login
2. Customer Login
3. Customer Register
0. Exit

Enter your choice: 1

[Staff Login]
Username: admin
Password: admin123
```

- Choose choice 1 to login to the staff page. The program will then ask the user to enter a staff username and password. Enter “admin” as username and “admin123” as password is the correct way to login.

```
[Staff Login]
Username: ali123
Password: 1234

Staff login failed

==== TARCIAN BOOKSTORE ORDERING SYSTEM ====

1. Staff Login
2. Customer Login
3. Customer Register
0. Exit

Enter your choice:
```

- However, if the user enters the wrong username or password while login as staff, it will prompt an error message and the user is asked to enter their choice again.

Staff Main Menu

```
Enter your choice: 4

===== STAFF MENU =====

1. View Book Catalogue
2. Menu Management
3. Stock
4. Sales Reporting
5. Order Reporting
6. Logout

Enter your choice:
```

View Book Catalogue

```
===== STAFF MENU =====

1. View Book Catalogue
2. Menu Management
3. Stock
4. Sales Reporting
5. Order Reporting
6. Logout

Enter your choice: 1

ID      TITLE                                CATEGORY        PRICE     STOCK    SOLD
-------------------------------------------------------------------------------------
1       Book A                               Fiction         29.50      0         7
2       The Last Kingdom                     Fiction         35.00      15        0
3       My Cat Is Secretly an Alien          Fiction         28.50      9         0
4       The Girl Who Spoke to Shadow         Fiction         32.90      5         1
5       How to Marry a Dragon                Fiction         26.40      1         4
6       The Science of Learning              Non-fiction     40.00      7         0
7       How to Lie with Statistics           Non-fiction     33.30      5         1
8       Understanding Human Behavior         Non-fiction     38.80      9         0
9       Why need to go to SCHOOL             Non-fiction     25.00      0         3
10      Adventures in Alphabet Land          Children        21.00      8         0
11      Why Monsters Need Do Homework        Children        18.90      10        0
12      The Magical Pencil                   Children        20.50      8         0
13      How to Train Your Teddy Bear         Children        19.99      6         0
14      A Thousand Autumns                   Novel           36.60      17        3
15      I Fell in Love with a Sandwich       Novel           27.70      7         0
16      The Silence Between Stars            Novel           34.20      6         0
17      My Ex Turned Into a Vampire          Novel           29.90      4         0
18      C++ Programming Handbook             Reference       50.00      8         2
19      My Code Works Only at Midnight       Reference       24.40      5         0
20      Advanced Algorithms                  Reference       48.80      0         6
21      C++ Projects for Lazy Students       Reference       31.10      6         1


Press 'Enter' to return to menu.....
```

- Enter 1 to view full book catalogue
- Press Enter to return to menu

Menu management

```
===== STAFF MENU =====

1. View Book Catalogue
2. Menu Management
3. Stock
4. Sales Reporting
5. Order Reporting
6. Logout

Enter your choice: 2

==== Manage Book Catalogue ====

1. Update Book Info
2. Add New Book
3. Remove Book
4. Return

Enter your choice:
```

- Enter 2 to enter menu management, then the program will show the selection of managing book catalogue, such as updating book info, adding new books and removing books.

Update Book Info

```
==== Manage Book Catalogue ====

1. Update Book Info
2. Add New Book
3. Remove Book
4. Return

Enter your choice: 1

================================== BOOK CATALOGUE ==================================

ID      TITLE                                CATEGORY        PRICE     STOCK    SOLD
------------------------------------------------------------------------------------
1       Book A                               Fiction         29.50      0         7
2       The Last Kingdom                     Fiction         35.00      15        0
3       My Cat Is Secretly an Alien          Fiction         28.50      9         0
4       The Girl Who Spoke to Shadow         Fiction         32.90      5         1
5       How to Marry a Dragon                Fiction         26.40      1         4
6       The Science of Learning              Non-fiction     40.00      7         0
7       How to Lie with Statistics           Non-fiction     33.30      5         1
8       Understanding Human Behavior         Non-fiction     38.80      9         0
9       Why need to go to SCHOOL             Non-fiction     25.00      0         3
10      Adventures in Alphabet Land          Children        21.00      8         0
11      Why Monsters Need Do Homework        Children        18.90      10        0
12      The Magical Pencil                   Children        20.50      8         0
13      How to Train Your Teddy Bear         Children        19.99      6         0
14      A Thousand Autumns                   Novel           36.60      17        3
15      I Fell in Love with a Sandwich       Novel           27.70      7         0
16      The Silence Between Stars            Novel           34.20      6         0
17      My Ex Turned Into a Vampire          Novel           29.90      4         0
18      C++ Programming Handbook             Reference       50.00      8         2
19      My Code Works Only at Midnight       Reference       24.40      5         0
20      Advanced Algorithms                  Reference       48.80      0         6
21      C++ Projects for Lazy Students       Reference       31.10      6         1

Enter book id to update book info:
```

- Choose 1 to update book info. The program will then show the book catalogue, letting admin choose the book they want to update the info.

```
Enter new title: Book B

Book info updated successfully

============================== CURRENT BOOK CATALOGUE ==============================

ID      TITLE                                CATEGORY        PRICE     STOCK    SOLD
--------------------------------------------------------------------------------------
1       Book B                               Fiction         29.50      0         7
2       The Last Kingdom                     Fiction         35.00      15        0
3       My Cat Is Secretly an Alien          Fiction         28.50      9         0
4       The Girl Who Spoke to Shadow         Fiction         32.90      5         1
5       How to Marry a Dragon                Fiction         26.40      1         4
6       The Science of Learning              Non-fiction     40.00      7         0
7       How to Lie with Statistics           Non-fiction     33.30      5         1
8       Understanding Human Behavior         Non-fiction     38.80      9         0
9       Why need to go to SCHOOL             Non-fiction     25.00      0         3
10      Adventures in Alphabet Land          Children        21.00      8         0
11      Why Monsters Need Do Homework        Children        18.90      10        0
12      The Magical Pencil                   Children        20.50      8         0
13      How to Train Your Teddy Bear         Children        19.99      6         0
14      A Thousand Autumns                   Novel           36.60      17        3
15      I Fell in Love with a Sandwich       Novel           27.70      7         0
16      The Silence Between Stars            Novel           34.20      6         0
17      My Ex Turned Into a Vampire          Novel           29.90      4         0
18      C++ Programming Handbook             Reference       50.00      8         2
19      My Code Works Only at Midnight       Reference       24.40      5         0
20      Advanced Algorithms                  Reference       48.80      0         6
21      C++ Projects for Lazy Students       Reference       31.10      6         1


Press 'Enter' to return to menu.....
```

- if the admin chooses to update the title, the admin is asked to enter the new title for the book. If book info is updated successfully, it will display a new book catalogue after updating.

Add new book

```
Enter your choice: 2
Enter book title: Book C
Enter book category: Non-Fiction
Enter book price: 20.00
Enter book stock: 15

Book added successfully!

============================== CURRENT BOOK CATALOGUE ==============================

ID      TITLE                                   CATEGORY        PRICE     STOCK    SOLD
--------------------------------------------------------------------------------------
1       Book B                                  Fiction         29.50      0         7
2       The Last Kingdom                        Fiction         35.00      15        0
3       My Cat Is Secretly an Alien             Fiction         28.50      9         0
4       The Girl Who Spoke to Shadow            Fiction         32.90      5         1
5       How to Marry a Dragon                   Fiction         26.40      1         4
6       The Science of Learning                 Non-fiction     40.00      7         0
7       How to Lie with Statistics              Non-fiction     33.30      5         1
8       Understanding Human Behavior            Non-fiction     38.80      9         0
9       Why need to go to SCHOOL                Non-fiction     25.00      0         3
10      Adventures in Alphabet Land             Children        21.00      8         0
11      Why Monsters Need Do Homework           Children        18.90      10        0
12      The Magical Pencil                      Children        20.50      8         0
13      How to Train Your Teddy Bear            Children        19.99      6         0
14      A Thousand Autumns                      Novel           36.60      17        3
15      I Fell in Love with a Sandwich          Novel           27.70      7         0
16      The Silence Between Stars               Novel           34.20      6         0
17      My Ex Turned Into a Vampire             Novel           29.90      4         0
18      C++ Programming Handbook                Reference       50.00      8         2
19      My Code Works Only at Midnight          Reference       24.40      5         0
20      Advanced Algorithms                     Reference       48.80      0         6
21      C++ Projects for Lazy Students          Reference       31.10      6         1
22      Book C                                  Non-Fiction     20.00      15        0


Press 'Enter' to return to menu.....
```

- Choose 2 to add a new book
- Then, enter the book title, category, price and stock for the new book
- The latest book catalogue displayed if the adding book action is successful

Remove Book

```
==== Manage Book Catalogue ====

1. Update Book Info
2. Add New Book
3. Remove Book
4. Return

Enter your choice: 3

================================== BOOK CATALOGUE ==================================

ID      TITLE                                CATEGORY        PRICE     STOCK    SOLD
------------------------------------------------------------------------------------
1       Book B                               Fiction         29.50      0         7
2       The Last Kingdom                     Fiction         35.00      15        0
3       My Cat Is Secretly an Alien          Fiction         28.50      9         0
4       The Girl Who Spoke to Shadow         Fiction         32.90      5         1
5       How to Marry a Dragon                Fiction         26.40      1         4
6       The Science of Learning              Non-fiction     40.00      7         0
7       How to Lie with Statistics           Non-fiction     33.30      5         1
8       Understanding Human Behavior         Non-fiction     38.80      9         0
9       Why need to go to SCHOOL             Non-fiction     25.00      0         3
10      Adventures in Alphabet Land          Children        21.00      8         0
11      Why Monsters Need Do Homework        Children        18.90      10        0
12      The Magical Pencil                   Children        20.50      8         0
13      How to Train Your Teddy Bear         Children        19.99      6         0
14      A Thousand Autumns                   Novel           36.60      17        3
15      I Fell in Love with a Sandwich       Novel           27.70      7         0
16      The Silence Between Stars            Novel           34.20      6         0
17      My Ex Turned Into a Vampire          Novel           29.90      4         0
18      C++ Programming Handbook             Reference       50.00      8         2
19      My Code Works Only at Midnight       Reference       24.40      5         0
20      Advanced Algorithms                  Reference       48.80      0         6
21      C++ Projects for Lazy Students       Reference       31.10      6         1
22      Book C                               Non-Fiction     20.00      15        0

Enter book id to remove:
```

- Choose 3 to remove book

```
Enter book id to remove: 22

============================== CURRENT BOOK CATALOGUE ==============================

ID      TITLE                                   CATEGORY        PRICE     STOCK    SOLD
-------------------------------------------------------------------------------------
1       Book B                                  Fiction         29.50      0         7
2       The Last Kingdom                        Fiction         35.00      15        0
3       My Cat Is Secretly an Alien             Fiction         28.50      9         0
4       The Girl Who Spoke to Shadow            Fiction         32.90      5         1
5       How to Marry a Dragon                   Fiction         26.40      1         4
6       The Science of Learning                 Non-fiction     40.00      7         0
7       How to Lie with Statistics              Non-fiction     33.30      5         1
8       Understanding Human Behavior            Non-fiction     38.80      9         0
9       Why need to go to SCHOOL                Non-fiction     25.00      0         3
10      Adventures in Alphabet Land             Children        21.00      8         0
11      Why Monsters Need Do Homework           Children        18.90      10        0
12      The Magical Pencil                      Children        20.50      8         0
13      How to Train Your Teddy Bear            Children        19.99      6         0
14      A Thousand Autumns                      Novel           36.60      17        3
15      I Fell in Love with a Sandwich          Novel           27.70      7         0
16      The Silence Between Stars               Novel           34.20      6         0
17      My Ex Turned Into a Vampire             Novel           29.90      4         0
18      C++ Programming Handbook                Reference       50.00      8         2
19      My Code Works Only at Midnight          Reference       24.40      5         0
20      Advanced Algorithms                     Reference       48.80      0         6
21      C++ Projects for Lazy Students          Reference       31.10      6         1


Press 'Enter' to return to menu.....
```

- Enter the book id you would like to remove
- If the book id exists, the remove action will be successful, then the program will display the current book catalogue after removing action.

```
Enter book id to remove: 1

============================== CURRENT BOOK CATALOGUE ==============================

ID      TITLE                                   CATEGORY        PRICE     STOCK    SOLD
-------------------------------------------------------------------------------------
1       The Last Kingdom                        Fiction         35.00      15        7
2       My Cat Is Secretly an Alien             Fiction         28.50      9         0
3       The Girl Who Spoke to Shadow            Fiction         32.90      5         0
4       How to Marry a Dragon                   Fiction         26.40      1         1
5       The Science of Learning                 Non-fiction     40.00      7         4
6       How to Lie with Statistics              Non-fiction     33.30      5         0
7       Understanding Human Behavior            Non-fiction     38.80      9         1
8       Why need to go to SCHOOL                Non-fiction     25.00      0         0
9       Adventures in Alphabet Land             Children        21.00      8         3
10      Why Monsters Need Do Homework           Children        18.90      10        0
11      The Magical Pencil                      Children        20.50      8         0
12      How to Train Your Teddy Bear            Children        19.99      6         0
13      A Thousand Autumns                      Novel           36.60      17        0
14      I Fell in Love with a Sandwich          Novel           27.70      7         3
15      The Silence Between Stars               Novel           34.20      6         0
16      My Ex Turned Into a Vampire             Novel           29.90      4         0
17      C++ Programming Handbook                Reference       50.00      8         0
18      My Code Works Only at Midnight          Reference       24.40      5         2
19      Advanced Algorithms                     Reference       48.80      0         0
20      C++ Projects for Lazy Students          Reference       31.10      6         6


Press 'Enter' to return to menu.....
```

- System will auto rearrange the book id after removing a book.

Stock

```
===== STAFF MENU =====

1. View Book Catalogue
2. Menu Management
3. Stock
4. Sales Reporting
5. Order Reporting
6. Logout

Enter your choice: 3

==== STOCK MANAGEMENT ====

1. View Stock
2. Update Stock
3. Return

Enter your choice:
```

- **Choose 3 to enter stock management page**

View Stock

```
==== STOCK MANAGEMENT ====

1. View Stock
2. Update Stock
3. Return

Enter your choice: 1

================================== VIEW STOCK ==================================

ID      TITLE                                 CATEGORY        PRICE     STOCK    SOLD
--------------------------------------------------------------------------------------
1       The Last Kingdom                      Fiction         34.90      15        7
2       My Cat Is Secretly an Alien           Fiction         28.50      9         0
3       The Girl Who Spoke to Shadow          Fiction         32.90      5         0
4       How to Marry a Dragon                 Fiction         26.40      1         1
5       The Science of Learning               Non-fiction     40.00      7         4
6       How to Lie with Statistics            Non-fiction     33.30      5         0
7       Understanding Human Behavior          Non-fiction     38.80      9         1
8       Why need to go to SCHOOL              Non-fiction     25.00      0         0
9       Adventures in Alphabet Land           Children        21.00      8         3
10      Why Monsters Need Do Homework         Children        18.90      10        0
11      The Magical Pencil                    Children        20.50      8         0
12      How to Train Your Teddy Bear          Children        19.99      6         0
13      A Thousand Autumns                    Novel           36.60      17        0
14      I Fell in Love with a Sandwich        Novel           27.70      7         3
15      The Silence Between Stars             Novel           34.20      6         0
16      My Ex Turned Into a Vampire           Novel           29.90      4         0
17      C++ Programming Handbook              Reference       50.00      8         0
18      My Code Works Only at Midnight        Reference       24.40      5         2
19      Advanced Algorithms                   Reference       48.80      0         0
20      C++ Projects for Lazy Students        Reference       31.10      6         6


Press 'Enter' to return to menu.....
```

- Choose 1 to view stock

Update Stock

```
==== STOCK MANAGEMENT ====

1. View Stock
2. Update Stock
3. Return

Enter your choice: 2

==================================== STOCK ======================================

ID      TITLE                                CATEGORY        PRICE     STOCK    SOLD
--------------------------------------------------------------------------------
1       The Last Kingdom                     Fiction         34.90      15       7
2       My Cat Is Secretly an Alien          Fiction         28.50      9        0
3       The Girl Who Spoke to Shadow         Fiction         32.90      5        0
4       How to Marry a Dragon                Fiction         26.40      1        1
5       The Science of Learning              Non-fiction     40.00      7        4
6       How to Lie with Statistics           Non-fiction     33.30      5        0
7       Understanding Human Behavior         Non-fiction     38.80      9        1
8       Why need to go to SCHOOL             Non-fiction     25.00      0        0
9       Adventures in Alphabet Land          Children        21.00      8        3
10      Why Monsters Need Do Homework        Children        18.90      10       0
11      The Magical Pencil                   Children        20.50      8        0
12      How to Train Your Teddy Bear         Children        19.99      6        0
13      A Thousand Autumns                   Novel           36.60      17       0
14      I Fell in Love with a Sandwich       Novel           27.70      7        3
15      The Silence Between Stars            Novel           34.20      6        0
16      My Ex Turned Into a Vampire          Novel           29.90      4        0
17      C++ Programming Handbook             Reference       50.00      8        0
18      My Code Works Only at Midnight       Reference       24.40      5        2
19      Advanced Algorithms                  Reference       48.80      0        0
20      C++ Projects for Lazy Students       Reference       31.10      6        6

Enter book id to update stock:
```

- Choose 2 to update stock

```
Enter book id to update stock: 4
Enter new stock: 10

================================== CURRENT STOCK ==================================

ID      TITLE                                CATEGORY        PRICE     STOCK    SOLD
-----------------------------------------------------------------------------------
1       The Last Kingdom                     Fiction         34.90      15       7
2       My Cat Is Secretly an Alien          Fiction         28.50      9        0
3       The Girl Who Spoke to Shadow         Fiction         32.90      5        0
4       How to Marry a Dragon                Fiction         26.40      10       1
5       The Science of Learning              Non-fiction     40.00      7        4
6       How to Lie with Statistics           Non-fiction     33.30      5        0
7       Understanding Human Behavior         Non-fiction     38.80      9        1
8       Why need to go to SCHOOL             Non-fiction     25.00      0        0
9       Adventures in Alphabet Land          Children        21.00      8        3
10      Why Monsters Need Do Homework        Children        18.90      10       0
11      The Magical Pencil                   Children        20.50      8        0
12      How to Train Your Teddy Bear         Children        19.99      6        0
13      A Thousand Autumns                   Novel           36.60      17       0
14      I Fell in Love with a Sandwich       Novel           27.70      7        3
15      The Silence Between Stars            Novel           34.20      6        0
16      My Ex Turned Into a Vampire          Novel           29.90      4        0
17      C++ Programming Handbook             Reference       50.00      8        0
18      My Code Works Only at Midnight       Reference       24.40      5        2
19      Advanced Algorithms                  Reference       48.80      0        0
20      C++ Projects for Lazy Students       Reference       31.10      6        6


Press 'Enter' to return to menu.....
```

- Enter the book id to update stock, then enter the new stock for it
- Current stock for each book displayed if the updating action is successful

Sales Report Menu

```
===== STAFF MENU =====

1. View Book Catalogue
2. Menu Management
3. Stock
4. Sales Reporting
5. Order Reporting
6. Logout

Enter your choice: 4

========== Sales Report ==========
1. Daily Sales Report
2. Monthly Sales Report
3. Return

Enter your choice:
```

- Choose 4 to display sales report menu

Daily Sales Report

```
========== Sales Report ==========
1. Daily Sales Report
2. Monthly Sales Report
3. Return

Enter your choice: 1

Date (YYYY-MM-DD): 2025-08-31

===== Daily Sales Report =====

TITLE                                SOLD           SALES
-------------------------------------------------------------------

-------------------------------------------------------------------
Sales not found on 2025-08-31


Press 'Enter' to return to menu.....
```

- Choose 1 to display daily sales report
- If the date entered is not found, it will display error message

```
Enter your choice: 1

Date (YYYY-MM-DD): 2025-09-05

===== Daily Sales Report =====

TITLE                                SOLD           SALES
-------------------------------------------------------------------
Book A                                2             59.00
C++ Projects for Lazy Students        1             31.10
Advanced Algorithms                   2             97.60
C++ Programming Handbook              2             100.00
A Thousand Autumns                    3             109.80
Why need to go to SCHOOL              3             75.00
The Girl Who Spoke to Shadow          1             32.90
How to Lie with Statistics            1             33.30

-------------------------------------------------------------------

Total Sold: 15
Total Daily Sales: 538.70


Press 'Enter' to return to menu.....
```

- If the date entered is found, the program will display a daily sales report with total sold and total daily sales.

Monthly sales report

```
========== Sales Report ==========
1. Daily Sales Report
2. Monthly Sales Report
3. Return

Enter your choice: 2
```

- Choose 2 to display monthly sales report

```
Month (YYYY-MM): 2025-09

===== Monthly Sales Report =====

TITLE                                SOLD           SALES
-------------------------------------------------------------------------
Book A                                7             206.50
C++ Projects for Lazy Students        1             31.10
Advanced Algorithms                   6             292.80
C++ Programming Handbook              2             100.00
A Thousand Autumns                    3             109.80
Why need to go to SCHOOL              3             75.00
The Girl Who Spoke to Shadow          1             32.90
How to Lie with Statistics            1             33.30
Book C                                1             12.00
How to Marry a Dragon                 4             105.60

-------------------------------------------------------------------------

Total Sold: 29
Total Monthly Sales: 999.00


Press 'Enter' to return to menu.....
```

- If the date entered is found, the program will display monthly sales report with total sold and total monthly sales.

```
Month (YYYY-MM): 2025-07

===== Monthly Sales Report =====

TITLE                                SOLD           SALES
-------------------------------------------------------------------------

-------------------------------------------------------------------------
Sales not found on


Press 'Enter' to return to menu.....
```

- If the date entered is not found, it will prompt an error message.

Order Reporting

```
===== STAFF MENU =====

1. View Book Catalogue
2. Menu Management
3. Stock
4. Sales Reporting
5. Order Reporting
6. Logout

Enter your choice: 5
```

- Choose 5 to display order reporting

```
Date (YYYY-MM-DD): 2025-09-05

============= Order Report on 2025-09-05 ==============
Order ID: 1
Customer Name: ali123
Title: Book A
Price: 29.50
Quantity: 1
Subtotal: 29.50
Total: 34.22
Payment method: E-Wallet
Order Date: 2025-09-05

------------------------------------------------------------------------

Order ID: 2
Customer Name: lily890
Title: C++ Projects for Lazy Students
Price: 31.10
Quantity: 1
Subtotal: 179.90
Total: 187.82
Payment method: Debit Card
Order Date: 2025-09-05

------------------------------------------------------------------------
```

- If the date entered is found, the program will display the order report.

5.1.3 Program Testing with Input Validation (Customer Part)

Test Data 3:

Step	Test Data Input	Expected Output
1	Main Menu: Enter abc	System displays "Invalid input. Please enter a number."
2	Main Menu: Enter 9	System displays "Your selection is invalid."
3	Customer Login - Username: ali123 Password: ali0123	System displays "Invalid username or password."
4	Customer Menu: Enter 9	System displays “Invalid choice. Please try again.”
5	Order Entry - Book ID: Enter abc	System displays "Invalid input. Please enter a number."
6	Order Entry - Book ID: Enter 999	System displays "Invalid Book ID, Please Enter Again."
7	Order Entry - Quantity: Enter -5 / 5	System displays "Invalid quantity" and shows available stock
8	Order Entry - Quantity: Enter 0	System accepts and continues without adding to cart and displays “Anymore Order (Y/N)? ”
9	Order Entry - Anymore order: Enter 5	System displays "Invalid input. Please try again. “
10	Order Entry - Anymore order: Enter c	System displays "Invalid input. Please try again. “
11	Process to Checkout - confirm checkout: Enter 7	System displays "Please enter Y or N”
12	Process to Checkout - confirm checkout: Enter a	System displays "Please enter Y or N”
13	Process to Checkout - Payment Method: Enter 5	System displays "Invalid payment method. Please try again.”
14	Process to Checkout - Payment Method: Enter X	System displays "Invalid payment method. Please try again.”
15	Payment Amount: Enter 10.00 (less than total)	System displays "Not enough payment amount. Please try again."

Step 1: Step 2:

```
==== TARCIAN BOOKSTORE ORDERING SYSTEM ====

1. Staff Login
2. Customer Login
3. Customer Register
0. Exit

Enter your choice: abc

Invalid input. Please enter a number.

==== TARCIAN BOOKSTORE ORDERING SYSTEM ====

1. Staff Login
2. Customer Login
3. Customer Register
0. Exit

Enter your choice: 9

Your selection is invalid.
```

Step 3:

```
========== Customer Login ==========
Enter username: ali123
Enter password: ali0123

Invalid username or password.
```

Step 4:

```
Dear customer. Welcome to TARCIAN Book Ordering System!

1. Order Entry
2. Browse all Books
3. View Shopping Cart
4. Process to Checkout
5. View Order History
6. Logout

Choice: 9

You have selected option 9.

Invalid choice. Please try again.

Dear customer. Welcome to TARCIAN Book Ordering System!

1. Order Entry
2. Browse all Books
3. View Shopping Cart
4. Process to Checkout
5. View Order History
6. Logout

Choice:
```

Step 5: Step 6: Step 7: Step 8: Step 9: Step 10:

```
================================ BOOK CATALOGUE ================================

ID      TITLE                                CATEGORY        PRICE     STOCK    SOLD
--------------------------------------------------------------------------------
1       Book A                               Fiction         29.50     2        16
2       The Last Kingdom                     Fiction         35.00     17       11
3       My Cat Is Secretly an Alien          Fiction         28.50     29       0
4       The Girl Who Spoke to Shadow         Fiction         32.90     15       1
5       How to Marry a Dragon                Fiction         26.40     15       9
6       The Science of Learning              Non-fiction     40.00     7        7
7       How to Lie with Statistics           Non-fiction     33.30     5        1
8       Understanding Human Behavior         Non-fiction     38.80     9        0
9       Why need to go to SCHOOL             Non-fiction     25.00     25       3
10      Adventures in Alphabet Land          Children        21.00     8        0
11      Why Monsters Need Do Homework        Children        18.90     10       0
12      The Magical Pencil                   Children        20.50     8        0
13      How to Train Your Teddy Bear         Children        19.99     10       0
14      A Thousand Autumns                   Novel           36.60     17       3
15      I Fell in Love with a Sandwich       Novel           27.70     7        0
16      The Silence Between Stars            Novel           34.20     26       0
17      My Ex Turned Into a Vampire          Novel           29.90     9        0
18      C++ Programming Handbook             Reference       50.00     8        2
19      My Code Works Only at Midnight       Reference       24.40     15       0
20      Advanced Algorithms                  Reference       48.80     11       6
21      C++ Projects for Lazy Students       Reference       31.10     21       1
22      Book C                               Reference       12.00     15       0

Order 1
Enter the book ID to add to shopping cart: abc
Invalid input. Please enter a number.

Enter the book ID to add to shopping cart: 999
Invalid Book ID, Please Enter Again.

Enter the book ID to add to shopping cart: 1
Enter quantity: -5
Invalid quantity. Available stock: 2

Enter quantity: 5
Invalid quantity. Available stock: 2

Enter quantity: 0

Anymore order (Y/N)? 5
Invalid input. Please try again.


Anymore order (Y/N)? c
Invalid input. Please try again.
```

Step 11: Step 12: Step 13: Step 14: Step 15:

```
Invoice:

                                TARCIAN BOOKSTORE
                            No. K666, Block K TARUMT,
                        Jalan Genting Klang, Setapak, 53000
                                Kuala Lumpur.
                              Tel : 03 - 1314520

Invoice No. INV: 24                                                  Date: 2025-09-09
--------------------------------------------------------------------------------
ID   Title                              Qty       Price (RM)     Total (RM)
--------------------------------------------------------------------------------
1    Book A                              2         29.50          59.00

Subtotal :                                                        RM59.00
Discount (5%) :                                                 - RM 2.95
Service Charge (6%) :                                             RM 3.36
SST Tax (6%) :                                                    RM 3.36

Total Payable:                                                    RM62.78


(W = e-Wallet, C = Credit card, D = Debit card, A = Cash)
Select payment method: 5
Invalid payment method. Please try again.


(W = e-Wallet, C = Credit card, D = Debit card, A = Cash)
Select payment method: X
Invalid payment method. Please try again.


(W = e-Wallet, C = Credit card, D = Debit card, A = Cash)
Select payment method: w

Please enter you payment amount (Eg.Rm 120.80): RM 10.00
Not enought payment amount. Please try again.
```

6.0 PROGRAM LISTING

data.h

```
//data.h
#ifndef DATA_H
#define DATA_H
#include <map>
#include <string>
#include <vector>
using namespace std;

struct book_catalogue {
	int id;
	string title;
	string category;
	double price;
	int stock;
};

struct customer {
	string username;
	string password;
};

struct order {
	int order_id;
	string username;
	string book_title;
	double book_price;
	int quantity;
	double subtotal;
	double total;
	string payment_method;
	string order_date;
};

struct sales {
	string key;
	string title;
	int quantity = 0;
	double revenue = 0.0;
};

extern sales daily_sales[1000];
extern int dailyCount;

extern sales monthly_sales[1000];
extern int monthlyCount;

const int MAX_BOOKS = 100;
extern book_catalogue bookcatalogue[MAX_BOOKS];
extern int bookCount;

const int MAX_CUSTOMER = 100;
extern customer customer_info[MAX_CUSTOMER];
extern int customerCount;

extern vector<order> order_detail;
```

```
extern int total_sold[MAX_BOOKS];

void loadBook();
void saveBook();
void displayBook();
void loadOrder();
void saveOrder();
void loadCustomer();
void saveCustomer();
void clearScreen();

#endif
```

data.cpp

```
//data.cpp
#include "data.h"
#include <fstream>
#include <iostream>
#include <string>
#include <iomanip>
#include <map>
using namespace std;

book_catalogue bookcatalogue[MAX_BOOKS]; // set the maximum number for book catalogue
int bookCount = 0; // number of book

customer customer_info[MAX_CUSTOMER]; // set the maximum number for customer
int customerCount = 0; // number of customer

vector<order>order_detail; // store order detail with no maximum number

sales daily_sales[1000];
int dailyCount = 0;

sales monthly_sales[1000];
int monthlyCount = 0;

int total_sold[MAX_BOOKS] = { 0 };

void clearScreen() { // clear screen
	for (int i = 0; i < 50; i++) {
		cout << endl;
	}
}
```

- loadBook() / saveBook() / displayBook()

```
// read book catalogue
void loadBook() {
	// declare file stream variables
	ifstream inData;
	// open input file
	inData.open("books.txt");
	if (!inData) {
		cout << "Error opening file"; // promp error msg if can't open file
		return;
	}
	int id, stock;
	string title, category;
	double price;

	// read data
	while (inData >> id){
		inData.ignore();
		getline(inData, title,'|');
		inData >> category >> price >> stock;
		bookcatalogue[bookCount].id = id;
		bookcatalogue[bookCount].title = title;
		bookcatalogue[bookCount].category = category;
		bookcatalogue[bookCount].price = price;
		bookcatalogue[bookCount].stock = stock;

		bookCount++;
	}
	inData.close(); //close input file
}

// save book catalogue
void saveBook(){
	ofstream outData;
	// open output file
	outData.open("books.txt");
	if (!outData) {
		cout << "Error saving books";
		return;
	}
	// write the data into the file
	for (int i = 0; i < bookCount; i++) {
		outData << bookcatalogue[i].id << " "
			<< bookcatalogue[i].title << "|"
			<< bookcatalogue[i].category << " "
			<< fixed << setprecision(2) << bookcatalogue[i].price << " "
			<< bookcatalogue[i].stock << endl;
	}
	outData.close(); // close output file
}

// display book catalogue
void displayBook() {
	cout << "\nID\t"<<left<<setw(35)<<"TITLE"<<left<<setw(16)<<"CATEGORY"<< left <<
setw(10) << "PRICE" << left << setw(9) << "STOCK" << "SOLD" << endl;
```

```
	cout << "-------------------------------------------------------------------------------" << endl;
	for (int i = 0; i < bookCount; i++) {
		cout << bookcatalogue[i].id << "\t" << left << setw(35) << bookcatalogue[i].title << left << setw(16) << bookcatalogue[i].category << left << setw(11) << fixed << setprecision(2) << bookcatalogue[i].price << left << setw(10) << bookcatalogue[i].stock << total_sold[i] << endl;
	}
}
```

- loadOrder() / saveOrder()

```
// read order_detail
void loadOrder() {
	ifstream inData;
	inData.open("order.txt");
	if (!inData) {
		cout << "Error opening file";
		return;
	}
	int order_id,quantity;
	double subtotal,total,book_price;
	string username, book_title, payment_method, order_date;

	while (inData >> order_id>> username){
		inData.ignore();
		getline(inData, book_title, '|');
		inData >> book_price >> quantity >> subtotal >> total;
		inData.ignore();
		getline(inData, payment_method, '|');
		inData >> order_date;
		order_detail.push_back({ order_id, username, book_title, book_price,quantity,subtotal,total,payment_method,order_date }); //push order detail into vector
		for (int j = 0; j < bookCount; j++) {
			if (bookcatalogue[j].title == book_title) {
				total_sold[j] += quantity;
				break;
			}
		}
		// daily sales
		bool found = false;
		for (int i = 0; i < dailyCount; i++) {
			if (daily_sales[i].key == order_date && daily_sales[i].title == book_title) {
				found = true;
				daily_sales[i].quantity += quantity;
				daily_sales[i].revenue += book_price * quantity;
				break;
			}
		}
		if (!found) {
			daily_sales[dailyCount].key = order_date;
			daily_sales[dailyCount].title = book_title;
			daily_sales[dailyCount].quantity = quantity;
			daily_sales[dailyCount].revenue = book_price * quantity;
			dailyCount++;
		}
```

```
                    // monthly_sales
                    string month = order_date.substr(0, 7);
                    found = false;
                    for (int i = 0; i < monthlyCount; i++) {
                            if (monthly_sales[i].key == month && monthly_sales[i].title == book_title) {
                                    found = true;
                                    monthly_sales[i].quantity += quantity;
                                    monthly_sales[i].revenue += book_price * quantity;

                                    break;
                            }
                    }
                    if (!found) {
                            monthly_sales[monthlyCount].key = month;
                            monthly_sales[monthlyCount].title = book_title;
                            monthly_sales[monthlyCount].quantity = quantity;
                            monthly_sales[monthlyCount].revenue = book_price * quantity;
                            monthlyCount++;
                    }

            }
            inData.close();
}

// save order_detail
void saveOrder() {
            ofstream outData;
            outData.open("order.txt");
            if (!outData) {
                    cout << "Error saving order";
                    return;
            }
            for (int i = 0; i < order_detail.size(); i++) {
                    outData << order_detail[i].order_id << " " << order_detail[i].username << " " <<
order_detail[i].book_title << "|" << fixed << setprecision(2) << order_detail[i].book_price << " " <<
order_detail[i].quantity << " " << fixed << setprecision(2) << order_detail[i].subtotal << " " << fixed <<
setprecision(2) << order_detail[i].total << " " << order_detail[i].payment_method << "|" <<
order_detail[i].order_date << endl;
            }
            outData.close();
}
```

- loadCustomer() / saveCustomer()

```
// read customer info
void loadCustomer() {
            ifstream inData;
            inData.open("customer.txt");
            if (!inData) {
cout << "Error opening file";
                    return;
            }
            string username, password;

            while (inData >> username >> password) {
```

```
                customer_info[customerCount].username = username;
                customer_info[customerCount].password = password;

                customerCount++;
        }
        inData.close();
}

// save customer info
void saveCustomer() {
        ofstream outData;
        outData.open("customer.txt");
        if (!outData) {
                cout << "Error saving customer";
                return;
        }
        for (int i = 0; i < customerCount; i++) {
                outData << customer_info[i].username << " " << customer_info[i].password << endl;
        }
        outData.close();
}
```

main.cpp

```
#include <iostream>
#include <string>
#include "data.h"
using namespace std;

int mainMenu();
void staffLogin();
void customerLogin();
void customerRegister();

int main() {
        loadBook();
        loadOrder();
        loadCustomer();
        int choice;
        do {
                choice = mainMenu();
                switch (choice) {
                case 1:
                        staffLogin();
                        break;
                case 2:
                        customerLogin();
                        break;
                case 3:
                        customerRegister();
                        break;
                case 0:
                        cout << "\nExiting system..." << endl;
```

```
                        break;
                default:
                        cout << "\nYour selection is invalid." << endl;
                        break;
                }
        } while (choice != 0);
        return 0;
}

//display main menu
int mainMenu() {
        int choice;
        while (true) {
                cout << "\n==== TARCIAN BOOKSTORE ORDERING SYSTEM ====" << endl;
                cout << "\n1. Staff Login" << endl;
                cout << "2. Customer Login" << endl;
                cout << "3. Customer Register" << endl;
                cout << "0. Exit" << endl;
                cout << "\nEnter your choice: ";
                if (cin >> choice) {
                        return choice;
                }
                else {
                        cin.clear();
                        cin.ignore(100, '\n');
                        cout << "\nInvalid input. Please enter a number." << endl;
                }
        }
}
```

staff.cpp

```
#include <iostream>
#include <string>
#include "data.h"
#include <iomanip>
using namespace std;

void clearScreen();
void backToMenu();
void staffLogin();
int mainMenu();
void staffMain();
int staffMenu();
void bookCatalogue();
void menuManage();
void updateBook();
void addBook();
void removeBook();
void stock();
void salesReport();
void orderReport();

void backToMenu() {
        char key;
```

```
        cout << "\n\nPress 'Enter' to return to menu..... ";
        cin.ignore();
        cin.get(key);
        clearScreen();
}
```

- staffLogin()

```
//staff login
void staffLogin() {
        string username, password;
                cout << "\n[Staff Login]" << endl;
                cout << "Username: ";
                cin >> username;
                cout << "Password: ";
                cin >> password;
                if (username == "admin" && password == "admin123") {
                        cout << "\nSatff login successfull" << endl;
                        clearScreen();
                        staffMain();
                }
                else {
                        cout << "\nStaff login failed" << endl;
                }
}
```

- staffMain() / staffMenu()

```
void staffMain() {
        int choice;
        do {
                choice = staffMenu();
                switch (choice) {
                case 1 :
                        bookCatalogue();
                        break;
                case 2:
                        menuManage();
                        break;
                case 3:
                        stock();
                        break;
                case 4:
                        salesReport();
                        break;
                case 5:
                        orderReport();
                        break;
                case 6:
                        clearScreen();
```

```
                        return;
                default:
                        cout << "\nYour selection is invalid" << endl;
                        break;
                }
        } while (choice != 6);
}

// staff menu
int staffMenu() {
        int choice;
        while (true) {
                cout << "\n===== STAFF MENU =====" << endl;
                cout << "\n1. View Book Catalogue" << endl;
                cout << "2. Menu Management" << endl;
                cout << "3. Stock" << endl;
                cout << "4. Sales Reporting" << endl;
                cout << "5. Order Reporting" << endl;
                cout << "6. Logout" << endl;
                cout << "\nEnter your choice: ";
                if (cin >> choice) {
                        return choice;
                }
                else {
                        cin.clear();
                        cin.ignore(100, '\n');
                        cout << "\nInvalid input. Please enter a number." << endl;
                }
        }
}
```

- bookCatalogue()

```
// view book catalogue
void bookCatalogue() {
        displayBook();
        backToMenu();
        return;
}
```

- menuManage()

```
// manage book menu
void menuManage() {
        int choice;
        do {
                cout << "\n==== Manage Book Catalogue ====" << endl;
                cout << "\n1. Update Book Info" << endl;
                cout << "2. Add New Book" << endl;
                cout << "3. Remove Book" << endl;
```

```
                cout << "4. Return" << endl;
                cout << "\nEnter your choice: ";
                if (!(cin >> choice)) {
                        cin.clear();
                        cin.ignore(100, '\n');
                        cout << "\nInvalid input. Please enter a number." << endl;
                        continue;
                }
                switch (choice) {
                case 1:
                        updateBook();
                        break;
                case 2:
                        addBook();
                        break;
                case 3:
                        removeBook();
                        break;
                case 4:
                        return;
                        break;
                default:
                        cout << "\nInvalid choice" << endl;
                        break;
                }
        } while (choice !=4);
}
```

- updateBook()

```
// eidt book info
void updateBook() {
        int id, choice, i;
        bool found = false;
        do {
                cout << "\n==================================== BOOK CATALOGUE
==================================" << endl;
                displayBook();
                cout << "\nEnter book id to update book info: ";
                if (!(cin >> id)) {
                        cin.clear();
                        cin.ignore(100, '\n');
                        cout << "\nInvalid input. Please enter a number.\n" << endl;
                        continue;
                }
                for (i = 0; i < bookCount; i++) {
                        if (bookcatalogue[i].id == id) {
                                found = true;
                                do {
                                        cout << "\n1. Update Title" << endl;
                                        cout << "2. Update Category" << endl;
                                        cout << "3. Update Price" << endl;
                                        cout << "4. Return" << endl;
                                        cout << "\nSelect field to update: ";
```

```
                                        // check if the input is invalid
                                        if (!(cin >> choice)) {
                                                cin.clear();
                                                cin.ignore(100, '\n');
                                                cout << "\nInvalid input. Please enter a number." <<
endl;
                                                continue;
                                        }
                                        switch (choice) {
                                        case 1:
                                                cout << "\nEnter new title: ";
                                                cin.ignore();
                                                getline(cin, bookcatalogue[i].title);
                                                break;
                                        case 2:
                                                cout << "\nEnter new category: ";
                                                cin.ignore();
                                                getline(cin, bookcatalogue[i].category);
                                                break;
                                        case 3:
                                                cout << "\nEnter new price: ";
                                                cin >> bookcatalogue[i].price;
                                                break;
                                        case 4:
                                                return;
                                                break;
                                        default:
                                                cout << "\nInvalid choice." << endl;
                                                break;
                                        }
                                        if (choice == 1 || choice == 2 || choice == 3) {
                                                saveBook();
                                                cout << "\nBook info updated successfully" << endl;
                                                cout << "\n==============================
CURRENT BOOK CATALOGUE ==============================" << endl;
                                                displayBook();
                                                backToMenu();
                                                return;
                                        }
                                } while (choice <1 || choice > 4);
                        }
                }
                if (!found) { // the book is not exist
                        cout << "\nBook not found." << endl;
                }
        } while (!found);
}
```

- addBook()

```
// add new book
void addBook() {
        if (bookCount >= MAX_BOOKS) {
                cout << "Book catalogue is full!" << endl;
```

```
                return;
        }
        book_catalogue newBook;
        // new id for new book
        newBook.id = (bookCount == 0) ? 1 : bookcatalogue[bookCount - 1].id + 1;
        cout << "Enter book title: ";
        cin.ignore();
        getline(cin, newBook.title);
        cout << "Enter book category: ";
        getline(cin, newBook.category);
        cout << "Enter book price: ";
        cin >> newBook.price;
        cout << "Enter book stock: ";
        cin >> newBook.stock;
        bookcatalogue[bookCount] = newBook;
        bookCount++;
        // save the new book into book catalogue
        saveBook();
        cout << "\nBook added successfully!" << endl;
        cout << "\n================================ CURRENT BOOK CATALOGUE
================================" << endl;
        displayBook();
        backToMenu();
        return;
}
```

- removeBook()

```
// remove book
void removeBook() {
        int id, i;
        bool found = false;
        do {
                cout << "\n================================== BOOK CATALOGUE
==================================" << endl;
                displayBook();
                cout << "\nEnter book id to remove: ";
                if (!(cin >> id)) {
                        cin.clear();
                        cin.ignore(100, '\n');
                        cout << "\nInvalid input. Please enter a number.\n" << endl;
                        continue;
                }
                for (i = 0; i < bookCount; i++) { //find the corresponding book
                        if (bookcatalogue[i].id == id) {
                                found = true; //book is exist
                                for (int j = i; j < bookCount - 1; j++) {
                                        bookcatalogue[j] = bookcatalogue[j + 1]; //replace the book
                                }
                                bookCount--;
                                for (int k = 0; k < bookCount; k++) {
                                        bookcatalogue[k].id = k + 1; //rearrange the id
                                }
                                cout << "Book removed successfully!" << endl;
                                saveBook();
```

```
                                        cout << "\n============================== CURRENT BOOK
CATALOGUE ==============================" << endl;
                                        displayBook();
                                        backToMenu();
                                        return;
                                }
                        }
                        if (!found) {
                                cout << "\nBook not found" << endl;
                        }
                } while (!found);
}
```

- stock()

```
void stock() {
        int choice, i,id;
        bool found = false;
        do {
                cout << "\n==== STOCK MANAGEMENT ====" << endl;
                cout << "\n1. View Stock" << endl;
                cout << "2. Update Stock" << endl;
                cout << "3. Return" << endl;
                cout << "\nEnter your choice: ";
                if (!(cin >> choice)) {
                        cin.clear();
                        cin.ignore(100, '\n');
                        cout << "\nInvalid input. Please enter a number." << endl;
                        continue;
                }
                switch (choice) {
                case 1: // view stock
                        cout << "\n================================== VIEW STOCK
==================================" << endl;
                        displayBook();
                        backToMenu();
                        return;
                        break;
                case 2: // edit stock
                        do {
                                cout << "\n====================================== STOCK
========================================" << endl;
                                displayBook();
                                cout << "\nEnter book id to update stock: ";
                                if (!(cin >> id)) {
                                        cin.clear();
                                        cin.ignore(100, '\n');
                                        cout << "\nInvalid input. Please enter a number." << endl;
                                        continue;
                                }
                                for (i = 0; i < bookCount; i++) {
                                        if (bookcatalogue[i].id == id) {
                                                found = true;
                                                cout << "Enter new stock: ";
```

```
                                                cin >> bookcatalogue[i].stock;
                                                saveBook(); //save new stock for the book
                                                cout <<
"\n==================================== CURRENT STOCK
==================================" << endl;
                                                displayBook();
                                                backToMenu();
                                                return;
                                        }
                                }
                                if (!found) {
                                        cout << "\nBook not found" << endl;
                                }
                        } while (!found);
                        break;
                case 3:
                        return;
                        break;
                default:
                        cout << "\nInvalid choice" << endl;
                        break;
                }
        } while (choice != 3);
}
```

- salesReport()

```
// generate sales report
void salesReport() {
        int choice;
        string date, month;
        bool found;
        double total_revenue;
        int total_quantity;
        do {
                cout << "\n========== Sales Report ==========" << endl;
                cout << "1. Daily Sales Report" << endl;
                cout << "2. Monthly Sales Report" << endl;
                cout << "3. Return" << endl;
                cout << "\nEnter your choice: ";
                if (!(cin >> choice)) {
                        cin.clear();
                        cin.ignore(100, '\n');
                        cout << "\nInvalid input. Please enter a number." << endl;
                        continue;
                }
                switch (choice) {
                case 1: // daily sales report
                        total_revenue = 0;
                        total_quantity = 0;
                        found = false;
                        cout << "\nDate (YYYY-MM-DD): ";
                        cin >> date;
                        cout << "\n===== Daily Sales Report =====" << endl;
                        cout << "\n" << left << setw(35) << "TITLE" << left << setw(15) << "SOLD" <<
```

```
"SALES" << endl;
                        cout << "------------------------------------------------------------------------" << endl;
                        for (int i = 0; i < dailyCount; i++) {
                                if (daily_sales[i].key == date) {
                                        found = true;
                                        cout << left << setw(36) << daily_sales[i].title << left <<
setw(14) << daily_sales[i].quantity << fixed << setprecision(2) << daily_sales[i].revenue << endl;
                                        total_quantity += daily_sales[i].quantity; // calculate the total
quantity
                                        total_revenue += daily_sales[i].revenue; // calculate the total
revenue
                                }
                        }
                        cout << "\n------------------------------------------------------------------------" << endl;
                        if (found) {
                                cout << "\nTotal Sold: " << total_quantity << endl;
                                cout << "Total Daily Sales: " << total_revenue << endl;
                        }
                        if (!found){
                                cout << "Sales not found on " << date << endl;
                        }
                        backToMenu();
                        return;
                        break;
                case 2: // monthly sales report
                        total_revenue = 0;
                        total_quantity = 0;
                        found = false;
                        cout << "\nMonth (YYYY-MM): ";
                        cin >> month;
                        cout << "\n===== Monthly Sales Report =====" << endl;
                        cout << "\n" << left << setw(35) << "TITLE" << left << setw(15) << "SOLD" <<
"SALES" << endl;
                        cout << "------------------------------------------------------------------------" << endl;
                        for (int i = 0; i < monthlyCount; i++) {
                                if (monthly_sales[i].key == month) {
                                        found = true;
                                        cout << left << setw(36) << monthly_sales[i].title << left <<
setw(14) << monthly_sales[i].quantity << fixed << setprecision(2) << monthly_sales[i].revenue << endl;
                                        total_quantity += monthly_sales[i].quantity; // calculate the total
quantity
                                        total_revenue += monthly_sales[i].revenue; // calculate the total
revenue
                                }
                        }
                        cout << "\n------------------------------------------------------------------------" << endl;
                        if (found) {
                                cout << "\nTotal Sold: " << total_quantity << endl;
                                cout << "Total Monthly Sales: " << total_revenue << endl;
                        }
                        if (!found) {
                                cout << "Sales not found on " << date << endl;
                        }
                        backToMenu();
                        return;
                        break;
                default:
                        cout << "Invalid selection." << endl;
                }
```

```
        } while (choice != 3);
}
```

- orderReport()

```
// generate order report
void orderReport() {
        bool found = false;
        string date;
        cout << "\nDate (YYYY-MM-DD): ";
        cin >> date;
        cout << "\n============== Order Report on " << date << " ==============" << endl;
        for (int i = 0; i < order_detail.size(); i++) {
                if (order_detail[i].order_date == date) {
                        found = true;
                        cout << "Order ID: " << order_detail[i].order_id << "\n" << "Customer Name: "
<< order_detail[i].username << "\n" << "Title: " << order_detail[i].book_title << "\n" << "Price: " <<
order_detail[i].book_price << "\n" << "Quantity: " << order_detail[i].quantity << "\n" << "Subtotal: " <<
order_detail[i].subtotal << "\n" << "Total: " << order_detail[i].total << "\n" << "Payment method: " <<
order_detail[i].payment_method << "\n" << "Order Date: " << order_detail[i].order_date << endl;
                        cout << "\n------------------------------------------------------------------------\n" << endl;
                }
        }
        if (!found) {
                cout << "Order not found on " << date << endl;
        }
        backToMenu();
        return;
}
```

Customer.cpp

```
#include <iostream>
#include <string>
#include "data.h"
#include <iomanip>
#include <ctime>
using namespace std;

string getCurrentDate();
void clearScreen();
int clearShoppingCart();
void backToMenuPage();
void customerLogin();
void customerRegister();
void customerMenu();
void order_Entry();
void addToCart(int, int);
void browseBooks();
void ViewShoppingCart();
void processCheckOut();
void orderHistory();

// shopping cart item information
struct CartItem {
    int id = 0;
    int quantity = 0;
};

const int MAX_CART_ITEMS = 20; // maximum number of items in the shopping cart
CartItem cart[MAX_CART_ITEMS];
int cartCount = 0; // number of items in the shopping cart
int order_id;
string username;

// get the current date
string getCurrentDate() {
    time_t timestamp = time(NULL);
    struct tm datetime;
    localtime_s(&datetime, &timestamp);
    char output[20];
    strftime(output, sizeof(output), "%Y-%m-%d", &datetime);
    return string(output);
}

int clearShoppingCart() {
    cartCount = 0;
    return cartCount;// clear shoppingc card to 0.
}

// back to customer menu
void backToMenuPage() {
    char key;
    cout << "\n\nPress 'Enter' to return to customer menu..... ";
    cin.ignore();
    cin.get(key);
    clearScreen();
}
```

```
// customer login
void customerLogin() {
    string password;
    bool loginSuccess = false;

    cout << "\n========== Customer Login ==========\n";
    cout << "Enter username: ";
    cin >> username;
    cout << "Enter password: ";
    cin >> password;

    for (int i = 0; i < customerCount; i++) {
        if (customer_info[i].username == username && customer_info[i].password == password) {
            cout << "\nCustomer login successful\n" << endl;
            clearScreen();
            customerMenu();
            loginSuccess = true;
            break;
        }
    }

    if (!loginSuccess) {
        cout << "\nInvalid username or password.\n" << endl;
    }
}

// register new customer
void customerRegister() {
    int i;
    bool found;
    if (customerCount >= MAX_CUSTOMER) { // check if the number of customer exceed the customer
accomoadation
        cout << "Registration is full!" << endl;
        return;
    }
    customer newCustomer; // add new customer
    do {
        cout << "\n========== CUSTOMER REGISTRATION ==========\n";
        cout << "Enter your username: ";
        cin >> newCustomer.username;
        found = false;
        // check if the username is already being used
        for (i = 0; i < customerCount; i++) {
            if (customer_info[i].username == newCustomer.username) {
                found = true;
                cout << "Username already exists! " << endl;
            }
        }
        if (!found) {
            break;
        }
    } while (found);
    // the username is not being used
    cout << "Enter your password: ";
    cin >> newCustomer.password;
    cout << "Registration successfully" << endl;
    customer_info[customerCount] = newCustomer; // add new customer
    customerCount++; // add the number of customer
    saveCustomer(); // save the new customer into customer info
```

```
}


//------------------------main customer page-----------------------------------------------------------
void customerMenu() {

    int customerChoice;

    cout << "\t\t\t  ______  _    ____   _____ ___    _   _   _                " << endl;
    cout << "\t\t\t |_   _|/ \\  |  _ \\ / ___|_ _|  / \\  | \\ | |         " << endl;
    cout << "\t\t\t   | | / _ \\ | |_) | |    | |  / _ \\ |  \\| |          " << endl;
    cout << "\t\t\t   | |/ ___ \\|  _ <| |___ | | / ___ \\| |\\  |          " << endl;
    cout << "\t\t\t  _|_/_/ __\\_\\_|_\\_\\\\____|___/_/___\\_\\_| \\_|_  _____ " << endl;
    cout << "\t\t\t | __ ) / _ \\ / _ \\| |/ / ___|_   _/ _ \\|  _ \\| ____|" << endl;
    cout << "\t\t\t |  _ \\| | | | | | | ' /\\___ \\ | || | | | |_) |  _|  " << endl;
    cout << "\t\t\t | |_) | |_| | |_| | . \\ ___) || || |_| |  _ <| |___ " << endl;
    cout << "\t\t\t |____/ \\___/ \\___/|_|\\_\\____/ |_| \\___/|_| \\_\\_____|" << endl;

    cout << "\n======================================== Customer Menu
========================================\n\n";
    cout << right << setw(53) << "TARCIAN BOOKSTORE" << endl;
    cout << "\t\t\t\tNo. K666, Block K TARUMT," << endl;
    cout << right << setw(63) << "Jalan Genting Klang, Setapak, 53000" << endl;
    cout << right << setw(50) << "Kuala Lumpur." << endl;
    cout << "\n";
    // discount announcement
    cout << " _____" << endl;
    cout << "|                                       |" << endl;
    cout << "| Special OFFER !!!                     |" << endl;
    cout << "| Spend RM50 to get a 5% discount!!     | " << endl;
    cout << "| Spend RM100 to get a 10% discount!! |" << endl;
    cout << "|_____|" << endl;

    do {
        cout << "\nDear customer. Welcome to TARCIAN Book Ordering System!\n" << endl;
        cout << "1. Order Entry" << endl;
        cout << "2. Browse all Books" << endl;
        cout << "3. View Shopping Cart" << endl;
        cout << "4. Process to Checkout" << endl;
        cout << "5. View Order History" << endl;
        cout << "6. Logout" << endl;
        cout << "\nChoice: ";
        if (!(cin >> customerChoice)) { // check the input validation
            cin.clear();
            cin.ignore(100, '\n');
            cout << "\nInvalid input. Please enter a number.\n" << endl;
            continue;
        }
        cout << "\nYou have selected option " << customerChoice << ".\n\n";
        switch (customerChoice) {
        case 1:
            order_Entry();
            break;
        case 2:
            browseBooks();
            break;
        case 3:
            ViewShoppingCart();
            break;
```

```
        case 4:
            processCheckOut();
            break;
        case 5:
            orderHistory();
            break;
        case 6:
            cout << "Thank you for visiting TARCIAN Book Ordering System!\n";
            clearScreen();
            return;
            break;
        default:
            cout << "Invalid choice. Please try again." << endl;
            break;
        }
    } while (customerChoice != 6);
}

// order for book
void order_Entry() {
    int id, quantity, month;
    char continueOrder;
    bool found;
    int orderNum = 1;

    cout << "\n================================== BOOK CATALOGUE
==================================" << endl;
    displayBook();
    do {
        cout << "\nOrder " << orderNum << endl;
        do {
            found = false;
            cout << "Enter the book ID to add to shopping cart: ";
            if (!(cin >> id)) {
                cin.clear();
                cin.ignore(100, '\n');
                cout << "\nInvalid input. Please enter a number.\n" << endl;
                continue;
            }
            for (int i = 0; i < bookCount; i++) {
                if (bookcatalogue[i].id == id) {
                    found = true;
                    int already_quantity = 0;
                    for (int k = 0; k < cartCount; k++) {
                        if (cart[k].id == bookcatalogue[i].id) {
                            already_quantity = cart[k].quantity;
                            break;
                        }
                    }
                    do {
                        cout << "Enter quantity: ";
                        if (!(cin >> quantity)) {
                            cin.clear();
                            cin.ignore(100, '\n');
                            cout << "\nInvalid input. Please enter a number.\n" << endl;
                            continue;
                        }

                        if (quantity > 0 && (quantity + already_quantity) <= bookcatalogue[i].stock) {
```

```
                    addToCart(id, quantity);
                    cout << "Added to cart: " << quantity
                        << " copy/copies of \"" << bookcatalogue[i].title << "\"" << endl;
                    already_quantity += quantity;
                    break;
                }
                else {
                    if (quantity == 0) {
                        break;
                    }
                    cout << "Invalid quantity. ";
                    cout << "Available stock: " << bookcatalogue[i].stock - already_quantity<< endl;
                    if (bookcatalogue[i].stock - already_quantity == 0) {
                        break;
                    }
                }
            } while (true);

            break;
        }
    }
    if (!found) {
        cout << "Invalid Book ID, Please Enter Again.\n" << endl;
    }

} while (!found);

// continue order or exit
do {
    cout << "\nAnymore order (Y/N)? ";
    cin >> continueOrder;
    if (!(continueOrder == 'Y' || continueOrder == 'y' || continueOrder == 'N' || continueOrder == 'n')) {
        cout << "Invalid input. Please try again.\n" << endl;
    }
} while (!(continueOrder == 'Y' || continueOrder == 'y' || continueOrder == 'N' || continueOrder == 'n'));

orderNum++;

} while (continueOrder == 'Y' || continueOrder == 'y');

backToMenuPage();
return;

}

// add the book to shopping cart
void addToCart(int id, int quantity) {
    // check if the book is already exist in the shopping cart
    for (int i = 0; i < cartCount; i++) {
        if (cart[i].id == id) { // if yes
            cart[i].quantity += quantity; // calculate the total quantity
            return;
        }
    }
    //if no,add new item to cart
    cart[cartCount].id = id;
    cart[cartCount].quantity = quantity;
    cartCount++; // shopping cart count +1
}
```

```
// view book catalogue
void browseBooks() {
   cout << "\n==================================== BOOK CATALOGUE
====================================" << endl;
   displayBook();
   backToMenuPage();
   return;
}

// view shopping cart
void ViewShoppingCart() {

   double total = 0.0;

   if (cartCount == 0) { // check if the cart is empty
      cout << "\n====================================== Hi I'm your Shopping Cart
======================================\n\n";
      cout << "!!! Your shopping cart is empty yet.!!!" << endl;
      cout << "Please add some books to your cart first." << endl;

      backToMenuPage();
      return;
   }
   // the cart is not empty
   cout << "\n============================ Hi I'm your Shopping Cart
============================\n\n";
   cout << fixed << setprecision(2); // Set decimal precision for prices
   cout << left << setw(5) << "No." << setw(35) << "Title" << setw(10) << "Qty" << setw(15) << "Price
(RM)" << setw(15) << "Total (RM)" << endl;
   cout << "--------------------------------------------------------------------------------" << endl;
   for (int i = 0; i < cartCount; i++) {
      //Find the book corresponding to this bookId
      for (int j = 0; j < bookCount; j++) {
         if (bookcatalogue[j].id == cart[i].id) {
            total = cart[i].quantity * bookcatalogue[j].price;
            cout << left << setw(5) << i + 1 << setw(36) << bookcatalogue[j].title << setw(10) <<
cart[i].quantity << setw(15) << bookcatalogue[j].price << setw(10) << total << endl;
         }
      }
   }
   // remove item from shopping cart
   char confirm;
   do {
      cout << "\nDo you want to remove an item from cart? (Y/N): ";
      if (!(cin >> confirm)) {
         cin.clear();
         cin.ignore(100, '\n');
         cout << "Invalid input." << endl;
         continue;
      }
      if (confirm != 'Y' && confirm != 'y' && confirm != 'n' && confirm != 'N') {
         cout << "Invalid input." << endl;
      }
   } while (confirm != 'Y' && confirm != 'y' && confirm != 'n' && confirm != 'N');

   if (confirm == 'N'||confirm=='n') { // no remove action
      cout << "Okay, nothing removed.\n";
      backToMenuPage();
```

```
        return;
    }
    // remove action
    int choice;
    do {
        cout << "Enter the item number to remove (0 to cancel): ";
        if (!(cin >> choice)) {
            cin.clear();
            cin.ignore(100, '\n');
            cout << "Invalid input.\n";
            continue;
        }
        if (choice == 0) { // cancel the remove
            cout << "Remove cancelled.\n";
            backToMenuPage();
            return;
        }

        if (choice < 1 || choice > cartCount) {
            cout << "Invalid item number.\n";
        }
    } while (choice < 1 || choice > cartCount);

    // remove the corresponding item
    for (int i = choice - 1; i < cartCount - 1; i++) {
        cart[i] = cart[i + 1]; //replace the item
    }
    cartCount--; //cart count decreased by 1

    cout << "Item removed successfully.\n";

    // display the current shopping cart
    if (cartCount == 0) { // shopping cart is empty
        cout << "\nNow your shopping cart is empty.\n";
        backToMenuPage();
        return;
    }
    else {
        cout << "\nUpdated Shopping Cart:\n";
        total = 0.0;
        cout << left << setw(5) << "No." << setw(35) << "Title" << setw(10) << "Qty" << setw(15) << "Price
(RM)" << setw(15) << "Total (RM)" << endl;
        cout << "---------------------------------------------------------------------------------" << endl;

        for (int i = 0; i < cartCount; i++) {
            for (int j = 0; j < bookCount; j++) {
                if (bookcatalogue[j].id == cart[i].id) {
                    double itemTotal = cart[i].quantity * bookcatalogue[j].price;
                    total += itemTotal;
                    cout << left << setw(5) << i + 1 << setw(36) << bookcatalogue[j].title << setw(10) <<
cart[i].quantity << setw(15) << bookcatalogue[j].price << setw(15) << itemTotal << endl;
                }
            }
        }
        backToMenuPage();
        return;
    }
}
```

```
// checkout
void processCheckOut() {
    char confirmCheckout;

    // check if the shopping cart is empty
    if (cartCount == 0) {
        cout << "Remainder: " << endl;
        cout << "!!! Your shopping cart is empty yet.!!!" << endl;
        cout << "Please add some books to your cart first." << endl;

        backToMenuPage();
        return;
    }
    // shopping cart is not empty
    do {
        cout << "Are you confirm to process checkout (Y/N): ";
        cin >> confirmCheckout;

        if (confirmCheckout == 'N' || confirmCheckout == 'n') {
            cout << "\n(return to customer menu)" << endl;
            clearScreen();
            return;
        }
        else if (confirmCheckout != 'Y' && confirmCheckout != 'y') {
            cout << "Please enter Y or N." << endl;
        }
    } while (confirmCheckout != 'Y' && confirmCheckout != 'y');

    cout << "Okay... Then let's goo!" << endl;


    //------------------------------------------------ invoice page ------------------------------------------------
    double Total = 0.0, subtotal = 0.0;
    string typePaymentMethod;
    char paymentMethod;
    string date = getCurrentDate(); // checkout date

    if (!order_detail.empty()) { // check if the order_detail is empty
        order_id = order_detail.back().order_id + 1; // if not, order_id + 1
    }
    else {
        order_id = 1; // if yes, set the order_id with 1
    }

    clearScreen();

    cout << "\n\nInvoice: \n" << endl;
    cout << right << setw(53) << "TARCIAN BOOKSTORE" << endl;
    cout << "\t\t\t\tNo. K666, Block K TARUMT," << endl;
    cout << right << setw(63) << "Jalan Genting Klang, Setapak, 53000" << endl;
    cout << right << setw(50) << "Kuala Lumpur." << endl;
    cout << right << setw(53) << "Tel : 03 - 1314520 " << endl;
    cout << "\n";

    cout << "Invoice No. INV: " << order_id << right << setw(53) << "Date: " << date << endl;

    cout << "--------------------------------------------------------------------------------" << endl;
    cout << fixed << setprecision(2); // Set decimal precision for prices
    cout << left << setw(5) << "ID" << setw(35) << "Title" << setw(10) << "Qty" << setw(15) << "Price
```

```
(RM)" << setw(15) << "Total (RM)" << endl;
    cout << "--------------------------------------------------------------------------------" << endl;


    for (int i = 0; i < cartCount; i++) {
        // find the corresponding book
        for (int j = 0; j < bookCount; j++) {
            if (bookcatalogue[j].id == cart[i].id) {
                Total = cart[i].quantity * bookcatalogue[j].price;
                subtotal += Total;

                cout << left << setw(5) << i + 1 << setw(36) << bookcatalogue[j].title << setw(10) <<
cart[i].quantity << setw(15) << bookcatalogue[j].price << setw(10) << Total << endl;
            }
        }
    }
    // calculate discount, tax and total payable
    double serviceTax = 0.0, SST = 0.0, totalPayable = 0.0, discountPrice = 0.0;
    string discountPercentage;

    if (subtotal > 0 && subtotal < 50) {
        discountPercentage = "(0%) : ";
        discountPrice = subtotal * 0.00;
    }
    else if (subtotal >= 50 && subtotal < 100) {
        discountPercentage = "(5%) : ";
        discountPrice = subtotal * 0.05;
    }
    else if (subtotal >= 100) {
        discountPercentage = "(10%) :";
        discountPrice = subtotal * 0.10;
    }

    SST = round((subtotal - discountPrice) * 0.06 * 100) / 100.0;
    serviceTax = round((subtotal - discountPrice) * 0.06 * 100) / 100.0;
    totalPayable = subtotal - discountPrice + serviceTax + SST;
    totalPayable = round(totalPayable * 100) / 100.0;

    cout << endl;
    cout << left << setw(66) << "Subtotal : " << "RM" << right << setw(6) << subtotal << endl;
    cout << "Discount " << discountPercentage << right << setw(52) << "- RM" << right << setw(6) <<
discountPrice << endl;
    cout << left << setw(66) << "Service Charge (6%) :" << "RM" << right << setw(6) << serviceTax << endl;
    cout << left << setw(66) << "SST Tax (6%) :  " << "RM" << right << setw(6) << SST << endl;
    cout << left << setw(67) << "\nTotal Payable: " << "RM" << right << setw(6) << totalPayable << endl;

    //payment method selection
    do {
        cout << "\n\n(W = e-Wallet, C = Credit card, D = Debit card, A = Cash)" << endl;
        cout << "Select payment method: ";
        cin >> paymentMethod;

        // check valid payment method
        if (paymentMethod != 'W' && paymentMethod != 'C' && paymentMethod != 'D' &&
paymentMethod != 'A' && paymentMethod != 'w' && paymentMethod != 'c' && paymentMethod != 'd' &&
paymentMethod != 'a') {
            cout << "Invalid payment method. Please try again." << endl;
        }
    } while (paymentMethod != 'W' && paymentMethod != 'C' && paymentMethod != 'D' &&
```

```
paymentMethod != 'A' && paymentMethod != 'w' && paymentMethod != 'c' && paymentMethod != 'd' &&
paymentMethod != 'a');

    switch (paymentMethod) {
    case 'W': case'w':
        typePaymentMethod = "E-Wallet";
        break;
    case 'C': case'c':
        typePaymentMethod = "Credit Card";
        break;
    case 'D': case'd':
        typePaymentMethod = "Debit Card";
        break;
    case 'A': case'a':
        typePaymentMethod = "Cash";
        break;
    }

    double paymentAmount = 0.0;

    do {
        cout << "\nPlease enter you payment amount (Eg.Rm 120.80): RM ";
        cin.clear();
        cin >> paymentAmount;
        // check if the payment amount is enough
        if (paymentAmount < totalPayable) {
            cout << "Not enought payment amount. Please try again.";
        }

    } while (paymentAmount < totalPayable);


    // ------------------------------------------receipt page----------------------------------------------
    cout << "\n\nReceipt: \n" << endl;
    cout << right << setw(53) << "TARCIAN BOOKSTORE" << endl;
    cout << "\t\t\t\tNo. K666, Block K TARUMT," << endl;
    cout << right << setw(63) << "Jalan Genting Klang, Setapak, 53000" << endl;
    cout << right << setw(50) << "Kuala Lumpur." << endl;
    cout << right << setw(53) << "Tel : 03 - 1314520 " << endl;
    cout << "\n";

    cout << "Receipt No.RCP " << order_id << right << setw(55) << "Date: " << date << endl;

    cout << "--------------------------------------------------------------------------------" << endl;
    cout << fixed << setprecision(2);
    cout << left << setw(5) << "ID" << setw(35) << "Title" << setw(10) << "Qty" << setw(15) << "Price
(RM)" << setw(15) << "Total (RM)" << endl;
    cout << "--------------------------------------------------------------------------------" << endl;

    for (int i = 0; i < cartCount; i++) {
        // find the corresponding book
        for (int j = 0; j < bookCount; j++) {
            if (bookcatalogue[j].id == cart[i].id) {
                Total = cart[i].quantity * bookcatalogue[j].price;

                cout << left << setw(5) << i + 1 << setw(36) << bookcatalogue[j].title << setw(10) <<
cart[i].quantity << setw(15) << bookcatalogue[j].price << setw(10) << Total << endl;
            }
        }
```

```
    }

    double change = 0.0;
    change = paymentAmount - totalPayable; // calculate change

    cout << right << setw(81) << "-----------------" << endl;
    cout << left << setw(66) << "Subtotal : " << "RM" << right << setw(6) << subtotal << endl;
    cout << left << "Discount " << discountPercentage << right << setw(52) << "- RM" << right << setw(6) <<
discountPrice << endl;
    cout << left << setw(66) << "Service Charge (6%) :" << "RM" << right << setw(6) << serviceTax << endl;
    cout << left << setw(66) << "SST Tax (6%) :  " << "RM" << right << setw(6) << SST << endl;

    cout << left << setw(67) << "\nTotal Payable: " << "RM" << right << setw(6) << totalPayable << endl;

    cout << right << setw(81) << "-----------------" << endl;
    cout << left << setw(68) << "\n\nPayment Method: " << typePaymentMethod << endl;
    cout << left << setw(66) << "Payment Amount: " << "RM" << paymentAmount << endl;

    cout << left << setw(66) << "Change: " << "RM" << change << endl;

    cout << "\nThank you for using TARCIAN BOOKSTORE! See you next time! >_<" << endl;


    // save the order details and book catalogue
    for (int i = 0; i < cartCount; i++) {
        for (int j = 0; j < bookCount; j++) {
            if (bookcatalogue[j].id == cart[i].id) {
                bookcatalogue[j].stock -= cart[i].quantity;
                order newOrder;
                newOrder.order_id = order_id;
                newOrder.username = username;
                newOrder.book_title = bookcatalogue[j].title;
                newOrder.book_price = bookcatalogue[j].price;
                newOrder.quantity = cart[i].quantity;
                newOrder.subtotal = subtotal;
                newOrder.total = totalPayable;
                newOrder.payment_method = typePaymentMethod;
                newOrder.order_date = date;

                order_detail.push_back(newOrder);
            }
        }
    }
    saveBook();
    saveOrder();
    // Clear the shopping cart after successful checkout
    clearShoppingCart();

    backToMenuPage();
    return;
}

// view order history
void orderHistory() {
    int i;
    bool found = false;
    cout << "\n====================== ORDER HISTORY ====================" << endl;
    cout << "\n";
    for (i = 0; i < order_detail.size(); i++) {
```

```
        if (order_detail[i].username == username) { // check if the customer has order before
            found = true;
            cout << "Order ID: " << order_detail[i].order_id << endl;
            cout << "Date    : " << order_detail[i].order_date << endl;
            cout << "Book    : " << order_detail[i].book_title << endl;
            cout << "Quantity: " << order_detail[i].quantity << endl;
            cout << "Price   : RM " << fixed << setprecision(2) << order_detail[i].book_price << endl;
            cout << "Subtotal: RM " << order_detail[i].subtotal << endl;
            cout << "Total   : RM " << order_detail[i].total << endl;
            cout << "Payment : " << order_detail[i].payment_method << endl;
            cout << "\n";
        }
    }
    if (!found) { //if no
        cout << "You have no order yet" << endl;
    }
    backToMenuPage();
    return;
}
```